

SparshCare — Stakeholder Interview Log

Requirements elicitation • Six stakeholder interviews • Complete record

HIT701 — Software Development & Lifecycle Management • PGDM HHM 2025–27

Analyst: Dr. Gunjan Singhal • All interviews conducted 17 August 2026

Project: **SparshCare** — a field-screening, teleconsultation and referral system for the early detection of leprosy, at Kondagaon PHC, Bastar, Chhattisgarh, under the National Leprosy Eradication Programme.

How to read this document

Six stakeholder interviews were conducted as AI role-play simulations using Claude (Cowork), model claude-opus-5. Each is recorded to an identical structure so that findings can be compared across stakeholders and traced to source.

#	Stakeholder	Role	Reference
1	Sunita Devi	ASHA — frontline field worker	SPARSHCARE / INT-01 / SUNITA-ASHA
2	Dr. A. Rao	Medical Officer, Kondagaon PHC	SPARSHCARE / INT-02 / RAO-MO
3	Dr. Meera Iyer	Dermatologist / Leprologist, District Hospital	SPARSHCARE / INT-03 / MEERA-DERM
4	Mr. S. Verma	District Leprosy Officer, NLEP	SPARSHCARE / INT-04 / VERMA-DLO
5	Ramesh	Beneficiary — person screened and referred	SPARSHCARE / INT-05 / RAMESH-BENEF
6	Priya	IT / System Administrator	SPARSHCARE / INT-06 / PRIYA-SYSADMIN

Every interview contains the same seven parts: **1.** purpose and interview approach • **2.** the question-by-question log • **3.** the top three pains heard • **4.** contradictions and things to verify • **5.** the recorded conflict • **Appendix A**, a consolidated requirements register • and the design rules implied in the stakeholder's own framing.

Classification of captured lines

Every captured line in the question-by-question logs is marked with one or more of five categories:

Tag	Meaning	Register prefix
[Functional]	Something the system must do	FR
[Non-functional]	A quality the system must have — speed, offline working, security, usability	NFR
[Business rule]	An existing organisational or programme rule that constrains the design	BR
[Constraint]	A fixed environmental, physical, economic or regulatory limit	CON
[Emotional/cognitive]	Fear, trust, stigma, mental load, motivation	EC

Each answer additionally carries a roll-up tag line and is marked *need* (a problem to be solved) or *solution* (a proposed answer, to be traced back to the underlying need).

Requirement identifiers are unique across the whole document; each interview draws from its own reserved block, so no identifier is ever reused for two different requirements.

Interview	Functional	Non-functional	Business rule	Constraint	Emotional/cognitive
1 — Sunita Devi	FR-01-FR-10	NFR-01-NFR-10	BR-01-BR-03	CON-01-CON-05	EC-01
2 — Dr. Rao	FR-11-FR-29	NFR-11-NFR-16	BR-04-BR-09	CON-06-CON-11	EC-02-EC-04
3 — Dr. Meera	FR-30-FR-49	NFR-17-NFR-26	BR-10-BR-15	CON-12-CON-17	EC-05-EC-08
4 — Mr. Verma	FR-50-FR-69	NFR-27-NFR-36	BR-16-BR-21	CON-18-CON-23	EC-09-EC-12
5 — Ramesh	FR-70-FR-84	NFR-37-NFR-46	BR-22-BR-26	CON-24-CON-29	EC-13-EC-17
6 — Priya	FR-85-FR-104	NFR-47-NFR-60	BR-27-BR-33	CON-30-CON-36	EC-18-EC-20

Where a later interview reinforces a requirement already raised in an earlier one, the original identifier is cross-referenced rather than a new one issued.

A note on the personas and the figures

Names, the facility and all numbers in this document are illustrative, produced through structured role-play for the purposes of this studio. They are close to real NLEP field conditions but must not be quoted as official statistics. Items that would need checking against records before being used as a basis for design are listed in section 4 of each interview.

STAKEHOLDER INTERVIEW LOG

SparshCare — Requirements elicitation · Interview 1 of 6

Field	Value
Analyst	Dr. Gunjan Singhal
Date of interview	17 August 2026
Stakeholder role	ASHA (Accredited Social Health Activist)
Stakeholder (persona)	Sunita Devi, 34 — 9 years in service
Location / facility	Kondagaon PHC, Bastar, Chhattisgarh
Project	SparshCare — leprosy screening, teleconsultation & referral system
AI tool used	Claude (Cowork) — model claude-opus-5, role-play persona “Sunita Devi, ASHA”
Interview mode	Semi-structured, single session, 21 questions with follow-up probes
Duration (in-persona)	~30 min (stakeholder left for a delivery case)
Chat link / screenshot ref	SPARSHCARE / INT-01 / SUNITA-ASHA — [paste chat link or screenshot filename]

1. PURPOSE AND INTERVIEW APPROACH

Objective: understand how leprosy screening and referral actually work in the field today, before any feature of SparshCare is specified — and in particular to surface the non-functional and social constraints that would decide whether a field worker adopts or abandons the app.

Question flow used (deliberate sequence — rapport → current process → evidence → pain → constraints → conflict → solution ideas last):

- Q1–Q2 Rapport and role: tenure, weekly cycle, how she came to leprosy work, training received.
- Q3–Q5 Baseline and evidence: volume of suspected vs confirmed cases, caseload per day, characteristics of late-presenting cases.
- Q6–Q8 Current ‘as-is’ process end to end: one house walkthrough, what is written and when, where the paper goes afterwards.
- Q9–Q11 Existing digital experience and failure stories: what apps exist, what works, what frustrates, a concrete data-loss incident.
- Q12–Q14 Referral drop-off, outcome visibility and connectivity — the three functional gaps.
- Q15–Q18 Environment and confidentiality: who is in the room, who can see the screen, stigma, hidden cases, device ownership.

- Q19 Deliberate conflict probe: the district's request for ~25 data fields per person.
- Q20–Q21 Only now: open solution question, then a read-back and forced prioritisation to confirm understanding.

Tag key: Each captured line is marked with one or more of five categories — **[Functional]** (something the system must do) · **[Non-functional]** (a quality it must have) · **[Business rule]** (an existing organisational rule) · **[Constraint]** (a fixed environmental, economic or regulatory limit) · **[Emotional/cognitive]** (fear, trust, stigma, mental load). Each answer additionally carries a roll-up tag and is marked *need* (a problem to be solved) or *solution* (a proposed answer, to be traced back to the underlying need). In Appendix A these categories are carried forward as ID prefixes: FR = functional, NFR = non-functional, BR = business rule, CON = constraint, EC = emotional/cognitive.

2. QUESTION-BY-QUESTION LOG

Q1: Tell me about yourself — how long have you been an ASHA, and what all does your work cover in a normal week?

A1:

- Nine years in service; start year given uncertainly as “2017 — no, before that, maybe 2016”. **[Emotional/cognitive]**
- Weekly cycle: Monday at the sub-centre collecting the ANM's due-list (immunisation, antenatal); Tuesday–Wednesday house-to-house across her own village plus two hamlets, the farther one a 40-minute walk “if the road is dry”. **[Business rule]** **[Constraint]**
- Routine work = maternal check-ins, infant weighing, vaccination reminders, tablet compliance, referring fevers to the PHC. **[Business rule]**
- Leprosy skin screening is NOT part of the weekly cycle — it happens only “when the block sends word” for a campaign. **[Business rule]**
- Saturday is spent re-copying registers at home — unpaid, invisible administrative work. **[Constraint]** **[Emotional/cognitive]**

“Saturday, mostly I am sitting with my registers, copying. That is the day nobody sees me working but I am working.”

→ Tags: **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | *need*

Q2: How did you come to do leprosy screening — who trained you, and when was the last campaign?

A2:

- Training was 1–1.5 days at the block office, roughly three years ago; delivered as pictures projected on a cloth screen plus a demonstration of the cotton test, explained in Hindi. **[Business rule] [Constraint]**
- One half-day refresher last year, which she experienced mainly as being reprimanded for low case numbers. **[Business rule] [Emotional/cognitive]**
- Last campaign: after Holi — March, spilling into April. Instruction was to check every household and every member. **[Business rule]**
- Design implication: clinical vocabulary cannot be assumed. She describes the disease entirely through observable field signs, never clinical terminology. **[Non-functional]**

“Then last year they called us again for one morning, a refresher they said, but mostly they scolded us about the numbers being low.”

→ Tags: **[Non-functional] [Business rule] [Constraint] [Emotional/cognitive]** | *need*

Q3: In all your years of work, how many people have you found with these pale patches? And how many turned out to be the disease?

A3:

- Flags 8–10 people per campaign; most turn out to be fungal marks, vitiligo-type white spots, burn scars — she screens these out largely by whether the patch itches. **[Functional]**
- Confirmed cases across nine years: “four. Maybe five” — then revised down to “maybe four, maybe three”. She keeps no record of confirmed outcomes. **[Constraint]**
- Recalls three vividly by name/description; one was lost to follow-up when the woman moved to her mother’s village. **[Functional] [Emotional/cognitive]**
- Notable: no one has ever asked her for this figure before, and she has no way of producing it. **[Functional] [Constraint]**

“The fourth one I am not sure about — she went to her mother’s village and I never heard again. So maybe four, maybe three. I don’t keep a separate count of that.”

→ Tags: **[Functional] [Constraint] [Emotional/cognitive]** | *need*

Q4: On a campaign day, how many houses do you cover? Out of those, how many do you feel worried about?

A4:

- 25–30 households per day — but a household is 6–7 people, so “sometimes I am looking at a hundred and something people in a day”. **[Constraint]**

- Screening is not orderly: men are in the fields until evening, young women will not expose their arms in front of others so she takes them behind the kitchen, locked houses need a second visit. **[Constraint]**
- 2–3 people per day cause concern; roughly 1 per day she is confident about — “I feel it in my stomach”. **[Emotional/cognitive]**
- She volunteers the deeper worry unprompted: the people she never sees at all because they conceal symptoms. **[Constraint] [Emotional/cognitive]**

“The ones I worry about most are the ones I never see. People hide it. If it is on the back, on the thigh, they simply don’t tell me.”

→ Tags: **[Constraint] [Emotional/cognitive]** | *need*

Q5: Among the ones who were confirmed — how many already had bent hands or a foot they drag? What was different about those people?

A5:

- Two of four confirmed cases already had visible, irreversible damage at detection: clawed fingers, a dragging foot with an unnoticed ulcer on the sole bandaged in a dirty cloth; a woman with a weak grip who tied her sickle to her hand. **[Functional] [Constraint]**
- Common factors she identifies herself: (a) both elderly — families dismiss skin and nerve changes as ageing; (b) both socially isolated — widowed, or living in a household nobody visits, so no one prompts them to seek care. **[Functional] [Emotional/cognitive]**
- Symptoms had been present 2–3 years before detection and were self-attributed to hard manual work. **[Constraint] [Emotional/cognitive]**
- Contrast case: the young boy detected early was brought to her by his mother — “somebody was watching him”. **[Emotional/cognitive]**
- This is a usable risk-stratification rule of thumb generated from field observation, not from training. **[Functional]**

“Two years, sir. In two years I had walked past his door many times.”

→ Tags: **[Functional] [Constraint] [Emotional/cognitive]** | *need*

Q6: How are you managing screening and referral today? Walk me through one house from start to finish.

A6:

- Arrives early, and deliberately conceals the purpose — announces a “government health survey”, never the disease, because a neighbour sitting outside will overhear. **[Functional] [Emotional/cognitive]**

- Builds rapport first (water, family news), then asks to see everyone's hands and feet as 'routine'. Men roll up sleeves in the open; women are examined indoors. **[Business rule]** **[Constraint]**
- Examination sequence: inspect arms, legs, shoulders → if a pale patch, ask duration and whether it itches → cotton test with eyes closed, comparing patch against adjacent normal skin → press finger on palm and sole → ask about tingling, weakness, footwear slipping off unnoticed. **[Functional]** **[Business rule]**
- Records name, age, father's/husband's name and findings in the register — or on a loose sheet if the register is not in her bag. **[Functional]** **[Constraint]**
- Referral is verbal: "go to the PHC, Kondagaon, meet the doctor"; she tells them Tuesday and Friday for the skin doctor, "though he is not always there". Sometimes she writes an unsealed slip saying 'went for skin check, please see', which is often lost. **[Business rule]** **[Constraint]**
- Process then simply ends. There is no follow-up step of any kind. **[Functional]** **[Business rule]**

"And then, sir — then it is finished. I walk to the next house. That is the part that troubles me. I never know what happened. I tell twenty people go to the PHC and maybe six go."

→ Tags: **[Functional]** **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | *need*

Q7: What exactly do you write on the form, and at what point do you write it — in the house, or later at night?

A7:

- In the house she writes only name and age — because names are easily confused ("in one hamlet there are four Sukhram"). **[Functional]** **[Constraint]**
- She deliberately does NOT write the clinical finding in the house: people lean over and read it, and she cannot write 'no feeling in patch' while the man's brother reads over her shoulder. She instead makes a private tick or small circle only she understands. **[Functional]** **[Emotional/cognitive]**
- The full record is completed at night by lamplight from memory, amid family noise — and she admits she sometimes guesses which arm, or which household a symptom belonged to. **[Non-functional]** **[Constraint]**
- Paper failure: monsoon damage. Loose slips are worse — one was found washed and blank in her clothing two weeks later. **[Constraint]**
- This is a self-invented confidentiality workaround that any digital design must preserve, not remove. **[Non-functional]** **[Emotional/cognitive]**

"I make a small mark, a tick or a small circle in the corner, something only I understand, and I move on... Then at night I sit with the lamp and fill it properly."

→ Tags: [Functional] [Non-functional] [Constraint] [Emotional/cognitive] | need

Q8: Once the form is filled, where does it go and who sees it after that?

A8:

- Her own register stays with her at home (now kept in a plastic bag after a rain incident). [Constraint]
- Campaign forms listing suspected persons go to the ANM at the sub-centre, who tallies numbers; from there to the block supervisor, who sometimes collects all ASHAs' forms together. Beyond the block she does not know where the data goes. [Business rule] [Constraint]
- There is no return path. In nine years she has never once been told how many of the people she flagged attended, or how many were confirmed. [Functional] [Business rule]
- The only feedback she has ever received was pressure: a phone call asking why her numbers were lower than a neighbouring panchayat's ASHA. [Business rule] [Emotional/cognitive]
- Her reply to that call is itself a data-quality warning. [Emotional/cognitive]

"I told him — should I write false names to make the number big?"

→ Tags: [Functional] [Business rule] [Constraint] [Emotional/cognitive] | need

Q9: Has the department given you any app or website for this — for screening, or for sending someone to the PHC? What do you use it for?

A9:

- An existing maternal-and-child app is installed (set up for her by the ANM). She uses it because her incentive payments are linked to it — compliance is financial, not voluntary. [Business rule] [Emotional/cognitive]
- Recurring failures she names: slow to open, repeated OTP prompts, unexpected logouts requiring the ANM's help to recover. [Non-functional]
- Zero connectivity in the interior hamlet, so she writes on paper and later stands on a hill behind the school where one bar of signal appears, to enter data. [Non-functional] [Constraint]
- Critical failure mode: the app reports "failed" and the typed data is gone — she must re-enter it. [Non-functional]

- For leprosy: nothing at all. Entirely paper, on pre-printed forms sent from the block. No digital referral mechanism exists. **[Functional] [Business rule]**

“Sometimes it says ‘failed’, and whatever I typed is gone. Then I type it again.”

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | need

Q10: What was good about those apps, and what was frustrating? Do you still use them?

A10:

- Valued: the due-list. It tells her which children are due this week; previously she searched the register page by page and missed children. **[Functional]**
- Valued strongly: proof of work. When visits are recorded, her payment follows and a supervisor can no longer deny she made the visit — “that gave me some strength”. Digital record as protection, not surveillance. **[Functional] [Business rule]**
[Emotional/cognitive]
- Frustrations: slow typing (“my thumbs are not for this”); the English keyboard and English-spelling name entry; 5–7 minutes per entry. **[Non-functional]**
- Mandatory fields she does not understand block progress, so she enters whatever passes validation — a direct, self-reported source of bad data. **[Non-functional]**
- She still uses it, but only at home at night, transcribing from paper — never live in the household. **[Constraint]**

“There are boxes I don’t even understand what they want, but the app will not go forward until I fill something, so I fill anything. Whatever passes.”

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | need + solution — one embedded solution (due-list pattern worth reusing)

Q11: Tell me about a time a form got wet or went missing. What did you do, and what did that cost you?

A11:

- Last July, final week of a campaign. Returning from the far hamlet beyond the nala after 12–13 households, she was caught in heavy rain; pages stuck together, ink ran, three pages became unreadable. **[Constraint]**
- She re-walked the route two days later (delayed by a bad knee) — 40 minutes each way, twice, entirely unpaid. **[Constraint]**
- Households were annoyed at being asked again; one man asked whether she was paid per visit. Repeat visits damaged her credibility. **[Emotional/cognitive]**

- Worst consequence: a girl with a patch near her elbow whose worried mother had cooperated the first time now refused a second examination, having been told by a neighbour it was nothing. The girl never reached the PHC and later married into another village. **[Constraint] [Emotional/cognitive]**
- Total cost she names herself: a day of unpaid time, some loss of standing, and one lost case. **[Constraint] [Emotional/cognitive]**

“That is what it cost. One day of my time, my reputation a little, and one girl.”

→ Tags: **[Constraint] [Emotional/cognitive]** | *need*

Q12: Tell me about the last person you sent to the PHC who never went. What do you think stopped them?

A12:

- ‘Phulwanti’, ~26, two children. Large pale patch on the forearm, no sensation on cotton test — the ASHA was confident. She counselled her: curable, medicine free, some months of treatment. **[Functional]**
- The woman agreed with folded hands and never went. Excuses on follow-up (child’s fever, bus fare) then avoidance — “not at home” while audibly inside. **[Constraint] [Emotional/cognitive]**
- The ASHA rejects cost as the real barrier (bus fare is ₹30 and affordable). The actual reason: the woman has a 12-year-old daughter, and if the diagnosis becomes known the girl’s marriage prospects and the family’s social standing are destroyed — “they will not eat from their house”. **[Constraint] [Emotional/cognitive]**
- A second, structural barrier: attending the PHC is not private. The bus carries villagers, the queue is public, staff have local relatives — “by evening the village knows”. **[Constraint] [Emotional/cognitive]**
- Two years on, the patch remains untreated. The ASHA states she understands and might do the same, which is why exhortation alone will not fix this. **[Functional] [Emotional/cognitive]**

“So she chose her daughter. And I understand it, sir, that is the terrible part — I understand what she did. But that patch is still on her arm and it is two years now.”

→ Tags: **[Functional] [Constraint] [Emotional/cognitive]** | *need*

Q13: After you send someone, how do you come to know whether they actually met the doctor?

A13:

- She does not, in the main. Whatever she learns comes informally from the village, not from the health system. **[Functional] [Business rule]**

- Informal channels: the person volunteers it; she spots the blister pack of tablets in the house; a neighbour mentions it (forcing her to feign ignorance to protect confidentiality); or the ANM happens to mention it at the monthly meeting. **[Constraint]** **[Emotional/cognitive]**
- Nothing arrives from the PHC — no slip, no call, no list. To check she would have to travel to Kondagaon and be told the register-keeper is on leave. **[Business rule]** **[Constraint]**
- Her own estimate: she refers ~20 and learns the outcome for 5–6. And silence is not neutral — silence usually means non-attendance. **[Functional]** **[Emotional/cognitive]**
- This is the clearest unmet functional need in the interview and she raises it unprompted three separate times. **[Functional]**

“So mostly it is like posting a letter and never hearing back... And the silence is not a good silence, sir — mostly silence means they did not go.”

→ Tags: **[Functional]** **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | *need*

Q14: In the interior villages, what happens with the network? And if you’d entered ten people and the phone lost all of it — what then?

A14:

- The interior hamlet has no coverage — not slow, absent. Intermittent single-bar signal only on a ridge near the mahua trees; local children know which rock to stand on. **[Constraint]**
- The main village is workable, better in the evening. The correlation is adverse: coverage is worst exactly where screening need is highest. **[Non-functional]** **[Constraint]**
- On data loss she is unequivocal: losing ten entries means losing a full day’s walking and having to face ten families again — which, per Q11, destroys credibility. **[Non-functional]** **[Emotional/cognitive]**
- This has already happened at a smaller scale (four entries lost in the MCH app); she stopped touching the phone for two days. **[Non-functional]** **[Emotional/cognitive]**
- Explicit adoption threat: if the app can lose data she will keep paper in parallel, and will then abandon the app rather than do double work. **[Non-functional]** **[Emotional/cognitive]**
- Her comparison of paper vs app is the sharpest statement of the requirement. **[Non-functional]**

“The paper is stupid, it gets wet — but the paper does not say ‘failed’ and eat my whole morning.”

→ Tags: [Non-functional] [Constraint] [Emotional/cognitive] | *need*

Q15: When you're sitting in someone's house with your phone, who else is in the room? What can they see on your screen?

A15:

- There is effectively no private room. She works on a verandah or in a courtyard with the mother-in-law present, neighbours who arrive out of curiosity, children climbing on her, and a husband on a cot listening. **[Constraint]**
- The phone is a focus of attention: children grab it, non-literate elders stare at the bright screen, and school-educated teenagers read it faster than she can. **[Constraint]** **[Emotional/cognitive]**
- Current MCH content is harmless (name, vaccine date) — nothing there is secret. A leprosy screen is categorically different. **[Non-functional]**
- She names the exact failure mode: a disease name in the heading or a red 'suspected case' marker visible in the courtyard would spread through the lane by night and cost the family social inclusion — caused not by her or them but by a 12-year-old reading over her shoulder. **[Non-functional]** **[Emotional/cognitive]**
- Crucially, concealment gestures are themselves disclosure: shielding the screen or turning away signals that something is wrong. "Turning away is itself a message." **[Non-functional]** **[Emotional/cognitive]**
- She states the design constraint in her own terms: the app must let her do what she already does with the register — turn it away and mark it privately. **[Functional]** **[Non-functional]**

"That word on my screen in that courtyard, and by night the whole lane knows, and by next week the family is not invited to the marriage."

→ Tags: [Functional] [Non-functional] [Constraint] [Emotional/cognitive] | *need*

Q16: How do people react when you say they should see a doctor about a patch? Has anyone ever hidden one from you?

A16:

- Three reaction patterns: casual dismissal ("haan haan didi, achha" — she knows these will not attend); visible fear (sleeve pulled down, "it is nothing, it is from the stove", often before the disease is even mentioned); and anger as displaced fear (one man nearly pushed her out for an implied accusation she had not made). **[Emotional/cognitive]**

- Her cover story is transparent to villagers: an ASHA examining arms with cotton is understood — “we are not fooling anybody with our ‘health survey’”. **[Constraint]** **[Emotional/cognitive]**
- Two confirmed concealment cases: a neighbour who wore full-sleeve blouses through May and stopped inviting her in, whose large abdominal patch had been self-treated with private-shop ointment for years; and a schoolteacher’s son whose family, considering themselves socially above an ASHA, treated it privately — discovered only via the medical-shop owner. **[Constraint]** **[Emotional/cognitive]**
- She therefore discounts her own case count as covering only those who allowed themselves to be examined — the denominator is unknown. **[Constraint]**

“That is why my counting of four or five confirmed — I don’t believe that number, sir. It is only the ones who let me see.”

→ Tags: **[Constraint]** **[Emotional/cognitive]** | *need*

Q17: If word went round the village that you’re the one who marks people as ‘leprosy people’ — what would happen to your other work?

A17:

- Her assessment: total collapse across all her work, not just leprosy. Her roles are one door — a woman who fears her will not call her in labour, will not bring the child for immunisation, and will conceal a pregnancy until the fifth month. **[Constraint]** **[Emotional/cognitive]**
- Her access rests entirely on personal trust earned through non-official acts (sitting with a woman being beaten, taking a child to hospital by motorcycle at 2 a.m.). She has no salary, uniform or authority — “I have that, and that is the only thing I have.” **[Emotional/cognitive]**
- Failure is fast and irreversible: one incident, one week, and doors close. Politeness continues; truthful disclosure stops. “I will still be walking, still writing in the register, but the register will be lies.” **[Non-functional]** **[Emotional/cognitive]**
- She explicitly rejects framing confidentiality as a minor setting, and marks the asymmetry of risk between herself and the project team. **[Non-functional]** **[Emotional/cognitive]**

“This is not a small ‘privacy setting’ for me. This is my whole standing in the village. I am the one who has to live there afterwards. You will go back to your office.”

→ Tags: **[Non-functional]** **[Constraint]** **[Emotional/cognitive]** | *need*

Q18: Whose phone is it? Who else uses it, and where does it stay at night?

A18:

- Nominally hers, practically the household's: her husband takes it to market (his own phone has no internet), her 15-year-old son uses it for video and cricket, the younger child plays games, and neighbours borrow it for video calls to relatives in Raipur. **[Constraint]**
- At night it sits charging on a shelf near the kitchen or by the bed. "Nobody hides a phone in a village house. There is no cupboard with a lock." **[Constraint]**
- The screen-lock pattern is known to both her son and her husband — he watches her draw it daily. Device-level security therefore provides no protection for sensitive records. **[Non-functional] [Constraint]**
- Additional environmental constraint she raises: multi-hour power cuts. She returns from a full day's walking to a phone at 10% and still has entries to make — a battery-hungry app is unusable. **[Non-functional] [Constraint]**
- Implication: sensitive data needs app-level protection independent of the phone lock, and must be lightweight enough not to reintroduce the OTP problem of Q9. **[Non-functional]**

"So if you are thinking my phone is a safe box, no. It is a family phone that mostly stays in my hand."

→ Tags: **[Non-functional] [Constraint]** | *need*

Q19: The district office would like about 25 details recorded for every person you screen. What would that mean for your day? And which of those details actually help the person in front of you? [deliberate conflict probe]

A19:

- Rejected outright and emphatically — "please write down that I said no". **[Emotional/cognitive]**
- Her own arithmetic: ~5 minutes of typing per person × ~100 people = ~500 minutes, i.e. 8 hours of data entry on top of walking, examining, and the rapport-building she regards as the real work. Not feasible. **[Non-functional] [Constraint]**
- Predicted consequence is not refusal but data fabrication: the first few entries done properly, then arbitrary values — producing a full, credible-looking report that is largely worthless. She states this already happens with the MCH app. **[Non-functional] [Emotional/cognitive]**
- Second-order harm: a ten-minute interrogation about one individual signals to the household and neighbours that something is wrong. The form length itself is a confidentiality breach. **[Non-functional] [Emotional/cognitive]**

- Fields she considers genuinely useful — about six: (1) name; (2) house/locator; (3) father's or husband's name (disambiguation — “four Sukhram”); (4) patch site; (5) sensation present or absent — “that one question is the disease”; (6) numbness/weakness in hands or feet, plus unnoticed wounds and duration of symptoms. **[Functional]**
- Fields she names as useless to the patient: age, caste, ration-card number, education, toilet, head-of-household mobile number — “that is for somebody's report in the district. It does not help Phulwanti. Not one bit.” **[Business rule] [Emotional/cognitive]**
- Her stated trade-off: if 25 fields are mandated, then pay for a full day and halve the household target. **[Business rule] [Constraint]**

“Otherwise it is just a way to make honest people write false things.”

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | need — direct conflict with district reporting requirement;
 see §5

Q20: Now tell me freely — if we build this app for you, what should it do so that your day becomes easier and no person gets missed?

A20:

- Offline first, with guaranteed retention and silent automatic sync when signal returns — never a ‘failed’ message. One data-loss event and she reverts to paper permanently. **[Non-functional]**
- Complete on-screen discretion: no disease name, no red marker, no audio. It should look like the ordinary health survey. Any warning should reach her privately and quietly, later. Her son must not be able to open that part of the app — a separate but lightweight lock, “not another OTP business”. **[Functional] [Non-functional]**
- Speed: under a minute per person. Tap a body diagram rather than type. Hindi/Devanagari, not English spelling. Voice entry in her own words would be “the best thing you could give me”. **[Functional] [Non-functional]**
- Referral closure: tell her whether the person attended. “Right now I am shouting into a well.” If she knew someone had not gone she would revisit and, if needed, accompany them on the bus. **[Functional]**
- A recall list of persons she flagged earlier who have not been re-checked — “because I forget. Truly. I have hundreds of names in my head.” **[Functional] [Emotional/cognitive]**
- A neutral, government-branded reassurance screen for families (curable, free, duration) — more persuasive coming from the phone than from her — but still without naming the disease on screen. **[Functional]**

- Her work in the app must count towards her incentive without her having to petition the supervisor, otherwise it is one more unpaid task. **[Functional] [Constraint]**
- Closing request: pilot it in the far hamlet in June, walking with her — not demonstrated on a big screen at the block office. **[Non-functional]**

“Come to the village when you make it, sir. Don’t just show it to us in the block office on the big screen. Bring it and walk with me to the far hamlet, in June, and see if it works there. If it works there, it will work everywhere.”

→ **Tags: [Functional] [Non-functional] [Constraint] [Emotional/cognitive]** | *solution*
— *each item traced back to the need in §6 (Appendix A)*

Q21: Read-back: it must work with no signal and never lose entries, take two or three minutes not ten, reveal nothing on screen, and tell you whether the person reached the doctor — have I got that right, and which matters most?

A21:

- Confirmed with two corrections. First, the time target is one minute, not two or three — “you said three because you are sitting in a chair”. Two minutes only if the person is talkative. **[Non-functional] [Emotional/cognitive]**
- Second, and important: discretion is not only about the screen. Nothing she does at a flagged house should look different from a routine visit — screen content, time taken, and questions asked. “If my behaviour changes at that house, the neighbours read it.” **[Non-functional] [Emotional/cognitive]**
- Priority 1: confidentiality/discretion. Rationale — data loss makes her angry and she reverts to paper, which she has survived for nine years; a disclosure incident is irreversible and ends her standing in the village permanently. **[Non-functional] [Emotional/cognitive]**
- Priority 2 (a close second): guaranteed data retention — because without it nobody will use the app long enough for the confidentiality design to matter. **[Non-functional]**
- Priority 3: referral outcome feedback. She is explicit that this is not what makes the app viable — it is what she personally wants most, and the only need nobody has ever offered to address. **[Functional] [Emotional/cognitive]**
- Her reasoning here is the single most useful prioritisation input from the session. **[Emotional/cognitive]**

“If it loses my data, I am angry and I go back to paper. That is bad, but I survive... If it shows that word on my screen in Phulwanti’s courtyard — I don’t survive that. I don’t get that back.”

→ **Tags: [Functional] [Non-functional] [Emotional/cognitive]** | *need — prioritisation confirmed by stakeholder*

3. TOP 3 PAINS I HEARD

1. Stigma makes the whole process covert — and any disclosure risk is career-ending for her

People conceal symptoms from her, refuse referral to protect a daughter's marriage, and treat privately or not at all. She has invented her own concealment workarounds (a 'health survey' cover story, private tick marks, examining women indoors). If SparshCare displays anything identifiable on screen — or merely makes her behave differently at a flagged house — she loses village trust, and with it her access for immunisation, antenatal and delivery work too. Ranked #1 by the stakeholder herself, on the grounds that it is the only failure that cannot be recovered from. (Q12, Q15, Q16, Q17, Q21)

2. The referral black hole — she refers and never learns the outcome

Roughly 20 referrals yield about 6 attendances, and she finds out about outcomes only by gossip, spotting tablet packs, or a chance remark at a monthly meeting. Nine years, no feedback from the PHC, ever; the one message she has received from the system was pressure to raise her numbers. This blocks the one intervention she is well placed to make — revisiting and personally accompanying a non-attender. It is also the pain she carries emotionally: "those four names, and the girl with the elbow — I still carry them." (Q6, Q8, Q13, Q20, Q21)

3. Fragile records and unpaid rework — on paper today, and in the existing app too

Wet registers, lost slips, night-time re-copying from memory, and one documented incident (July) where three ruined pages cost her two unpaid 40-minute walks, some credibility, and one girl who was never screened again. The digital version of the same pain is worse in her eyes: the MCH app's 'failed' message deletes typed work outright. Because she is paid per task, this lands directly on her income — which is why she will run paper in parallel and then abandon any app she does not trust. (Q1, Q7, Q9, Q11, Q14)

Honourable mentions (real, but second tier)

- Late detection concentrated among the elderly and socially isolated — 2 of 4 confirmed cases already had irreversible hand/foot damage, symptoms having gone unremarked for 2–3 years (Q5).
- No memory aid across campaigns: she carries hundreds of names in her head, keeps no record of confirmed cases, and cannot revisit whom she flagged last year (Q3, Q13, Q20).
- Data-entry burden falls entirely outside paid hours — Saturdays re-copying registers, nights transcribing into the app (Q1, Q7, Q10).
- The device is a shared family phone with a pattern lock her son and husband both know, and daily multi-hour power cuts (Q18).

4. CONTRADICTIONS AND THINGS TO VERIFY

Items to triangulate against records or other stakeholders (PHC medical officer, ANM, block NLEP supervisor) before writing requirements.

- **Confirmed-case count is unreliable** — Stated “four, maybe five”, then revised to “maybe four, maybe three”, and finally discounted her own figure as covering only those who allowed examination. Do not use as a baseline. Verify against PHC/NLEP case registers for the block.
- **Tenure and training dates are approximate** — “2017 — no, before that, maybe 2016”; training “two days, I think — no, one and a half”, “maybe three years back”. Check block training records if training adequacy becomes a design input.
- **Time-per-person figures conflict across sources** — She reports 5–7 minutes per entry in the existing MCH app, demands ≤ 1 minute for SparshCare, and rejected the analyst’s 2–3 minute read-back. A field time-and-motion observation is needed before committing to a target.
- **‘Voice entry’ request contradicts her own confidentiality requirement** — She calls Hindi voice input “the best thing you could give me” (Q20), but speaking findings aloud in a shared courtyard is exactly the disclosure she fears in Q15 and Q17. Needs resolution — possibly voice only for non-identifying fields, or only when alone.
- **‘Reassurance screen for the family’ versus ‘nothing on screen’** — She wants a government-branded screen explaining that the condition is curable and free (Q20) while also requiring that the disease never appear on screen (Q15). The content of any patient-facing screen must be co-designed with her, not assumed.
- **Phulwanti timeline is inconsistent** — Described as the most recent non-attender, yet “it is two years now”. Clarify how recent the last referral drop-off actually was, and whether non-attendance has changed since.
- **‘The PHC sends nothing back’ — absence of mechanism, or unused mechanism?** — Verify with the PHC and ANM whether a formal feedback or referral-tracking register already exists and is simply not operated. This decides whether FR-05 is a new integration or the digitisation of a dormant process.
- **Skin OPD schedule** — She tells patients Tuesday and Friday, “though he is not always there”. Confirm the actual specialist roster — any in-app referral scheduling that sends people on a day the doctor is absent will destroy trust faster than no scheduling at all.
- **Incentive linkage may not exist for screening** — She expects app-recorded screening to drive her payment (Q10, Q20). Verify whether NLEP screening carries an ASHA incentive line at all; if not, this expectation must be managed explicitly rather than silently disappointed.

- **'No network at all' needs technical confirmation** — Distinguish true zero coverage from intermittent/degraded coverage in the interior hamlets via a signal survey. Design must assume zero regardless, but sync strategy depends on the real picture.
- **Self-reported data fabrication** — Her claim that she and others enter arbitrary values to pass mandatory fields is a strong assertion. Corroborate by auditing existing MCH app data for the block — it is the most persuasive evidence available against the 25-field proposal.
- **Stakeholder's own bias to note** — She is paid per task, so she has a legitimate interest in minimising recorded fields. Her arguments are credible and field-grounded, but the minimal field set should still be validated clinically with the PHC medical officer before being fixed.
- **Campaign-day volume figures do not reconcile** — A3 states she flags 8–10 people per campaign overall, while A4 states 2–3 people per campaign day cause concern; across a multi-day campaign the day-level figure implies a far higher campaign total than the 8–10 given in A3. These two figures cannot both hold and need reconciling — with her directly, or against campaign records — before either is used as a volume estimate.

5. RECORDED CONFLICT: DISTRICT '~25 FIELDS PER PERSON' VERSUS FIELD FEASIBILITY

Logged separately because it is an open decision for the project, not a resolved requirement.

- District/reporting position: approximately 25 attributes captured per person screened, for programme monitoring and demographic analysis.
- Stakeholder position: infeasible at a load of 100+ individuals per campaign day. Predicted outcome is not non-compliance but fabricated data — a complete-looking dataset of low validity. She also identifies a confidentiality harm: a long interrogation about one household member is itself visible to neighbours.
- Her counter-proposal: about six fields at point of contact — name, household locator, father's/husband's name, patch site, sensation present/absent, numbness or weakness plus unnoticed wounds and symptom duration.
- Her stated trade-off if 25 fields are mandated: pay a full day's wage and halve the household target.
- Options to take forward: (a) minimal mandatory set at point of contact with optional fields deferred to a private completion step (FR-10); (b) auto-populate demographic fields from existing household records instead of re-asking; (c) capture the long form only for confirmed or referred cases, not for every person screened; (d) escalate the trade-off explicitly to the district before build. Recommend (a) + (b) + (c) combined; requires district sign-off.

APPENDIX A — CONSOLIDATED REQUIREMENTS REGISTER (FROM THIS INTERVIEW ONLY)

Derived from the tags above. Priorities are this analyst's recommendation, with Must-have status on NFR-04 and NFR-02 following the stakeholder's own ranking in Q21. Every line traces to a source question; nothing here is inferred beyond what was said.

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-01	Capture one screening record with a minimal field set only: person name, household locator, father's/husband's name, patch site, loss of sensation (Y/N), numbness or weakness in hands/feet, unnoticed wound, duration of symptom	Functional	Q6, Q7, Q19	Must have
FR-02	Record patch location by tapping a body diagram rather than typing a description	Functional	Q19, Q20	Should have
FR-03	Record the cotton (sensation) test as a single binary choice — this is the decisive field	Functional	Q6, Q19	Must have
FR-04	Raise a referral for a flagged person without the disease name appearing on screen or on any slip the person carries	Functional	Q6, Q12, Q15	Must have
FR-05	Return the referral outcome to the referring ASHA — minimally 'attended / not attended' with a date	Functional	Q8, Q13, Q21	Must have
FR-06	Show the ASHA a list of persons she herself flagged earlier who have no recorded outcome, so she can revisit them	Functional	Q3, Q5, Q13	Should have
FR-07	Prompt extra attention for higher-risk profiles observed in the field: elderly, living alone / low household attention, symptoms present over a year	Functional	Q5	Could have
FR-08	Provide a neutral, government-branded reassurance screen for the family (curable, free, duration) that does not name the disease	Functional	Q20	Should have
FR-09	Count each completed screening towards the ASHA's task / incentive	Functional	Q10, Q20	Should have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	record automatically			
FR-10	Allow the sensitive clinical detail to be saved as an incomplete draft in the house and completed later in private	Functional	Q7	Should have
NFR-01	Full offline capture — every screening function works with zero mobile network	Non-functional	Q9, Q14	Must have
NFR-02	Zero data loss: entries persist locally and sync automatically in the background; a failed sync must never discard typed data	Non-functional	Q9, Q11, Q14	Must have
NFR-03	No session expiry, no repeated OTP re-login while in the field	Non-functional	Q9, Q10	Must have
NFR-04	On-screen discretion: no disease name, no red 'suspected case' banner, no audio cue; the screen must be indistinguishable from a routine health survey	Non-functional	Q15, Q17, Q21	Must have
NFR-05	Screening module protected by its own lightweight lock, separate from the phone's screen lock (device is shared with husband, teenage son, neighbours)	Non-functional	Q18	Must have
NFR-06	≤ 1 minute of device time per person screened	Non-functional	Q4, Q19, Q21	Must have
NFR-07	Devanagari / Hindi labels and input; tap and pictorial selection in place of free typing wherever possible	Non-functional	Q10, Q20	Must have
NFR-08	Low battery draw; usable at low charge (multi-hour daily power cuts)	Non-functional	Q18	Should have
NFR-09	Behavioural discretion: time spent and questions asked at a flagged house must not visibly differ from a normal visit	Non-functional	Q15, Q17, Q19	Should have
NFR-10	Usable by a class-10-educated, non-technical worker; no clinical vocabulary anywhere in the interface	Non-functional	Q2, Q10	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
BR-01	Leprosy screening happens only during campaigns declared by the block — it is not part of the routine weekly cycle	Business rule	Q1, Q2	Constraint on design
BR-02	Paper flow is ASHA → ANM at sub-centre → block supervisor. No return path to the ASHA exists today	Business rule	Q8	Constraint on design
BR-03	Skin OPD nominally sits Tuesday and Friday at Kondagaon PHC; attendance is unreliable	Business rule	Q6	Verify
CON-01	Interior hamlets have no mobile coverage at all (not slow — absent)	Constraint	Q9, Q14	Absolute
CON-02	The phone is a shared household device; the screen-lock pattern is known to family members	Constraint	Q18	Absolute
CON-03	Field load is 25–30 households, often 100+ individuals examined, in one campaign day	Constraint	Q4	Absolute
CON-04	The ASHA is paid per task; any unpaid added time is a direct income loss and will be resisted	Constraint	Q11, Q19	Absolute
CON-05	Stigma is material: disclosure threatens marriage prospects and social standing of the whole family	Constraint	Q12, Q16, Q17	Absolute
EC-01	Loss of village trust is the stakeholder's single greatest fear and the largest adoption risk in the project	Emotional/cognitive	Q17, Q21	Must design for

Design rules implied, in the stakeholder's own framing

- “Feeling or no feeling” is the decision. Optimise the entire interaction around that one field; everything else is supporting data.
- The app must let her keep doing what she already does — turn the record away and mark it privately — not remove that ability in the name of completeness.
- Discretion covers behaviour as well as pixels: identical interaction time and question pattern at flagged and unflagged houses.
- Losing data once is fatal to adoption. Paper is inferior in every respect except that it never says ‘failed’.

- Any unpaid minute added is a real income loss to a task-paid worker, and will be met with workarounds rather than compliance.
- Pilot acceptance test proposed by the stakeholder: usable in the far hamlet, in June, walking with her — not demonstrated at the block office.

STAKEHOLDER INTERVIEW LOG

SparshCare — Requirements elicitation · Interview 2 of 6

Field	Detail
Analyst	Dr. Gunjan Singhal
Date of interview	17 August 2026
Stakeholder role	Medical Officer (MBBS) — PHC in-charge, confirms diagnosis and starts MDT
Stakeholder (persona)	Dr. A. Rao — ~8 years in public health, 4 years at this PHC
Location / facility	Kondagaon PHC, Bastar, Chhattisgarh
Project	SparshCare — leprosy screening, teleconsultation & referral system
AI tool used	Claude (Cowork) — model claude-opus-5, role-play persona “Dr. A. Rao, Medical Officer”
Interview mode	Semi-structured, single session, 16 questions with follow-up probes
Duration (in-persona)	~25 min, twice interrupted by the OPD queue
Chat link / screenshot ref	SPARSHCARE / INT-02 / RAO-MO — [paste chat link or screenshot filename]

1. PURPOSE AND INTERVIEW APPROACH

Objective: establish what a referral must contain to be clinically actionable, test the six-field minimum proposed by the ASHA in Interview 1 against clinical validity, and surface the receiving end of the referral chain — capacity, availability, reporting burden and the limits of what software may decide. This is the counterpart interview to Interview 1: the ASHA is the sender, this stakeholder is the receiver, and the requirement that matters most is the one they both need and neither can create alone.

Question flow used (rapport → baseline evidence → current process → cost → pain → capacity → constraint → conflict → solution and read-back last):

- Q1–Q2 Rapport, role and baseline evidence: tenure, where leprosy sits in the caseload, annual case volume and disability-at-detection rate.
- Q3–Q5 Current “as-is” referral process from the receiving end: what arrives, what it costs to act on it, and the confirmation rate.
- Q6–Q7 The availability problem and realistic clinic capacity — the two hardest scheduling constraints in the project.
- Q8 The non-attendance gap: who notices, whose job is follow-up, and where escalation must and must not go.

- Q9 Existing reporting burden and the failure story of the previous digital tool — the adoption condition.
- Q10 Late-detection case narrative, used to surface the liability position (a deliberate probe).
- Q11–Q13 Constraint definition: what evidence would justify acting on a risk flag, the referral wrapper, and confidentiality inside the facility.
- Q14 The treatment tail — what happens after confirmation and where this stakeholder's responsibility stops being recorded anywhere.
- Q15 Deliberate conflict probe: the ASHA's "the form must be short" against this stakeholder's "the form must be detailed".
- Q16 Read-back and forced prioritisation.

Tag key: Each captured line is marked with one or more of five categories — **[Functional]** (something the system must do) · **[Non-functional]** (a quality it must have) · **[Business rule]** (an existing organisational rule) · **[Constraint]** (a fixed environmental, economic or regulatory limit) · **[Emotional/cognitive]** (fear, trust, stigma, mental load). Each answer additionally carries a roll-up tag and is marked *need* (a problem to be solved) or *solution* (a proposed answer, to be traced back to the underlying need). In Appendix A these categories are carried forward as ID prefixes: FR = functional, NFR = non-functional, BR = business rule, CON = constraint, EC = emotional/cognitive.

2. QUESTION-BY-QUESTION LOG

Q1: Tell me about yourself — how long at Kondagaon PHC, and where does leprosy work sit among everything else you handle?

A1:

- MBBS (Raipur), mandatory rural posting, then continuously in public health — ~8 years total, 4 of them at this PHC. **[Business rule]**
- Leprosy is a small share of volume. Daily OPD is fevers, diarrhoea, anaemia in pregnancy, hypertension, seasonal snakebite, TB follow-up; the MDT register carries only a handful of active cases at a time. **[Constraint]**
- But it carries disproportionate clinical worry, for a reason he states precisely: every other condition self-corrects for a missed diagnosis — the patient returns worse and is caught. Early leprosy does not, because the lesion is painless, so the patient never returns. **[Emotional/cognitive]**

- He positions himself as the endpoint, not the detector: “I confirm, I start MDT, I monitor for reactions” — and states that if a case does not reach him early, his competence is irrelevant. The ASHA is the decisive actor. **[Business rule]**
- Opened the session unprompted with scepticism of “another app”, citing a tablet reporting tool ~2 years ago that duplicated data entry and returned nothing. This framing runs through the whole interview and should be treated as the project’s primary adoption risk on the clinical side (see EC-03). **[Emotional/cognitive]**

“They come back four years later with a clawed hand and a foot they can’t feel, and at that point I can cure the infection but I cannot give them their nerve back.”

→ Tags: **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | need

Q2: In a year, how many confirmed leprosy cases do you see here? And of those, how many arrive already with a deformity?

A2:

- ~30 new confirmed cases per year (range given: 25–40). One year exceeded 40 following an active case-detection campaign — which he reads as evidence the cases were always present and simply not being found. **[Constraint]**
- Grade 2 disability (visible deformity) at first presentation: 3–5 per year, i.e. roughly 10–12%. **[Constraint]**
- He immediately qualifies his own figure: the national programme reports 1–2%, but that is the rate among cases that were found. His denominator excludes everyone still concealing symptoms — the same denominator problem the ASHA raised in Interview 1, Q16. **[Constraint]** **[Emotional/cognitive]**
- Volunteers a second indicator he considers more damning: 1–2 child cases per year. A child case implies a prolonged undiagnosed infectious adult in the same household — a whole family missed, not one patient. **[Functional]** **[Constraint]**
- Explicitly cautioned the analyst not to treat these numbers as validated (“I don’t want to give you a figure you’ll put in a slide”). Logged in §4 for verification. **[Constraint]**

“It says nothing about the man in Bade Dongar who has been hiding his hand for three years because he thinks people will stop eating with him.”

→ Tags: **[Functional]** **[Constraint]** **[Emotional/cognitive]** | need — baseline figures unverified

Q3: Walk me through what happens today when an ASHA refers someone to you. How does that person reach you, and what do you have in your hand when they arrive?

A3:

- There is no advance notification of any kind. The patient simply appears at the OPD window on an unpredictable day, having travelled 20–30 km — up to half a day in monsoon — and joins a queue of ~40 people. **[Constraint]**
- What arrives with them: a handwritten chit, sometimes a torn notebook page, typically reading “patch on back, please examine” plus the ASHA’s name and occasionally a date. No clinical content. **[Functional]**
- Consequence: he starts from zero. Duration of the lesion, whether sensation was tested and how, household contacts, any nerve concern — all of it evaporates between the house and his table. He repeats an examination the ASHA has already performed in rough form. **[Functional] [Constraint]**
- Compounding failure: on any given day he may be on immunisation duty, at a block meeting, or on leave. The PHC is open and staffed, but a leprosy suspect specifically requires him — sensation testing and nerve palpation cannot be delegated to the pharmacist or ANM (BR-08). The patient is told to return another day. **[Business rule] [Constraint]**
- The deepest gap, stated in his own ranking: he never knows a field case exists until the person appears. If a referral is made in March and the person never comes, that person does not exist in his world — not a defaulter, not a missed case, nothing. No list of “people who were supposed to come” exists anywhere. **[Functional] [Emotional/cognitive]**

“That’s the part I’d most want fixed, if you’re asking. Not the form. The invisibility.”

→ Tags: **[Functional] [Business rule] [Constraint] [Emotional/cognitive]** | need — ranked #1 by stakeholder at Q16

Q4: When a chit like that reaches you, what’s the first thing you do? How long does re-examining from the beginning take, and what does it cost you across an OPD day?

A4:

- First action is triage of timing, not examination — he decides when he can see the person properly, because it cannot be done at the window with forty people listening. **[Functional]**
- Full examination sequence he describes: whole-body surface inspection in adequate (preferably natural) light, because a faint hypopigmented patch on dark skin is not visible under his room’s bulb; sensation testing on each patch with cotton, eyes closed; nerve palpation — ulnar bilaterally, lateral popliteal, posterior tibial, great auricular; voluntary muscle testing if indicated; inspection of hands and feet for trophic change; eyes for lagophthalmos. **[Constraint]**
- Time: 15–20 minutes done properly. Longer with a slit-skin smear, or when a female attendant must be located before examining a woman. **[Constraint]**

- Against this: 80–120 patients per OPD session, i.e. under 4 minutes per patient before any emergency. A leprosy suspect is therefore a 20-minute hole in a day with no 20-minute holes. **[Constraint]**
- He is explicit about who pays that cost — not him, but the ~30 patients at the back of the queue whom he then rushes. **[Non-functional]**
- This is where the vague chit hurts most: with structured findings in advance he could triage — not diagnose remotely, triage — deciding which cases need his 20 minutes urgently and which can be scheduled onto a lighter day. **[Functional]**

“I’ve spent twenty minutes and displaced a queue to conclude ‘this is a fungal patch, here’s some clotrimazole.’”

→ Tags: **[Functional]** **[Non-functional]** **[Constraint]** | *need*

Q5: Out of every ten people an ASHA sends you, how many turn out to be actual cases?

A5:

- Two, occasionally three. So ~70–80% of referrals are something else — tinea, vitiligo, post-inflammatory hypopigmentation, pityriasis alba. **[Constraint]**
- He refuses to let this be recorded as an indictment of the ASHAs: an ASHA has no dermatology training, and over-referral is the clinically correct failure direction. “I would rather have seven wasted examinations than one clawed hand in four years’ time.” This directly corroborates the ASHA’s account in Interview 1, Q3. **[Business rule]**
- The finding he considers more useful: the ratio varies sharply between workers. The Interview 1 stakeholder (“Sunita”) runs ~3–4 in 10 because she now tests sensation before referring; others are at 1 in 10 or worse and appear to refer every pale patch. **[Constraint]** **[Functional]**
- At least one panchayat sends him almost no referrals at all — which he reads not as absence of disease but as absence of screening. **[Constraint]** **[Functional]**
- He has no systematic way to know any of this. He knows it because he has been here four years and recognises handwriting. No report tells him referral volume or hit rate by worker, or which villages have gone silent. “Right now my supervision is anecdote.” **[Functional]** **[Emotional/cognitive]**
- Raised the conflict with the ASHA’s short-form position himself, unprompted, and offered the reframing that resolves it: the list needed to triage is much smaller than the list needed to diagnose (see Q11, Q15). **[Functional]** **[Business rule]**

“And there is at least one panchayat area where I get almost no referrals at all, which does not mean there’s no leprosy there — it means nobody is looking.”

→ Tags: **[Functional]** **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | *need*

Q6: What happens when someone travels four hours to see you and you're on field duty that day?

A6:

- They are turned away. The reaction he describes is not anger but resignation — “this is what government offices do”. **[Emotional/cognitive]**
- Quantified cost to the patient: ₹60–80 fare, a lost day of wage labour (materially significant during tendu-leaf collection or harvest), and the return jeep may not run until 2 p.m. or at all. **[Constraint]**
- Set against a painless lesion, months of treatment, and severe stigma (he confirms marriages being called off, and the term kodhi still in use) — every incentive already points to not coming. A wasted trip removes the last one. **[Emotional/cognitive]** **[Constraint]**
- The cost he considers most damaging is not his own: the ASHA spends her personal credibility persuading a frightened family, promises the doctor will see them, and the system breaks her promise. Word travels; the next ten referrals from that hamlet become harder. This corroborates Interview 1, Q17 from the receiving end. **[Emotional/cognitive]** **[Functional]**
- Critically: his absences are almost entirely predictable to him. Immunisation days are fixed, block meetings scheduled, leave applied for weeks ahead. That information exists only in his head and in a block-office register. The ASHA, 30 km away, cannot reach it — the PHC landline has been dead for over a year and there is no mobile signal in half the hamlets. **[Constraint]** **[Functional]**
- States that a calendar alone — no screening forms, no risk scores — would justify the project. **[Functional]**
- Caveat he attaches immediately: availability is not fully predictable. Outbreaks, maternal emergencies and unannounced district inspections pull him out. The system must assume a booked slot can break and must reach the patient before they board the jeep. **[Functional]** **[Non-functional]**

“Otherwise you’ve just automated the same disappointment with better record-keeping.”

→ Tags: **[Functional]** **[Non-functional]** **[Constraint]** **[Emotional/cognitive]** | need — ranked #2 by stakeholder at Q16

Q7: How much of your week could you realistically protect for these referral appointments?

A7:

- Weekly shape: OPD every morning six days (non-negotiable, over-runs routinely); Tuesday and Friday are immunisation and field days; a block meeting most months;

afternoons absorbed by the MDT register, district reporting formats and TB follow-up. **[Constraint]**

- Realistic protected capacity: two afternoons per week, ~2–4 p.m., i.e. ~12 slots at 20 minutes, which he discounts to ~10 because something always intrudes. **[Constraint]**
- His own arithmetic on demand: ~30 confirmed cases/year at a 2-in-10 confirm rate implies ~150 referrals/year $\approx$ 3 per week. Ten slots is nominally generous. **[Constraint]** **[Functional]**
- Two reasons the arithmetic misleads. (a) Active case-detection campaigns produce ~30 suspects in a fortnight; the system must absorb surges and let him open extra capacity by choice, not force him into a fixed grid. **[Functional]** **[Constraint]**
- - (b) More important — those slots are for new suspects only. Each confirmed case returns monthly for supervised dosing and reaction checks for 6 or 12 months, so 25–30 people are on follow-up at any time. That recurring load exceeds the new-referral load. If new bookings crowd out follow-ups, the system trades early detection for treatment default — which he considers in some respects a worse outcome (resistance, plus the nerve damage still arriving). **[Functional]** **[Business rule]**
- Hard condition: the system must propose and hold slots, never auto-book him. Partly because he holds context the system cannot (a wedding across a village cluster means nobody will attend), but principally as a trust matter — “the first time it double-books me I’ll abandon it entirely and go back to paper. You get exactly one chance at that.” **[Functional]** **[Emotional/cognitive]**

“Do not let the system book me automatically. Propose slots, hold them, let me confirm — even if that’s one tap in the evening.”

→ Tags: **[Functional]** **[Constraint]** **[Business rule]** **[Emotional/cognitive]** | *need + embedded solution (propose-and-confirm pattern)*

Q8: When a person you were expecting never turns up, how do you come to know — and whose job is it to follow up?

A8:

- Two distinct situations, and the difference is the requirement. After confirmation, the patient is on the MDT register with a card and a due date, so an uncollected blister pack is visible and can be chased. Before confirmation there is nothing at all. **[Functional]** **[Business rule]**
- Pre-confirmation, a non-attender leaves no trace: no name, no record, no gap for anyone to notice. “Not a defaulter. Just absent from a list that was never made.” **[Functional]** **[Emotional/cognitive]**

- Formally the follow-up is the ASHA's. Practically she cannot do it: she has told ~15 people this quarter, holds the list in her head, has no way to learn who attended, and people tell an ASHA "yes I went" to close the conversation — so even her own follow-up is unreliable at source. Directly corroborates Interview 1, Q13. **[Business rule]**
[Constraint]
- Accountability therefore falls between the two stakeholders: she assumes he has seen them; he does not know they exist. He names this as the single largest hole, and as a bookkeeping problem rather than a technology problem. **[Functional]**
[Emotional/cognitive]
- Hard boundary on the fix: the chasing must NOT default to him. A daily list of twelve non-attenders arriving on the Medical Officer's screen is precisely how the previous tablet system died — it generated work for the person with the least time. He will stop opening it within a month. **[Functional]** **[Non-functional]** **[Emotional/cognitive]**
- What he wants instead is oversight, not tasks: an aggregate, periodic, self-initiated view — e.g. on the first of the month, "four screened in that panchayat, two never came" — to raise at the monthly ASHA meeting. **[Functional]** **[Non-functional]**
- One explicit exception, which he insists on: a case the ASHA flagged as high-concern (evident nerve thickening, weakness, multiple patches) that does not attend must be escalated to him actively and immediately, not left in a monthly list. Two escalation channels, and he warns that blurring them destroys both. **[Functional]** **[Business rule]**

"If you blur that line and send me everything, you'll have built a system that I ignore — and then the alerts that actually matter get ignored along with the rest."

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Emotional/cognitive]** | *need*
— with an explicit anti-requirement

Q9: What registers or reports do you have to fill for leprosy at the moment, and what does that take out of your day?

A9:

- Current record set: a patient card per confirmed case (PB/MB classification, lesion count, nerve involvement, disability grade at diagnosis, MDT start date, monthly dosing record, disability grade at release); the MDT stock register; a monthly progress report to the block, disaggregated by new cases, child cases, Grade 2 disability, female cases, on treatment, released, defaulted; plus separate campaign formats when a campaign runs. **[Business rule]** **[Constraint]**
- He does not resent the patient card — it is clinical, it belongs to the patient, and he would keep it unprompted. He specifically defends disability grading at start and at release as the only honest measure of whether detection is getting earlier. **[Business rule]** **[Emotional/cognitive]**

- The monthly reporting is the burden: 2–3 hours at month end transcribing counts from the register into formats. A block-level data operator then re-enters the same figures into the national portal, so each number is written three times by three people. He states he would not vouch for the accuracy of what reaches the state. **[Non-functional]** **[Constraint]**
- Failure story of the previous system, given in full: it added a fourth transcription and removed no register, because nobody at district authorised dropping the paper format. Entry was done by the ANM at the end of a twelve-hour day on a tablet that would not hold charge, and nothing ever came back — no summary, no feedback. Degraded within four months, abandoned within eight. **[Emotional/cognitive]** **[Non-functional]**
- His stated acceptance test for SparshCare, which is a replacement condition rather than a feature: if the ASHA records screening and he records confirmation in the system, the monthly progress report should already exist — counted, disaggregated, ready to check and sign. “Not a new form. A replacement.” **[Functional]** **[Non-functional]**
- Deferred out of role, as instructed: whether the block and district will accept digital in place of paper is for the district leprosy officer and procurement, not for him. But he warns that if the paper is not dropped, the system is a net addition regardless of design quality. **[Business rule]** **[Constraint]**

“I’d rather you solved that politically before you write a line of code than build something beautiful that dies the way the last one did.”

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Constraint]**

[Emotional/cognitive] | need — adoption precondition, partly outside project control

Q10: Has a case ever reached you that someone had screened earlier and missed? What happened to that person?

A10:

- Narrated one case in detail (“Budhram”, late thirties, still on his register). Presented ~2 years ago with clawing of the fourth and fifth fingers after months of dropping tools; ulnar nerve palpably thickened at the elbow, sensory loss in that distribution, and two hypopigmented patches on the back he had never mentioned because they did not trouble him. **[Emotional/cognitive]**
- Duration: patches present 3–4 years. He had been examined during a house-to-house survey; a worker looked at his back, said words to the effect of “it’s nothing, put some oil on it”, and moved on. He took that as reassurance. **[Constraint]** **[Emotional/cognitive]**
- Outcome: completed 12 months MDT, infection cured, counts as a treatment success in the monthly report. The ulnar nerve damage is permanent — incomplete grip,

anaesthetic fingers, recurrent burns and cuts, lifelong ulcer risk. He is a manual labourer and now earns less. **[Business rule] [Emotional/cognitive]**

- The analytically important part is not the clinical story but the record: nothing was written, so no decision could ever be reviewed, and no one knew a decision had been made. The screening simply vanished. **[Non-functional] [Emotional/cognitive]**
- He then turns this directly onto the project, and this is the liability position: SparshCare changes that by making the negative finding durable. A recorded “low risk” carries a timestamp, the ASHA’s name, and the software’s name. When such a person presents four years later with a claw hand, that record will be retrieved. **[Business rule] [Non-functional]**
- Position stated as non-negotiable: the software may flag, prioritise, and say “these findings warrant urgent examination”. It must never conclude, and must never print “low risk — no action needed” and close a case, because that sentence is a clinical judgement. If it does, liability lands in practice on an untrained ASHA who did what the screen said, or on him. **[Business rule] [Non-functional]**
- What he would accept: every screening recorded including apparently negative ones; negatives given a review path (re-check at ~3 months, or surfacing on a list he can glance at) but never closed; anything borderline or above always reaching a human — a specialist on a video call is acceptable, an algorithm is not — who examines, decides, and whose name attaches to the decision. **[Business rule] [Non-functional]**

“That worker who looked at his back was not negligent, in her mind. She looked, it didn’t seem serious, she said so. There was no record. Nothing was written, so nothing could be reviewed.”

→ Tags: **[Non-functional] [Business rule] [Constraint] [Emotional/cognitive]** | need
— defines an absolute design boundary (BR-04)

Q11: What would you need to see in front of you before you’d act on a risk level that a phone had calculated?

A11:

- He would not act on the score at all — he acts on the findings beneath it. Any score should be collapsed to a word or colour at the top, with raw findings one tap away. The rule must be transparent and explainable (“if X and Y then flag”), not opaque: “the model said so’ is not an answer I can give anyone.” **[Functional] [Non-functional]**
- THE PATCH — present or not; count banded (1 / 2–5 / >5); approximate site; and duration even if crude (weeks / months / years). He singles out duration as the cheapest question in the form and among the most valuable, because “three years” changes urgency completely. **[Functional]**

- SENSATION — the single highest-value item, and the one most likely to be performed badly. He wants not just loss yes/no but confirmation of method: cotton wisp, eyes closed, compared against adjacent normal skin. If it is tapped with a finger while the patient watches, the result is meaningless. He states plainly that a blank field is workable and a wrong field is dangerous. **[Functional] [Non-functional]**
- NERVE — palpation for thickening or tenderness at elbow and behind the knee. He concedes this is demanding for an untrained hand: some of his ASHAs can do it, most cannot reliably. Requires training, and honest recording of “not checked”. **[Functional] [Business rule] [Constraint]**
- FUNCTION AND DAMAGE — three items needing no clinical skill: can both eyes close fully; can the fingers spread and grip a stick or chappal, any curling; any wound or ulcer on hand or foot the person did not feel occurring. He rates the painless wound above any score. **[Functional]**
- HOUSEHOLD — anyone else in the house with a patch, and any prior leprosy treatment in the family. He calls contact information the highest-yield case-finding available and notes it currently reaches him only if he remembers to ask. **[Functional]**
- PHOTOGRAPH — useful with caveats: faint hypopigmentation on dark skin in a dim hut photographs poorly. It supports, never substitutes; nobody, including a remote specialist, should decide from a photo alone. **[Functional] [Non-functional]**
- FREE TEXT / VOICE — argued for explicitly despite being unstructured: one sentence in the ASHA’s own words (“the mother is worried, the patch is spreading, I don’t like this one”). He regards her situated judgement as a real signal and states he has seen her be right when the boxes said nothing. Note the tension with her own confidentiality constraint (Interview 1, Q15) — carried to §4. **[Functional] [Emotional/cognitive]**
- Two conditions on all of it. (a) A blank must reach him as blank; “not checked” and “checked, normal” are different information and merging them is actively unsafe. (b) He must be able to disagree with the system in the record — to close a high flag as tinea, or pull up a low-flagged person because the photograph or her sentence bothers him — without fighting the software, with his reasoning recorded, and with the override never treated as an error. **[Functional] [Non-functional] [Business rule]**

“That’s what I’d need. Not certainty — I don’t expect that from a phone. Enough to know who to see first.”

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | need + solution — field set proposed by stakeholder

Q12: What’s the minimum clinical detail you need arriving with a referral so that you’re not starting blind?

A12:

- Declined to repeat the findings list from Q11 and instead distinguished the clinical payload from the referral wrapper — a useful separation the analyst had not made. **[Functional]**
- IDENTITY AND CONTACT — name, age, sex, village and hamlet, and a phone number that actually reaches the person (theirs, a neighbour's, or the ASHA's). He attributes about half his follow-up failures to register numbers belonging to phones sold years earlier. **[Functional] [Constraint]**
- PROVENANCE — who screened, and when. Not for blame: so he can telephone her before telephoning the patient when something in her account puzzles him. Today he identifies the referring worker by recognising handwriting. **[Functional] [Business rule]**
- PRIOR HISTORY — the item he says people forget and rates above the photograph. Has this person been screened before and told it was nothing (the Budhram pattern)? Have they ever taken and completed MDT? A patch in a previously treated person is a different clinical problem — relapse, or a late reaction requiring urgent steroids rather than MDT. Without it he can start the wrong treatment. **[Functional] [Business rule]**
- DISCLOSURE STATE — what the family has already been told, and specifically whether the ASHA used the word “leprosy” in that house. A family that has spent a week believing the diagnosis requires a different consultation from one that thinks it is a fungus, and he does not want to be blindsided in front of them. **[Functional] [Emotional/cognitive]**
- PRACTICAL WRAPPER — which slot the person is booked into; approximate travel distance (someone who travelled four hours will not be sent away — he will stay late); whether a female attendant will be needed; and whether the person is more comfortable in Halbi or Gondi than Hindi, so an interpreter can be arranged. **[Functional] [Non-functional]**
- Rule he attaches: the whole packet must arrive before or with the patient, structured — not as a chit in their hand. **[Functional] [Business rule]**

“None of that is clinical detail in the strict sense. But it's the difference between an examination that works and one that ends with a frightened person leaving before I've finished.”

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | need

Q13: Inside the PHC, who else handles these records — and how do you keep a person's status from getting around?

A13:

- Facility staff: ANM, pharmacist, lab technician, one multipurpose worker, one attendant; plus a block-level data entry operator who transcribes his monthly figures. The ANM and

pharmacist necessarily know every case — she assists with dressings and examinations, he dispenses MDT and is accountable for stock. **[Constraint]** **[Business rule]**

- Honest assessment of current confidentiality: poor. The register sits on his table in a room whose door does not close properly, with a queue of forty outside. Staff live locally — his pharmacist's wife is from the same block as many patients. No malice required; in a place this size, talk at home is sufficient. **[Constraint]** **[Emotional/cognitive]**
- The treatment itself discloses. The MDT blister pack is visually distinctive and widely recognised, and collection is monthly on a known day, so a patient's pattern of attendance is legible to the village. Patients ask for packs in a plain bag (he complies) or send a relative (which he should not permit and sometimes does, because the alternative is they stop coming). **[Constraint]** **[Emotional/cognitive]**
- Confirms the stigma stakes independently of the ASHA: he has had a father sit in his consulting room and ask him not to write it down anywhere. **[Emotional/cognitive]**
- Access requirement: role-scoped visibility. The ANM and pharmacist see what their work requires. The ASHA sees the people SHE screened in HER area — not a browsable block-wide case list. He explicitly warns against over-correcting the current invisibility into “a village-wide leprosy directory on her phone... a document that will end up being shown to somebody.” **[Non-functional]** **[Emotional/cognitive]**
- Notification requirement, stated sharply: a reminder reading “leprosy appointment Wednesday” lands on a household phone used by husband and son. That discloses a diagnosis the patient had not chosen to share — automatically, at scale, with no human deciding to. Messages must say “PHC appointment” and nothing more. **[Functional]** **[Non-functional]** **[Emotional/cognitive]**
- Goes further: the patient should be able to choose to receive nothing at all, with the ASHA informed instead and visiting in person. He wants this as a supported option, not an edge case. **[Functional]** **[Non-functional]**

“You will have disclosed a diagnosis to a family that the patient had not chosen to tell, and you'll have done it automatically, at scale, with no one deciding to.”

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | *need*

Q14: Once you've confirmed a case, what happens next — what do you have to do, and where does your part end?

A14:

- Confirmation is not a single decision. He classifies paucibacillary or multibacillary (lesion and nerve count, sometimes a slit-skin smear), which sets 6 or 12 months of treatment;

grades disability at that moment as the baseline for comparison at release; registers the case; and gives the first dose under observation. **[Business rule] [Functional]**

- Counselling takes longer than the examination: curable, not hereditary, non-infectious within days of starting treatment (a sentence he says changes how the family behaves toward the patient), and the difficult part — continue for the full course despite feeling well after two months. He locates most defaulting at exactly that point. **[Business rule] [Emotional/cognitive]**
- Household contact examination plus single-dose prophylaxis. He calls this the highest-yield case-finding he does all year, and the step most often skipped when the day is busy — a clear candidate for system prompting. **[Functional] [Constraint]**
- Monthly thereafter: supervised dose and reaction check. He stresses that the reaction — immune flare with nerve swelling — is the real danger during treatment, frequently occurs after treatment has started, requires steroids within days, and is when the permanent damage actually happens. **[Business rule] [Constraint]**
- At completion: release from treatment with disability graded again. **[Business rule]**
- Where his part ends — and he identifies this as the weak point. He can cure the infection; he cannot repair existing damage. Physiotherapy, protective footwear, ulcer care and reconstructive surgery do not exist at the PHC. He refers to the district hospital by writing a chit — the same failure mode he receives from the ASHA, now propagated upward. The patient travels further, often finds the surgeon absent, and most go once and never again. **[Constraint] [Functional] [Emotional/cognitive]**
- Reactions can present one to two years post-discharge, when the patient believes the episode is over and does not return until a nerve is already damaged. **[Constraint] [Emotional/cognitive]**
- Design guidance he offers: screening-to-confirmation alone is useful, but the disability is actually manufactured in the tail — the 12 months of adherence, the day-200 reaction, and the upward referral that goes nowhere. Build so the tail can be added later; do not wall it off. **[Functional]**

“So my part doesn’t end. It just stops being recorded anywhere.”

→ Tags: **[Functional] [Business rule] [Constraint] [Emotional/cognitive]** | need — scope/roadmap input

Q15: The ASHA says a long form is impossible in the field — she covers dozens of houses. You need more clinical detail. Where does the sensible minimum sit, and what must this system do and never do? [deliberate conflict probe]

A15:

- Conceded the point to the ASHA and asked for the concession to be recorded: “if she says it’s impossible, believe her over me — I sit in one room and people come to me; she walks.” Endorsed her Interview 1 prediction that an over-long form produces evening reconstruction from memory or fabrication, not refusal — and noted that invented data is worse than none because he will act on it. **[Business rule]**
[Emotional/cognitive]
- Rejected the framing of a single form and proposed a two-tier structure, which is the substantive resolution to the conflict. **[Functional]**
- TIER 1 — completed for every person screened, deliberately minimal: patch present; sensation reduced (properly tested); painless wound on hand or foot, finger curling, ability to close both eyes; anyone else in the household with a patch; plus her own sentence, spoken if typing is slow. Target ~1 minute, four taps and a sentence, workable in a courtyard with no signal and syncing later. This is compatible with the ≤1-minute target she set in Interview 1, Q21. **[Functional]** **[Non-functional]** **[Constraint]**
- TIER 2 — opens only when Tier 1 trips a flag: patch count and sites, duration, nerve palpation, prior treatment history, photograph. ~3 additional minutes, invested only for a person she is genuinely worried about — “nobody minds work that obviously matters”. **[Functional]** **[Non-functional]**
- His argument for the split: applying one long form to everyone yields the worst compliance on the most important cases, because ninety uninteresting screenings will have exhausted the worker before she reaches the one that counts. **[Functional]**
[Business rule]
- Unprompted budget guidance: spend on the ASHA’s hands, not the form. Sensation testing done well is the most informative act in the entire chain; done carelessly it is noise he will trust and should not. Half a day’s training with real patients, repeated annually, outranks every feature discussed — “if your budget forces a choice between a better app and training the ASHAs, train the ASHAs.” **[Business rule]**
[Emotional/cognitive]
- MUST DO, in his ordering: make a screened person exist immediately in a shared record; book only into genuinely available time and reach the patient if the day breaks; deliver structured findings before or with the patient, with blanks shown as blanks; ensure anything borderline or above reaches a human who decides; replace paperwork rather than add to it (monthly report auto-generated); and show him his area honestly — who was screened, who never came, which panchayat has sent nothing in a year.
[Functional] **[Non-functional]** **[Business rule]**
- MUST NEVER DO: conclude or close a case (“low risk, no action needed”); book his time without confirmation; route routine chasing to him; disclose the condition via a message on a shared family phone; or prevent any user — ASHA, MO or specialist — from overriding it and recording why. **[Business rule]** **[Functional]**

“Build them so Sunita would choose to use them even if nobody made her. She’s the one walking.”

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Constraint]**

[Emotional/cognitive] | *solution — stakeholder-authored resolution to the \$5A conflict*

Q16: Read-back: findings must arrive structured and before the patient; appointments only in time you actually have; software may flag but diagnosis stays yours and borderline cases reach a human; and no second register. Have I got that right, and which comes first?

A16:

- Confirmed with three corrections, all substantive. **[Functional]**
- CORRECTION 1 — the analyst had priced a wasted journey as costing him “an angry OPD”. He rejects this: he is used to angry queues. The cost is that the ASHA spent her credibility persuading a frightened person, the system broke her promise, and every subsequent referral from that hamlet is harder. “The damage is to her standing, not my morning... it changes what you’d optimise.” **[Emotional/cognitive]** **[Functional]**
- CORRECTION 2 — “the diagnosis stays mine” is not a need but a boundary: “needs can be traded against each other; that one can’t. It doesn’t sit on the list, it sits around the list.” Same for the no-second-register point, which he classes as the condition under which anything survives past month four, not as a requirement. **[Business rule]**
- CORRECTION 3 — the read-back omitted non-attendance entirely, which he identifies as the largest hole in the present system. **[Functional]** **[Emotional/cognitive]**
- PRIORITY 1 — make the screened person exist. A name, village, date and findings in a shared record from the moment of screening. Everything else is downstream: no appointment can be booked, no findings routed, and above all no absence detected, for a person with no record. “If you fix only this and nothing else, I’d still call it a success, because ‘didn’t attend’ stops being silence.” **[Functional]**
- PRIORITY 2 — the availability calendar with confirmed slots and a break-notification path. He rates it the least sophisticated item discussed and the highest return per rupee, and notes it is what will make the ASHA believe in the rest — “she’ll trust a system that stops embarrassing her in front of families”. **[Functional]** **[Non-functional]** **[Emotional/cognitive]**
- PRIORITY 3 — structured findings arriving with the referral. He is candid that this one is largely for his own convenience: the first two determine whether a person is seen at all; this determines how well he uses the twenty minutes once they are in front of him. “Important — not the same order of importance.” **[Functional]**
- Explicit build guidance: if only one thing is built this year, build the first; if two, the first two. **[Functional]**

- Closing request: written notes to be sent to him for correction before anything is finalised — “I’d rather do that on paper than discover it in a demo” — and the Tier-1 form to be negotiated with the ASHA present in the same room rather than each stakeholder being told separately that the other is wrong. **[Business rule]**
[Emotional/cognitive]

“You cannot book an appointment for someone who isn’t in the system. You cannot send me findings about a person who has no record. And you certainly cannot notice that they didn’t come.”

→ **Tags: [Functional] [Non-functional] [Business rule] [Emotional/cognitive] | need**
— *prioritisation confirmed by stakeholder*

3. TOP 3 PAINS I HEARD

1. Field cases are invisible to the clinician until the person walks in — and many never do

A referral exists only as a chit in a patient’s hand and a memory in an ASHA’s head. Nothing reaches the PHC in advance, so a person who is screened but never attends leaves no trace anywhere in the system — “not a defaulter, just absent from a list that was never made”. Accountability falls between the two stakeholders: she believes he has seen them, he does not know they exist. This is the pain the stakeholder himself ranked first, on the ground that every other requirement is downstream of it — appointments, routed findings, and non-attendance alerts are all impossible for a person with no record. Corroborated from the opposite end by Interview 1, Q13 (“posting a letter and never hearing back”). (Q3, Q8, Q16)

2. Referrals arrive with no clinical content, so every case is re-examined from zero — inside a four-minute-per-patient queue

The chit says “patch on back, please examine”. Everything the ASHA observed — duration, sensation testing and its method, household contacts, any nerve concern — is lost between the house and the consulting table. A proper examination takes 15–20 minutes against an OPD load of 80–120 patients per session, and at a confirm rate of roughly 2 in 10 the majority of that time resolves to tinea or vitiligo. The cost is not borne by the Medical Officer but by the thirty patients at the back of the queue whom he then rushes. His own reframing is the design insight: structured findings in advance would not let him diagnose remotely, but would let him triage — deciding whose twenty minutes is urgent. (Q3, Q4, Q5, Q12)

3. Appointments bear no relation to actual availability — and the wasted journey is charged to the ASHA’s credibility

Patients travel up to half a day, unannounced, and are turned away when the Medical Officer is on immunisation duty, at a block meeting, or on leave — absences that are almost entirely predictable to him but visible to no one else, since the PHC landline has been dead over a year and the hamlets have no signal. The direct cost is ₹60–80 and a lost day’s wage against a painless lesion. The compounding cost, which he insisted the analyst re-price, is that the ASHA spent her personal standing persuading the family and the system broke her promise, making

every subsequent referral from that hamlet harder. He states that a working calendar alone would justify the project. (Q6, Q7, Q16)

Honourable mentions (real, but second tier)

- Reporting duplication: 2–3 hours at month end transcribing counts that a block operator then re-keys into the national portal — the same figure written three times by three people (Q9).
- No supervisory visibility: referral volume and hit rate vary widely between ASHAs and at least one panchayat sends almost nothing, but he knows this only by recognising handwriting — “my supervision is anecdote” (Q5).
- Confidentiality inside the facility: no private consulting space, locally-resident staff, and a visually identifiable MDT blister pack collected monthly on a known day (Q13).
- The untracked treatment tail: 25–30 people on monthly MDT follow-up at any time, reactions presenting up to two years post-discharge, and upward referral for disability care that he makes by writing exactly the kind of chit he complains about receiving (Q7, Q14).
- Contact examination and prophylaxis — the highest-yield case-finding he does — is the step most often skipped on a busy day, and nothing currently prompts it (Q14).

4. CONTRADICTIONS AND THINGS TO VERIFY

Items to triangulate against records or other stakeholders (ANM, block NLEP supervisor, district leprosy officer) before requirements are fixed. Cross-references to Interview 1 (ASHA) are marked [X-REF].

- **[X-REF — HIGH PRIORITY] Direct contradiction on the OPD roster** — The ASHA tells patients to attend on “Tuesday and Friday, for the skin doctor” (Interview 1, Q6). The Medical Officer names Tuesday and Friday as his immunisation and FIELD days — the days he is out (Q7). If accurate, referrals are being routinely directed to the two days the clinician is least likely to be present, which would independently explain a large share of wasted journeys and non-attendance. Verify against the actual PHC duty roster immediately; this may be a live operational problem worth reporting back regardless of the software.
- **Case-load and disability figures are the stakeholder’s own estimates** — he explicitly disclaimed them (“I don’t want to give you a figure you’ll put in a slide”). ~30 new cases/year and 10–12% Grade 2 disability at detection should be checked against the PHC MDT register and block NLEP returns before use as a baseline. Note his own caveat that the figure describes cases found, not cases existing.
- **The 2-in-10 confirmation rate is an impression, not a counted statistic** — no denominator of referrals received is recorded anywhere today. It is currently

unmeasurable, which is itself a finding: the project cannot demonstrate improvement against a baseline that does not exist. Consider a manual count over one quarter.

- **Claimed variation in referral quality between ASHAs** (3–4 in 10 versus 1 in 10, and a near-silent panchayat) rests on handwriting recognition over four years. Plausible and important, but unevidenced. FR-25 would generate the data; it cannot be validated before build.
- **Minimum field set** — partial conflict with Interview 1, resolved but not closed. His Tier-1 list overlaps the ASHA's six fields closely. The genuine additions are symptom duration, prior-treatment history, simple function tests, and nerve palpation. Nerve palpation is the weak link: he concedes most ASHAs cannot perform it reliably, so placing it in Tier 2 assumes training that does not currently exist. Do not specify it as a decision input until training is funded.
- **[X-REF] Voice input remains unresolved across both interviews** — The ASHA calls Hindi voice entry “the best thing you could give me” (Interview 1, Q20); the Medical Officer independently argues for her free-text sentence as high-value signal (Q11). Both conflict with her own confidentiality constraint — speaking findings aloud in a shared courtyard is precisely the disclosure she fears (Interview 1, Q15, Q17). Needs co-design: possibly voice permitted only for non-identifying content, or only once she has left the house.
- **Capacity arithmetic is not yet costed against total demand** — Two protected afternoons (~10 slots/week) comfortably covers ~150 new referrals/year, but excludes 25–30 concurrent MDT follow-ups and campaign surges of ~30 suspects per fortnight. Extract the actual monthly follow-up volume from the MDT register before sizing the booking model.
- **[X-REF] Both stakeholders state that no feedback path from PHC to ASHA exists** — Still worth verifying with the ANM whether a referral or outcome register exists on paper and is simply not operated — this decides whether FR-22 is a new integration or the digitisation of a dormant process.
- **The liability position (BR-04) is stated by the stakeholder as absolute, but its resolution is not a design matter** — Who owns a recorded risk determination, and what a negative screening record implies for the ASHA and the Medical Officer, requires programme and legal review at district/state level. Escalate; do not resolve inside the design team.
- **Auto-generated monthly reporting (FR-24) depends entirely on the district accepting a digital return in place of the paper format** — The stakeholder explicitly deferred this as outside his authority. Until the district leprosy officer confirms, treat NFR-14 (net-zero paperwork) as an assumption, not a requirement met.
- **The 1–2 child cases per year figure is offered as a transmission indicator** — and would materially strengthen the case for contact-tracing features (FR-20, and the contact-

examination prompt in Q14). Verify against the register before it is used in any business case.

- **Stakeholder bias to note** — Every additional field in the referral saves the Medical Officer time and costs the ASHA time; his interest points toward more detail exactly as hers points toward less. His two-tier proposal is a good-faith attempt to reconcile this and both parties should be present when Tier 1 is fixed — which he requested himself.
- **The “Budhram” narrative (Q10) is a persona illustration produced in role-play, not a case record** — It is retained because it defines the liability requirement, but must not be cited as evidence.
- **Coverage gap in this session** — teleconsultation, one of SparshCare’s three stated functions, was deferred by the stakeholder to a second interview and is therefore unelicited. The only signal captured is his stated caution that teleconsults may add coordination burden on him, and his acceptance in Q10 that a remote specialist counts as “a human” for escalation purposes. Requirements for teleconsult must not be inferred from this log.

5. RECORDED CONFLICTS

Logged separately because these are open decisions for the project, not resolved requirements.

5A. Form length — ASHA (Interview 1) versus Medical Officer (Interview 2)

- ASHA position: ~100 individuals examined per campaign day makes anything beyond ~6 fields infeasible; the predicted outcome is fabricated data rather than refusal, and a long interrogation is itself a confidentiality breach (Interview 1, Q19).
- Medical Officer position: vagueness is precisely what forces him to re-examine from zero and consume a 20-minute slot inside a four-minute-per-patient queue (Q3, Q4).
- Resolution proposed by the Medical Officer, unprompted, and recorded as his own: a two-tier form. Tier 1 (~1 minute, 4 taps and a sentence) for every person screened; Tier 2 (~3 minutes) opening only when Tier 1 trips a flag. His reasoning — a single long form yields the worst compliance on the most important cases, because the ninety uninteresting screenings exhaust the worker before she reaches the one that matters.
- Status: strong candidate resolution, NOT yet agreed. It has not been put to the ASHA. Open questions — (a) who defines the Tier-1 trip condition, and is it conservative enough to satisfy the Medical Officer without opening Tier 2 on most screenings; (b) whether nerve palpation can sit in Tier 2 at all without funded training; (c) whether Tier 2 can be deferred to a private completion step, which would also serve Interview 1’s FR-10.
- Both stakeholders have asked to settle the Tier-1 field list in the same room. Recommend a joint working session rather than further separate elicitation.

5B. Clinical decision authority and recorded liability

- Stakeholder position, stated as non-negotiable: the software may flag, prioritise, sort and warn; it must never conclude, and must never render “low risk — no action needed” and close a case. Anything borderline or above must reach a human who examines, decides, and whose name attaches to that decision — a remote specialist is acceptable; an algorithm is not.
- His reasoning is about the durability of the record, not about model accuracy: today a dismissive screening leaves no trace and cannot be reviewed; a recorded negative carries a timestamp, the ASHA’s name and the software’s name, and will be retrieved when a missed case presents years later. Liability then lands on an untrained worker who followed the screen, or on him.
- Project implication: this constrains the product beyond the interface. It rules out auto-closure of negative screenings, requires a review path for apparently negative cases, and requires that every risk rule be transparent and explainable (“‘the model said so’ is not an answer I can give anyone”).
- Status: outside the design team’s authority to settle. Requires programme and legal sign-off on data ownership, record retention, and where responsibility sits for a recorded screening outcome. Escalate to the district leprosy officer and the programme’s legal advisor before the risk-flagging module is specified.

5C. Clinician workload — an explicit anti-requirement

- The stakeholder was emphatic that non-attendance chasing must NOT be routed to him as tasks. Routine follow-up belongs to the ASHA, who is already in the village; he wants aggregate, periodic, self-initiated oversight only.
- The single exception: a high-concern flagged case that fails to attend must escalate to him actively and immediately.
- He warns that blurring the two channels destroys both — “if you blur that line and send me everything, you’ll have built a system that I ignore, and then the alerts that actually matter get ignored along with the rest.” This is recorded as a conflict because programme monitoring will be tempted to route everything to the clinician as the accountable officer.

APPENDIX A — CONSOLIDATED REQUIREMENTS REGISTER (FROM THIS INTERVIEW ONLY)

Derived from the tags above. Numbering continues from Interview 1 (ASHA) so that the two registers merge without collision. Priorities are this analyst’s recommendation; Must-have status on FR-11 and FR-13 follows the stakeholder’s own ranking at Q16. Every line traces to a source question; nothing here is inferred beyond what was said.

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-11	Create a persistent, shared record of every person screened at the moment of screening — name, village, date, findings — visible to both the referring ASHA and the PHC	Functional	Q3, Q8, Q16	Must have
FR-12	Deliver structured screening findings to the Medical Officer before or with the patient's arrival, in a form that supports triage (not remote diagnosis)	Functional	Q3, Q4, Q11, Q12	Must have
FR-13	Publish the Medical Officer's confirmed availability and allow booking only into slots he has confirmed	Functional	Q6, Q7	Must have
FR-14	Propose-and-confirm booking: the system may propose and hold a slot; it must never commit clinician time automatically	Functional	Q7	Must have
FR-15	Notify the patient and the referring ASHA when a booked slot breaks — in time to stop the journey, not after it	Functional	Q6, Q7	Must have
FR-16	Record and display "not checked" distinctly from "checked, normal" in every clinical field; blanks must never render as normal findings	Functional	Q11	Must have
FR-17	Clinician override in both directions (escalate a low flag, close a high flag) with recorded reasoning; an override must never be treated as an error	Functional	Q11, Q15	Must have
FR-18	Two-tier screening form: Tier 1 (~1 min) for every person screened; Tier 2 (~3 min) opening only when Tier 1 trips a flag	Functional	Q15	Must have
FR-19	Free-text or voice note in the ASHA's own words, presented verbatim to the Medical Officer alongside the structured fields	Functional	Q11	Should have
FR-	Capture prior history: previously	Functional	Q12	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
20	screened and told it was nothing; previously took or completed MDT (distinguishes relapse and reaction from new disease)			
FR-21	Referral wrapper: identity, a working contact number, referring worker and date, booked slot, travel distance, language, need for a female attendant, and what the family has already been told	Functional	Q12	Should have
FR-22	Make non-attendance a visible state; route routine follow-up to the referring ASHA, never to the Medical Officer as a task list	Functional	Q8	Must have
FR-23	Actively escalate to the Medical Officer only high-concern flagged cases that fail to attend — a separate channel from routine non-attendance	Functional	Q8	Must have
FR-24	Auto-generate the monthly NLEP progress report, disaggregated, from data already captured — for the Medical Officer to check and sign, not re-key	Functional / Business rule	Q9	Must have
FR-25	Supervisory view for the Medical Officer: screening, referral and confirmation counts by worker and by panchayat, including areas with zero referrals	Functional	Q5	Should have
FR-26	Protect MDT follow-up capacity (25–30 concurrent patients, monthly) separately from new-suspect slots; allow the Medical Officer to open surge capacity during campaigns	Functional	Q7, Q14	Should have
FR-27	Photograph capture as supporting evidence only — explicitly never sufficient on its own for any decision, local or remote	Functional	Q11	Could have
FR-28	Prompt household contact examination and single-dose	Functional	Q14	Should have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	prophylaxis when a case is confirmed — the highest-yield case-finding step and the one most often skipped			
FR-29	Design for later extension into the treatment tail: monthly supervised dosing, reaction alerts, post-release reactions, and referral upward for disability care	Functional	Q14	Could have (roadmap)
NFR-11	Any risk rule must be transparent and explainable; the score is collapsed to a word or colour with raw findings one tap away	Non-functional	Q11	Must have
NFR-12	Role-scoped access: an ASHA sees only the people she screened in her own area — no browsable block-wide case list on any field device	Non-functional	Q13	Must have
NFR-13	Notifications must never disclose the condition — “PHC appointment” only — and the patient may opt out of all messaging in favour of an in-person visit from the ASHA	Non-functional	Q13	Must have
NFR-14	Net-zero paperwork: the system must replace existing registers and returns, not add a further transcription. Adoption condition, not a feature	Non-functional	Q1, Q9	Must have
NFR-15	Minimal clinician-facing load: no daily task lists to the Medical Officer; oversight is aggregate, periodic and self-initiated	Non-functional	Q8, Q9	Must have
NFR-16	A Tier-1 screening record must be completable in ~1 minute and a Tier-2 record in ~3 further minutes (consistent with NFR-06 from Interview 1)	Non-functional	Q15	Must have
BR-04	Clinical decision authority rests with a clinician. Software may flag, prioritise and warn; it must never conclude, and never close a case as “low risk —	Business rule	Q10, Q11, Q15, Q16	Absolute boundary

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	no action needed”			
BR-05	Every borderline or higher case must reach a human examiner — the Medical Officer or a specialist — whose name attaches to the decision. Negative screenings are given a review path, never closed	Business rule	Q10, Q15	Absolute boundary
BR-06	Classification PB / MB determines 6 or 12 months of MDT; disability is graded at diagnosis and again at release from treatment	Business rule	Q14	Constraint on design
BR-07	Monthly supervised dose plus reaction check for the full course; reactions require steroids within days and may present up to ~2 years after release	Business rule	Q14	Constraint on design
BR-08	Sensation testing and nerve palpation cannot be delegated to non-clinical PHC staff — a leprosy suspect requires the Medical Officer personally	Business rule	Q3	Constraint on design
BR-09	Reporting chain: MO monthly report → block → district → national portal, with a block data operator re-keying from paper	Business rule	Q9	Verify — see §4
CON-06	OPD load is 80–120 patients per session, i.e. under 4 minutes per patient; a proper leprosy examination takes 15–20 minutes and cannot be compressed	Constraint	Q4	Absolute
CON-07	Realistic protected capacity is ~2 afternoons per week, ~10 usable slots — against ~150 referrals/year plus 25–30 concurrent follow-ups and campaign surges	Constraint	Q7	Absolute
CON-08	Availability is not fully predictable: emergencies, outbreaks and unannounced inspections pull the Medical Officer out at no notice	Constraint	Q6, Q7	Absolute

ID	Requirement / observation (stated as a need)	Type	Source	Priority
CON-09	No private consulting space; staff are locally resident; the MDT blister pack is visually identifiable and collected monthly on a known day	Constraint	Q13	Absolute
CON-10	Disability care — physiotherapy, protective footwear, ulcer care, reconstructive surgery — is unavailable at the PHC; upward referral to district currently fails in the same way ASHA referrals fail	Constraint	Q14	Absolute
CON-11	Approximately 2 in 10 referrals are confirmed; over-referral is the clinically correct failure direction and must not be designed against	Constraint	Q5	Absolute
EC-02	Liability exposure: a recorded “low risk” that proves wrong lands on an untrained ASHA or on the Medical Officer. Primary source of clinical resistance to risk scoring	Emotional/cognitive	Q10, Q11	Must design for
EC-03	Residual distrust from a prior tablet system that added a fourth transcription, returned nothing, and was abandoned within eight months. Primary adoption risk on the clinical side	Emotional/cognitive	Q1, Q9	Must design for
EC-04	The cost of a broken referral is charged to the ASHA’s standing in the village, not to the clinician’s time — this changes what the system should optimise	Emotional/cognitive	Q6, Q16	Must design for

Design rules implied, in the stakeholder’s own framing

- “Make the screened person exist.” Everything else — booking, routed findings, non-attendance detection — is downstream of a record existing at the moment of screening.
- Blank is not normal. “Not checked” and “checked, normal” are different information; merging them is actively unsafe.
- Flag, prioritise, warn — never conclude. A negative screening gets a review path, not a closure.

- Propose slots; never commit clinician time automatically. One double-booking loses this user permanently — “you get exactly one chance at that”.
- Send the worrying ones. If everything reaches the clinician, nothing does.
- Replace a register or add nothing. A system that does not remove paperwork is a net addition regardless of how good it is.
- Optimise for the person walking: “build them so Sunita would choose to use them even if nobody made her.”
- Spend on hands before features: if the budget forces a choice between a better app and training the ASHAs in sensation testing, train the ASHAs.

Recommended next steps

- IMMEDIATE — verify the PHC duty roster against what the ASHA tells patients (§4, first item). If referrals are being sent on the Medical Officer’s field days, that is an operational finding worth reporting back to the block irrespective of the software.
- Second session with this stakeholder, already requested by him, covering teleconsultation — the function this interview did not reach — plus his review of these written notes before anything is finalised.
- Joint working session, Medical Officer and ASHA together, to fix the Tier-1 field list. Both have asked for this; further separate elicitation will not resolve §5A.
- ANM at the sub-centre — the paper flow between the two stakeholders, and whether any dormant referral-outcome register already exists (decides whether FR-22 is integration or digitisation).
- Block NLEP supervisor / district leprosy officer — whether a digital return can replace the paper monthly format (NFR-14, FR-24), the origin and negotiability of the 25-field request from Interview 1, and whether screening carries an ASHA incentive.
- Programme and legal review of BR-04 / BR-05 and EC-02 — ownership of a recorded screening determination — before the risk-flagging module is specified.
- District hospital — the disability-care referral path (CON-10), which is currently the same broken chit mechanism one level up.
- One dermatologist or NLEP specialist — clinical validation of the Tier-1 and Tier-2 field sets, and the design of the teleconsult escalation route accepted in principle at Q10.

STAKEHOLDER INTERVIEW LOG

SparshCare — Requirements elicitation · Interview 3 of 6

Label	Content
Analyst	Dr. Gunjan Singhal
Date of interview	17 August 2026
Stakeholder role	Dermatologist & Leprosy Specialist, District Hospital (sole specialist for the district; teleconsultant for high-risk cases flagged in the field)
Stakeholder (persona)	Dr. Meera Iyer
Location / facility	District Hospital, Bastar, Chhattisgarh
Project	SparshCare — leprosy screening, teleconsultation & referral system
AI tool used	Claude (Cowork) — model claude-opus-5, role-play persona “Dr. Meera Iyer, Dermatologist & Leprosy Specialist”
Interview mode	Semi-structured, single session, 13 questions with follow-up probes
Duration (in-persona)	~25 min
Chat link / screenshot ref	SPARSHCARE / INT-03 / MEERA-DERM — [paste chat link or screenshot filename]

1. PURPOSE AND INTERVIEW APPROACH

Objective: this interview set out to understand, from the specialist end of the teleconsultation link, what a remote opinion actually needs in order to be safe and useful — what context must arrive before a call is even joined, how consultation time can be scheduled around a fixed and already-overloaded clinical calendar, what image and data quality a diagnosis can tolerate given the field worker’s entry-level device and patchy network, and how a specialist’s opinion gets captured, actioned and tracked once it has been given. It also probed the informal WhatsApp-based consultation channel already in use and where those images end up, the language and household dynamics that shape what a remote call into a tribal home can actually learn, and — critically — where a case goes, and whether anyone ever finds out, when it cannot be resolved remotely.

Question flow used (deliberate sequence — rapport → current process → evidence → pain → constraints → conflict → solution ideas last):

- **Q1-Q2 (rapport / current state)** — coverage and caseload across the district, then the anatomy and cost of the last unstructured teleconsultation.
- **Q3-Q5 (evidence)** — what an in-person examination actually captures and precisely what a screen cannot carry; the informal image channel in use today and where those

images end up; a documented case in which an unusable image caused permanent harm.

- **Q6-Q7 (constraints)** — the information a consultation needs to be worth her time, down to a stated minimum floor; and the language, translation and household dynamics that shape what a remote call can actually learn.
- **Q8-Q9 (constraints)** — her real weekly clinical calendar and the windows it can spare; what a wasted or unattended slot costs, and how she would rather use that time.
- **Q10-Q11 (pain)** — what happens to a case she cannot resolve remotely, whether the next step is ever tracked, and what outcome feedback she would actually use.
- **Q12 (conflict)** — the image and data quality she needs to assess a patch, set directly against the ASHA's entry-level phone and absent network.
- **Q13 (solution ideas last)** — validating the analyst's playback of the whole interview and naming her single first priority.

Tag key: Each captured line is marked with one or more of five categories — **[Functional]** (something the system must do) · **[Non-functional]** (a quality it must have) · **[Business rule]** (an existing organisational rule) · **[Constraint]** (a fixed environmental, economic or regulatory limit) · **[Emotional/cognitive]** (fear, trust, stigma, mental load). Each answer additionally carries a roll-up tag and is marked *need* (a problem to be solved) or *solution* (a proposed answer, to be traced back to the underlying need). In Appendix A these categories are carried forward as ID prefixes: FR = functional, NFR = non-functional, BR = business rule, CON = constraint, EC = emotional/cognitive.

2. QUESTION-BY-QUESTION LOG

Q1: Tell me about your work — how large an area do you cover, and how much of it do you handle without the patient in front of you?

→ *Probe: how many leprosy cases a month, and how many already have nerve damage or deformity by the time you see them?*

A1: Coverage, caseload and the current state of remote working.

- Covers ~11 blocks, ~1.8 million population. **Only dermatologist with leprosy training in the government system in the district.** A second post has been sanctioned but vacant for three years. **[Constraint]**
- Based physically at District Hospital; patients travel to her. From far blocks that is a **three-hour bus journey each way** — “which is why half of them don’t come.” **[Constraint]**
- Remote work today is **unstructured**: MOs or supervisors phone or WhatsApp her ad hoc. Roughly **15-20 such contacts a month**, arriving at any time — during OPD, during a

biopsy, at 9 p.m. “That’s not teleconsultation, that’s a favour.” [Business rule]
[Emotional/cognitive]

- New confirmed cases reaching her: **25–40 per month**, seasonal (spike after ASHA survey rounds, then a lull). [Constraint]
- About **one-third are multibacillary with visible nerve involvement at presentation**. [Constraint]
- **Grade 2 disability at diagnosis: ~8–10%** (claw hand, foot ulcer, corneal involvement). “It should be under two.” A Grade 2 case has usually been symptomatic for a year or more — nobody caught the anaesthetic patch when it was still just a patch. [Constraint]
[Emotional/cognitive]

→ Tags: [Constraint] [Business rule] [Emotional/cognitive] | *need — capacity is a hard ceiling; the system must increase reach without increasing interruptions*

Q2: Tell me about the last remote consultation you did. How did it come to you, and what did you know before you joined?

→ Probe: *how long before you understood why you’d been called, and what did that cost?*

A2: Anatomy of the last unstructured teleconsultation, and what it cost.

- Last Thursday, ~11:15 a.m., **mid-OPD**. Unannounced WhatsApp video call from a PHC Medical Officer in an eastern block. [Constraint]
- What she knew before answering: **the caller’s name. Nothing else**. She answered only because “if I don’t answer, he stops calling, and then someone gets missed.” [Emotional/cognitive] [Functional]
- Call opened with apologies, then a hunt for the ASHA (who had stepped out), then the MO walking through a corridor with the camera. [Constraint]
- **It took nine minutes to establish why she had been called.** [Functional] [Non-functional]
- The ASHA had done a sensation test **three weeks earlier**, but her findings were in a **paper register at her home** — nobody present could report them. Dr. Iyer had to re-derive the history over the phone from a frightened patient answering in monosyllables. [Functional] [Business rule] [Emotional/cognitive]
- Cost: **11 patients waiting**, two of whom had travelled from far blocks that morning. She gave the call 22 minutes, of which “maybe six were clinically useful.” The last two patients on the list got four minutes each — “that is not medicine, that’s processing.” [Constraint] [Emotional/cognitive]

- Outcome: **no confident opinion possible**. Yellow fluorescent corridor light, video breaking into blocks; she could not see the border of the patch (the discriminator between leprosy, vitiligo, pityriasis alba and fungal patch). She referred the woman to district — the exact three-hour journey teleconsultation was meant to avoid. **[Non-functional] [Constraint]**

→ Tags: **[Functional] [Non-functional] [Emotional/cognitive]** | *need — pre-call context is missing; a structured record in place of the paper register also surfaced as a solution-shaped detail*

Q3: Walk me through how you assess a patch when the person is in front of you — what do you look at, touch, ask?

→ Probe: *which of those can you not do through a screen?*

A3: The in-person examination, and precisely which parts a screen cannot carry.

- **Look:** in daylight, never under tube light. Number and site of patches; **asymmetry** (leprosy is characteristically asymmetric); **margin definition and whether raised**; surface (dry, scaly, shiny); **loss of hair**; **absence of sweating** over the patch. **[Functional]**
- **Touch — “this is where the diagnosis actually lives”:** sensation testing over the patch versus adjacent normal skin **with the patient’s eyes closed** (cotton wisp, pin, sometimes warm/cold). “If a patch is anaesthetic, that is close to diagnostic in this country.” **[Functional] [Business rule]**
- **Nerve palpation**, both sides, with the back of the fingers: ulnar at elbow, median at wrist, radial cutaneous, common peroneal at fibular neck, posterior tibial behind the ankle, great auricular in the neck — feeling for **thickening, cord-like texture, tenderness, nodularity**. “A tender thickened nerve means a reaction and that person needs steroids today, not next month.” **[Functional] [Business rule]**
- **Motor testing:** little-finger abduction, finger spread against resistance, wrist lift, foot dorsiflexion, **full eye closure**. Then inspection of hands and feet for ulcers, cracks, callus, digit shortening. **[Functional]**
- **Ask:** duration; numbness; tingling/burning/weakness; **any sudden change in the last two weeks** (new lesions, existing lesions turning red/swollen, fever, nerve pain — “that’s a reaction and it’s the emergency in this disease”); household/neighbourhood contacts; previous MDT and whether completed; pregnancy; drug history. **[Functional] [Business rule]**
- **Cannot be done through a screen:** palpate a nerve (“the most important single piece of information for predicting whether this person ends up with a claw hand — no camera on earth gives it to me”); perform the sensation test; slit-skin smear; feel skin dryness or infiltration. **[Constraint]**

- Colour and margin are *theoretically* transmissible but practically lost: yellow indoor light erases hypopigmentation, compression smudges a defined border. [Constraint] [Non-functional]
- **Key reframe:** “I am not examining the patient. I am examining the health worker’s examination.” Consequence: quality of captured findings matters more than video quality. [Emotional/cognitive] [Functional]
- **Hard clinical boundary:** she will never authorise starting MDT from a screen alone. Teleconsult decides **who** needs in-person confirmation, **how fast**, and **whether it is an emergency**. It does not replace confirmation. [Business rule]

→ **Tags:** [Functional] [Business rule] [Constraint] | *need — defines the mandatory data model for sensation and nerve findings, and the governance boundary on treatment initiation*

Q4: When a colleague or PHC wants your view on a skin lesion today, outside any formal system — how does that reach you?

→ *Probe: afterwards, what happens to those images? Who else has them?*

A4: *The informal WhatsApp channel in use today, and where the images end up.*

- **Personal WhatsApp, personal phone.** Her number has circulated for years and is now in ~30 phones across the district. She is a member of **two WhatsApp groups she did not create**, one with **41 members, several of whom she cannot identify**. [Constraint] [Non-functional]
- Typical inbound: a photograph with no text or “Madam pls advise.” No age, duration, site or sensation test. Occasional voice notes recorded on a bus. She replies between patients, in incomplete sentences. [Functional] [Constraint]
- **Image sprawl:** every image persists on the ASHA’s phone, the MO’s phone, every group member’s phone, every forward, and every device’s camera roll auto-backing-up to consumer cloud. “I have no idea where a photograph goes after I’ve seen it. None.” [Non-functional] [Emotional/cognitive]
- Images frequently include the **patient’s face** (uncropped, or deliberately included for identification), and often intimate sites — breast, buttock — sometimes children. [Non-functional] [Emotional/cognitive]
- **Stigma is the real risk, not paperwork:** a confirmed diagnosis can end a marriage negotiation, get a woman returned to her parents’ house, cost employment. Patients ask her not to write the word in their file. “That could destroy someone’s life, and I would have been the reason it got taken.” [Emotional/cognitive]

- **No medical record exists.** Her opinion lives as a WhatsApp message — not in the PHC file, not retrievable, not defensible if a case is questioned later. “A chat history I might have cleared.” [Business rule] [Non-functional]
- Stated trade-off: “If you give me something where the image goes to a place instead of a person, and my written opinion lands in the patient’s actual record — **I will use it even if the video quality is worse. The record matters more to me than the picture.**” [Functional] [Emotional/cognitive]

→ Tags: [Non-functional] [Business rule] [Emotional/cognitive] | *need — governed image custody and a real medical record; she explicitly ranks the record above image fidelity*

Q5: Tell me about a time the image or video was too poor to assess. What did you do?

→ Probe: *what happened to that patient meanwhile — what did the delay cost?*

A5: *Two failures caused by unusable images — one routine, one permanent.*

- **The common failure:** photo taken indoors under tube light from ~4 feet, whole back in frame, patch the size of a fingernail on screen; or flash used, washing out the pigment difference; or blurred. She asks for a retake. “Sometimes a better one comes in two days. **Usually nothing comes at all,**” because the ASHA has moved to another hamlet. “The case just — evaporates.” [Non-functional] [Constraint]
- **The case she thinks about (~14 months ago):** male farm labourer, late thirties, eastern block. Video dropped four times; they fell back to audio. The MO reported “some numbness” and “a little weakness,” and on nerve palpation said “normal, maybe slightly thick.” **He had never been taught to palpate a nerve** — which she did not know at the time. [Constraint] [Business rule] [Emotional/cognitive]
- What the words concealed: **early clawing of the 4th and 5th fingers and a tender nerve — a Type 1 reaction**, requiring prednisolone within days. She advised starting MDT at the PHC and coming to district “when he can.” [Functional] [Emotional/cognitive]
- He arrived **eleven weeks later with fixed clawing** — not reversible with steroids. He will be cured of the infection but cannot grip. “A farm labourer who cannot close his hand around a handle.” [Emotional/cognitive]
- **Wider cost:** household lost his wage; and the village saw what happened. “The visible hand is the thing that produces the stigma, more than the diagnosis — so the next person with a numb patch in that hamlet is now *less* likely to come forward. Deformity is contagious in that sense. It teaches people to hide.” [Emotional/cognitive]
- **Her own conclusion:** the information was not unavailable, it simply never reached her. Three good daylight photographs, a marked-up sensation test and a nerve chart sent the day before would have let her read it “at seven in the morning with my chai” and call

the reaction. **“The live call is nice. The data arriving before the call is the thing that actually changes outcomes.”** [Functional] [Non-functional]

- Where the network will not carry a call — “and in a third of my district it won’t” — she would rather have slow photographs plus an audio note than a failed video call. [Functional] [Constraint]

→ Tags: [Functional] [Non-functional] [Constraint] [Emotional/cognitive] | *need — asynchronous store-and-forward as a first-class path; a competency gap in nerve palpation also surfaced as a hidden dependency*

Q6: Before you join a call, what do you need to know about the person for that time to be worth anything?

→ Probe: *what is the least you would accept and still give an opinion?*

A6: *The pre-call information packet: the full set, and the minimum floor.*

- Target: **open a case the evening before and read it in four minutes.** [Non-functional]
- **Who:** age, sex, village, distance from PHC, **and the name of the ASHA** — “I want to know whose eyes I’m borrowing; after six months I’ll know which of them are reliable.” [Functional] [Business rule]
- **Story:** duration; change over time; numbness; anything new in the last two weeks; household/neighbourhood contacts; previous MDT and completion; **pregnancy** (“unglamorous, and it changes what I can prescribe”). [Functional]
- **Examination:** number and site of patches; **sensation test with method stated and result per patch** (not “numbness present”); **nerve palpation of all main nerves, both sides, thickened Y/N and tender Y/N** — “tender is the word I look for first”; simple motor (eye closure, finger spread, foot drop); existing ulcer or deformity. [Functional] [Business rule]
- **Images:** 2-3, daylight, close, in focus, with a scale reference, **cropped to exclude the face.** [Functional] [Non-functional]
- **The referral question, in plain words, at the top:** “why am I being called, and what do you want from me?” e.g. *Is this leprosy or fungal? Is this a reaction? Can this person start MDT at the PHC or must they come to district?* “Half the value of a consultation is knowing what’s being asked.” [Functional]
- **Operational:** will the patient actually be present at the appointed time, and does she know a doctor will be looking at her. [Functional]
- **The minimum floor (six lines and a picture):** age & sex; duration; one sentence of history; sensation test result on the main patch; whether any nerve was tender or thickened; whether there is any weakness; one usable photograph; **and the question**

being asked. With that she can nearly always return one of three dispositions — *see urgently / can wait for the next outreach camp / not leprosy, treat locally.* [Functional] [Business rule]

- **Failure mode below the floor:** “I don’t refuse, I just default to ‘send them to district’” — which sounds safe but is the real failure, because a three-hour journey means half never arrive. **“Every missing field pushes me toward an unsafe referral dressed up as a safe one.”** [Emotional/cognitive] [Business rule]
- **Non-negotiable field design:** the sensation test must record **how** it was done and **whether the patient’s eyes were closed.** “Don’t let the form accept ‘sensation: normal’ without the method.” A test done with the patient watching yields a false negative — “and a false negative on an anaesthetic patch is how we miss the diagnosis entirely.” [Functional] [Business rule]
- **Timing:** the packet must arrive **the evening before**, not ten minutes before. “If it arrives ten minutes before, I read it in the corridor and I’m back to guessing.” [Non-functional]

→ Tags: [Functional] [Business rule] [Non-functional] | *need, stated with solution-level precision — usable almost directly as acceptance criteria*

Q7: When you’re speaking with someone from an interior tribal village, how do you and the patient understand each other, and who else is at their end?

→ Probe: *has anyone been reluctant to show you something with other people in the room?*

A7: *Language, translation loss, and who else is standing in the room.*

- She has **Hindi, Halbi, English.** Interior blocks speak **Halbi and Gondi** — “I catch perhaps a third and I mislead myself about the other two-thirds.” The **ASHA translates;** occasionally a schoolteacher or the patient’s son. [Constraint] [Emotional/cognitive]
- **Translation is not neutral in this examination.** There is no clean local equivalent for “numb.” She asks “can you feel this,” and what returns is “she says it’s fine” — “Fine is not an answer. Did she feel it *less* than the other arm, or is she being polite to a doctor on a screen?” She now insists on the patient’s actual words, which costs time over a bad line. [Non-functional] [Functional]
- **Translators soften.** “There may be nerve damage” is rendered gentler; she can hear from the length that it isn’t what she said. Conversely, **some ASHAs answer for the patient** from their own examination — sometimes more accurate, sometimes meaning the patient never spoke at all. [Emotional/cognitive] [Constraint]
- **Who is in the room:** never just the patient. ASHA; often MO or ANM; husband or **mother-in-law** (usually the one talking if the patient is a young woman); children; and typically four or five neighbours in the doorway watching. A Sarpanch has joined a call uninvited. [Constraint] [Emotional/cognitive]

- **Reluctance is routine and clinically decisive.** Patches occur on back, buttock, abdomen, breast, inner thigh. A woman will not lift her blouse with her father-in-law, neighbours and a male MO holding the phone. “So she shows me the one on her forearm, and I assess the forearm, and I don’t learn about the other three.” **[Emotional/cognitive]** **[Constraint]**
- **Concealment from the family:** a young married woman whose ASHA warned beforehand that the in-laws must not know. The call proceeded with the mother-in-law present and “we both pretended it was about a fungal infection.” She could not use the word, could not ask about contacts, could not explain twelve months of treatment; MDT was later collected under a different pretext. “That is a bad system producing a bad compromise, and I did it anyway, because the alternative was she takes nothing.” **[Emotional/cognitive]** **[Business rule]**
- Gender composition changes disclosure: a **female ASHA holding the phone with the men stepping out for two minutes** often solves it. “Nobody thinks to ask.” **[Functional]** **[Emotional/cognitive]**
- Design consequence: the far end must be able to **control the room**; the ASHA needs a **private channel to her before the call**; and for a patch in a private site, “the right answer is often not a video call at all — it’s the ASHA, alone with her, taking one careful photograph with consent.” Less intrusive, better image. **[Functional]** **[Non-functional]**

→ **Tags:** **[Non-functional]** **[Constraint]** **[Emotional/cognitive]** **[Functional]** | *need — a socio-cultural constraint that invalidates a naive live-video design*

Q8: How does your week actually work? When could you give time to these consultations without wrecking everything else?

→ *Probe: what does an unscheduled call cost the patients already in front of you?*

A8: *The weekly roster, the two windows she can protect, and the cost of interruption.*

- **Mon:** OPD from 9 a.m., worst day (weekend backlog, 70–80 patients), ward round after. **Tue a.m.:** procedures — biopsies, minor surgery, ulcer debridement; “gloved, sterile, cannot pick up a phone.” **Wed:** OPD; **first Wednesday** = district review meeting with the Leprosy Officer, “which eats the afternoon and produces nothing.” **Thu:** two Thursdays a month at outreach camps in a block — in a jeep or a school building **with no signal**; otherwise OPD. **Fri:** OPD plus the **MDT clinic**. **Sat:** OPD till 1 p.m. **Sun:** off, reachable for genuine emergencies. **[Constraint]**
- **Proposed windows: Tuesday and Friday, 15:30–17:00.** OPD has thinned, procedures done, she is at her desk “doing paperwork I resent.” **[Functional]**
- **Slot design in her own words:** 15 minutes per case = 6 slots, but **schedule 5 and leave one empty** — “if you fill every slot the first delay destroys the rest.” She will read all five packets **the previous evening on her own time** (~20 minutes) “because it’s the

difference between a useful consultation and a guessing game.” [Functional] [Non-functional]

- **Cost of interruption — attention:** “attention is the whole job.” Taking the call in front of a waiting patient “also tells her exactly how much her time is worth”; not taking it leaves her distracted. [Emotional/cognitive]
- **Cost of interruption — arithmetic:** a 20-minute unscheduled call in a 70-patient OPD does not cost 20 minutes, it costs the tail of the list. Four minutes per patient “is enough to write a prescription — not enough to notice a numb patch on someone who came in for something else. **And that is precisely how we find early cases.**” [Constraint] [Emotional/cognitive]
- **Tuesday morning is the sharp edge:** a call during a biopsy means either the patient lies there with an open wound, or she doesn’t take it. “Neither is acceptable and both have happened.” [Constraint]
- **Subtle cost:** context-switching produces quiet errors — “I forget to ask about pregnancy. I don’t check the other hand.” [Emotional/cognitive]
- **Condition 1 — the field end must honour the window:** a slot means the ASHA confirmed **that morning** that the patient will be present, at that place, at that time. If the patient is not there **within three minutes**, the system moves to the next case and that one re-queues — “the window must not collapse because one woman went to the fields.” She wants a **confirmed-count by lunchtime** so she can plan the afternoon. [Functional] [Business rule]
- **Condition 2 — a genuine urgent lane:** reactions do not wait for Friday. But urgent must mean **the packet reaches her within two hours** and she replies with a written opinion or a five-minute voice call — **not a surprise video**. “Ninety percent of urgent cases I can settle from the findings and a photograph.” [Functional] [Business rule]
- **Urgency must be checklist-driven** — tender nerve, new weakness, sudden change in lesions, eye involvement — “not from whoever’s most worried that day. If everything is urgent I’ll start ignoring the flag within a month, and then a real one will go past me.” [Business rule] [Emotional/cognitive]
- **Asynchronous availability:** most receptive to written input **early morning (~07:00)**, before OPD — relevant to when digests, packets and exception lists should land. [Non-functional]

→ Tags: [Functional] [Non-functional] [Business rule] [Constraint] | *need — a concrete solution proposal is included; the batching windows should be validated against her official duty roster*

Q9: What happens when you join and the patient isn’t ready, or doesn’t come at all?

→ Probe: what do you do with that time, and how often does it happen?

A9: No-shows and unprepared patients — frequency, and what the dead time costs.

- **Frequency (informal calls today):** someone is not properly ready roughly **1 in 3**; outright no-show **1 in 4 or 5**. [**Constraint**]
- **A third, overlooked category:** the patient is present but **not ready** — mid-cooking, sitting with the sun behind her, or with no idea who the doctor is or why. “Technically a successful call, and clinically a waste.” [**Non-functional**] [**Emotional/cognitive**]
- **What she does with the time: nothing.** “Ten or fifteen dead minutes is not enough to start anything.” She cannot begin a biopsy or take a proper history knowing she’ll be interrupted, so she checks the phone, rings the MO who doesn’t answer, and loses the thread of her paperwork. “**Fragmented time is worth less than no time.**” [**Emotional/cognitive**] [**Constraint**]
- **The real damage is behavioural:** “after the third no-show I stop protecting the slot” — she lets OPD run in, agrees to one more patient, and then **she** is the no-show when a case is ready. “An ASHA who walked four kilometres with a frightened woman sits waiting for a doctor who doesn’t come. She won’t try as hard next time. **That’s how these systems die — not with a complaint, just with everyone quietly stopping.**” [**Emotional/cognitive**] [**Business rule**]
- **Explicitly not asking for better attendance:** “Rain, funerals, wage labour, the bus. You will not fix attendance in a tribal block.” [**Constraint**]
- **What she wants instead — make the dead slot reusable:** if the 15:45 patient is absent, surface the **next reviewed store-and-forward case** so she can write an opinion immediately. “Turn fifteen wasted minutes into two disposed cases. Then a no-show costs the patient a delay, but it doesn’t cost me anything, and I never learn to resent the window.” [**Functional**]
- **Patient preparation script for the ASHA:** a doctor from the district will look at your skin; it will take ten minutes; sit near the door where the light is; wear something you can lift the sleeve of. “Two minutes of preparation, and half my bad calls become adequate ones.” [**Functional**]

→ Tags: [**Functional**] [**Non-functional**] [**Emotional/cognitive**] | *need — she reframed the problem from attendance to time-utilisation; the backfill queue is her own solution idea*

Q10: When you look at a case and simply cannot reach a conclusion remotely — what happens to that person?

→ Probe: how do you make sure the next step happens? Do you ever find out it didn’t?

A10: What happens to an inconclusive case, and whether the next step is ever tracked.

- **Three possible outcomes, only one good. [Business rule]**
- **(a) Refer to district** — a slip of paper and a three-hour bus journey. “Realistically fewer than half come.” Non-attenders are disproportionately **women and daily-wage workers** — a day’s travel is a day’s earnings plus permission from the household. **[Constraint]** **[Emotional/cognitive]**
- **(b) Hold for the next outreach camp** in that block — works, but camps recur only every few months. “If the case is a possible reaction, a few months is the whole ballgame.” **[Constraint]**
- **(c) Safety-netting** — “treat as fungal for four weeks; if unchanged and still numb, bring them back.” Sound reasoning, “completely dependent on somebody remembering in four weeks.” **[Business rule]** **[Emotional/cognitive]**
- **On follow-through: “No. I have no way to make sure it happens, and I usually never find out.”** There is no loop — nothing comes back, not *attended*, not *did not attend*, not *could not be traced*. The referral exists in the patient’s hand and nowhere else. The ASHA moves on, the MO has three hundred other things, “and I certainly am not [holding the thread], because I have no list of the people I told to come.” **[Functional]** **[Emotional/cognitive]**
- **How she does find out: by accident, always too late** — recognising patches from a photograph at a camp eight months later; someone arriving at district with a claw hand mentioning that a doctor on the phone once said something; learning about one man because his brother came in for something else. **[Emotional/cognitive]**
- On the Grade 2 case: **nobody chased him for eleven weeks** because no record existed saying *Dr. Iyer asked for this man to be seen*. “It existed as a sentence I spoke into a phone and then forgot, because I see thirty of these a month and **I am not a database.**” **[Functional]** **[Emotional/cognitive]**
- **Her single highest-value change:** “It isn’t image quality and it isn’t video. It’s my opinion and the recommended action **written down, attached to that person’s record, sent to the MO and the ASHA in language they can act on, with a date on it** — and then **something (the system, not me) asks on that date: did this happen?** And if it didn’t, it comes back to my list.” **[Functional]**
- “I would rather have a plain, ugly system that closes that loop than a beautiful one with high-definition video. Because right now I am giving opinions into a void, and I have no idea how many of them landed.” **[Emotional/cognitive]** **[Functional]**

→ **Tags: [Functional] [Business rule] [Emotional/cognitive]** | *need — the highest-value requirement she articulated; it closes the referral loop*

Q11: After you’ve given an opinion, how do you come to know what happened to that person?

→ Probe: does anyone tell you whether treatment was started? Would you want to know?

A11: Outcome feedback: what exists today (nothing per-patient) and what she would use.

- **No per-patient feedback exists.** What exists is the MDT register at each PHC and monthly NLEP returns, surfacing at the district review meeting as **aggregate figures** — new cases, MB, child cases, Grade 2. “I cannot look at that table and find the woman whose photograph I looked at in April. There is no link between an opinion I gave and a treatment that started. Those are two separate universes that happen to share a district.” [Functional] [Business rule] [Emotional/cognitive]
- **Would she want to know? Yes — but not as notifications.** “If you send me a notification every time something happens to every case I’ve touched, I’ll mute it in a fortnight and then you’ve lost me for the ones that matter.” [Non-functional] [Emotional/cognitive]
- **(1) Treatment started — Y/N with date,** visible on the case when she opens it, **pulled not pushed.** [Functional] [Non-functional]
- **(2) Exceptions only — the highest-value feedback:** “these six people you asked to be seen were not seen.” Short, actionable, “I’d take it once a week.” [Functional] [Non-functional]
- **(3) Urgent cases (suspected reaction):** within the week, was steroid treatment started or not. “Those are the ones where delay costs a hand.” [Functional] [Business rule]
- **(4) Tell her when she was wrong.** If she said “not leprosy, treat as fungal” and the person is confirmed and started on MDT three months later, she needs to see it — **“not for anyone’s audit; for my own calibration. I am making judgements from photographs and second-hand examinations, which is not something I was trained to do, and the only way I get better is by finding out which of my remote calls didn’t hold up. Right now I never find out, so I have no idea whether I’m good at this or dangerous at it.”** [Functional] [Emotional/cognitive]
- **Secondary use of the same signal:** if cases from one ASHA repeatedly turn out differently from her recorded findings, “that’s not her fault — it means she needs retraining on sensation testing, and somebody should know that. At the moment nobody does.” [Functional] [Business rule]

→ Tags: [Functional] [Non-functional] [Emotional/cognitive] | need — careful: the delivery mechanism must be pull-based plus a weekly exception digest, not per-event alerts

Q12: Here is my difficulty. Sunita works on an entry-level phone in a village with 2G or no signal, and you need images clear enough to assess. What must this system do for you? What must it never do? And how should we handle cases where the image just isn’t good enough?

A12: Harvest — must do, must never do, and handling an inadequate image.

- **Framing:** the phone and the network are fixed. “So the design has to move the burden off the pipe and onto everything around it.” **[Constraint]**
- **MUST — 1. Treat photographs as secondary evidence.** The primary evidence is what Sunita’s hands found: the sensation test (eyes closed) and nerve palpation. “Those are text. Text goes over 2G.” A good nerve chart with no image will get a decision; a beautiful photograph with no findings will not. **“Make the findings mandatory and the image optional — you’ve currently got that backwards.”** **[Functional]** **[Business rule]** **[Non-functional]**
- **MUST — 2. Do quality control while the patient is still standing there** (“the single highest-value thing you can build”). Coach at capture: go to the doorway/outside, no flash, get close, place a coin for scale, take **one wide shot for site and one tight shot of the border**; then **check focus on the handset before accepting**. “Every quality problem I’ve described is really a problem of feedback arriving three days late, after Sunita has walked to the next hamlet.” **[Functional]** **[Non-functional]**
- **MUST — 3. Send a tight crop at good quality rather than the whole back at bad quality.** “A small file with the detail where I need it beats a large file smeared evenly.” The crop also removes the face — solving privacy in the same stroke. **[Functional]** **[Non-functional]**
- **MUST — 4. Queue offline and upload on signal** — “she should never lose work because there was no tower.” Allow **voice capture in Hindi or Halbi** where typing is the barrier, **but keep structured fields for sensation and nerve results** — “I need to read those in ten seconds, not listen to two minutes.” **[Functional]** **[Non-functional]** **[Constraint]**
- **MUST — 5. Store-and-forward is the main road, not the fallback.** Case arrives as a packet → she reads it the evening before → writes an opinion. The **live call is an escalation** for cases she cannot settle on paper, held in the Tue/Fri window. “Build it that way round and the bandwidth problem mostly disappears.” **[Functional]** **[Business rule]**
- **MUST — 6. Close the loop:** written opinion into the record and back to the MO and the ASHA, with a date and an action, and something asks later whether it happened. **[Functional]**
- **NEVER** — put a case in front of her without the packet; make live video the only path; accept “sensation: normal” without the method and eyes-closed status; transmit an identifiable face where a cropped patch would do; let her opinion exist only as a voice note; let everything be flagged urgent. **[Business rule]** **[Non-functional]**
- **NEVER** — let a remote opinion be what starts a person on twelve months of MDT. “I decide who needs confirming and how fast; somebody trained confirms in person, and their hands sign for that treatment.” **This boundary must be enforced in the design, not**

left to convention — “if it drifts, someone will start treating from photographs and that is a disaster with a long fuse.” [Business rule]

- **NEVER — don’t use this to grade Sunita.** “If her image quality is poor that’s a training and equipment problem, and the moment she thinks a bad photo reflects on her, she’ll stop sending the difficult cases — which are the only ones I want.” [Emotional/cognitive] [Business rule]
- **When the image is still not good enough — (a) make “image inadequate” a first-class recordable outcome** with a reason from a short list (too dark, out of focus, too far, wrong site), returned to Sunita as **one specific instruction** — “Retake in daylight, 20 cm, show the edge of the patch” — not “please resend.” Third occurrence is a **retraining flag, not a blame flag.** [Functional] [Business rule]
- **(b) The rule she insists on: an inadequate image must never resolve as reassurance.** Default is “**unassessed — needs in-person,**” never “probably nothing, review later.” If the written findings are strong enough she will act on them and say so; otherwise the person goes on the outreach-camp list (or to district if red-flagged) and **the case stays open with her name on it until someone has physically looked at her.** “A photograph I couldn’t read must cost the system something. At the moment it costs nothing, which is exactly why those cases evaporate.” [Business rule] [Functional]
- **Closing principle:** “This thing supports the examination, it doesn’t perform it. What you can replace is the nine minutes I spend in a corridor finding out why I was called, and the eleven weeks a man waited with a hand that was still saveable.” [Emotional/cognitive] [Functional]

→ Tags: [Functional] [Non-functional] [Business rule] [Constraint]

[Emotional/cognitive] | *need + solution — needs stated with a solution design attached; separate the two before writing requirements*

Q13: Playback: consultations are worth your time only if findings and images reach you beforehand; they need set windows rather than interrupting OPD; when the picture is poor you’d rather have a slower clearer route than an inconclusive live call because a wrong reassurance is worse than a delay; and your opinion must reach the MO and ASHA in writing since you can’t confirm treatment from a screen. Correct? And which would you protect first?

A13: *Validation of the analyst’s playback, two corrections, and the stated first priority.*

- **Correction 1 — “wrong reassurance is worse than delay” holds for diagnosis, not for reactions.** For “is this leprosy or fungus,” a fortnight costs nothing. For **a tender nerve with new weakness, delay is the harm** — steroids in days or the hand is gone. “**Don’t build one speed.** Slow and certain for diagnosis, fast and decisive for reactions — and the **checklist** decides which lane a case is in, not the ASHA’s anxiety and not mine.” [Business rule] [Functional]

- **Correction 2 — the written opinion is not only about treatment authorisation.** “The bigger reason is that **an unrecorded opinion doesn’t produce an action. Nobody chases what nobody wrote down.** Eleven weeks.” [Functional] [Emotional/cognitive]
- Otherwise the playback was accepted: “You’ve heard me.” [Emotional/cognitive]
- **What she would protect first: the packet arriving before she joins** — findings, images and the referral question, the evening before. [Functional] [Business rule]
- **Her reasoning — and it is a sequencing argument, not a value one:** “Not because it’s the most valuable — it isn’t. **The loop closing is what actually changes outcomes for patients.** But the packet is **the gate.** Without it I’m back in a corridor at minute nine, and after six weeks of that I stop answering, and then there’s no system to close a loop on. **The packet is what makes me participate. The loop is what makes my participation matter.** You need both, and you have to win the first one to get the chance at the second.” [Business rule] [Emotional/cognitive]
- **The single field she will defend against every simplification:** the sensation test with **method recorded and whether the patient’s eyes were closed.** “It’ll look like a fussy detail on a form. **It’s the diagnosis.**” [Functional] [Business rule]

→ Tags: [Business rule] [Functional] [Constraint] | *need — confirmed priority: the pre-call packet is the Release 1 gate; the closed loop is the highest outcome value*

3. TOP 3 PAINS I HEARD

1. Blind, context-free calls that begin at zero.

- Nine minutes of a 22-minute call spent discovering why she had been called; “maybe six [minutes] were clinically useful.”
- ASHA’s findings existed — in a paper register in her house, three weeks old, unreachable by anyone on the call.
- Root cause: **no case data travels ahead of the conversation.** She is repeatedly asked to reason from a corridor, a monosyllabic patient and a name she half-recognises.
- Emotional weight: she answers anyway (“if I don’t answer, he stops calling, and then someone gets missed”), so the pain is absorbed silently and never registers as a complaint.

(Q1, Q2, Q6)

2. Unscheduled interruption and unusable slots wrecking a fixed clinical calendar.

- Sole specialist for ~1.8 m people; 70–80 patients on a Monday OPD; calls arrive mid-OPD, mid-biopsy, at 9 p.m.

- A 20-minute interruption costs the tail of the list — four minutes per patient, “enough to write a prescription, not enough to notice a numb patch on someone who came in for something else.”
- Patient not ready ~1 in 3; outright no-show ~1 in 4-5; “fragmented time is worth less than no time.”
- Second-order failure: after repeated no-shows she stops defending the slot, then **she** becomes the no-show — “that’s how these systems die — not with a complaint, just with everyone quietly stopping.”

(Q1, Q8, Q9)

3. Unassessable input feeding an open loop — so delay and error are invisible until they are permanent.

- Poor images (tube light, flash, four feet away, compression) make the patch border — the discriminator — unreadable; retakes usually never arrive and “the case just evaporates.”
- With missing data she defaults to “send them to district,” a three-hour journey fewer than half complete: “an unsafe referral dressed up as a safe one.”
- Her opinion survives only as a WhatsApp message — not in the record, not dated, not chased, not defensible. “I am not a database.”
- No outcome feedback of any kind, so she cannot tell whether her remote judgement is sound: “I have no idea whether I’m good at this or dangerous at it.”
- Human cost already realised: eleven weeks’ delay → fixed clawing in a farm labourer; district Grade 2 rate 8-10% against a target under 2%.

(Q5, Q6, Q10, Q11)

4. CONTRADICTIONS AND THINGS TO VERIFY

- **Image quality vs. device and network — the headline conflict.** She needs border, surface and colour fidelity; Sunita has an entry-level handset on 2G or no signal. Her own resolution (findings mandatory, image optional; capture-time coaching; tight high-quality crops) is plausible but **untested**. Verify: actual handset models and camera specs in use, realistic upload sizes at 2G, block-wise coverage, and whether a coached crop is genuinely diagnostic for a dermatologist. Run a bench test with real ASHA-captured images before committing.
- **“Findings beat images” assumes a competency that may not exist.** The whole design rests on the ASHA’s sensation test and nerve palpation — yet in her worst case an **MO** reported a nerve as “normal, maybe slightly thick” having **never been taught to palpate**

a nerve. Verify with the DLO/training cell: what nerve-palpation and sensation-testing training ASHAs have actually received, and whether it is refreshed or supervised. A structured form does not create a skill. **This is the largest hidden dependency in the project.**

- **“Sensation: normal” is only meaningful with method + eyes-closed recorded.** She calls this the single field she will defend, but it adds friction to a form filled by a low-literacy user on a small screen. Verify feasibility with ASHAs in the field; test whether a photo/short video of the test being performed is an acceptable alternative or additional evidence.
- **Two speeds, and only one checklist.** Diagnosis: delay is safer than false reassurance. Reactions: delay is the harm (steroids within days). Verify the reaction-flag checklist (tender nerve, new weakness, sudden lesion change, eye involvement) against current **NLEP guidance**, and confirm **who may initiate prednisolone at PHC level** without her physical review — this sits close to her “never authorise MDT from a screen” boundary.
- **Roster figures are from memory, in a corridor, at speed.** Tue/Fri 15:30–17:00 was proposed on the spot, yet she also has 2 outreach Thursdays a month, first-Wednesday district review, Tuesday-morning theatre and Saturday OPD. Verify the **official duty roster** with the hospital, and confirm the windows survive leave, camp season and emergencies.
- **Single point of failure.** She is the only leprosy-trained dermatologist in the district; the second sanctioned post has been vacant three years. Verify: who covers teleconsultation when she is on leave, at a camp with no signal, or transferred? A system designed around one person’s calendar inherits that person’s absence. Not addressed in this interview.
- **All quantitative claims are her estimates and must be triangulated** against NLEP district returns: 25–40 new cases/month; ~1/3 MB with nerve involvement at presentation; **8–10% Grade 2 at diagnosis**; 15–20 informal contacts/month; “a third of the district” without call-capable bandwidth; no-show ~1 in 4–5.
- **Privacy: current practice may already be non-compliant, and her preferred fix increases stored data.** Clinical images including faces and intimate sites circulate in a **41-member WhatsApp group with unidentified members**. She wants the system to hold images centrally instead. Verify the district/state position on clinical image storage, consent, retention and NLEP confidentiality, plus applicable data-protection obligations — and how consent is actually obtained in dialect, aloud, with the ASHA present (she flagged that a checkbox ticked on the patient’s behalf is not consent).
- **Governance risk her boundary depends on:** she will not authorise MDT remotely, but a written specialist opinion in a record may be read by an MO or ASHA as authorisation. Verify how the recommendation is worded and permissioned so “confirm in person before treatment” cannot drift.

- **The exception report requires data she does not own.** “These six people were not seen” implies linkage to PHC MDT registers / NLEP returns. Verify feasibility, data ownership, who updates attendance, and whether the DLO will sanction it — otherwise the loop she values most cannot be closed by this system alone.
 - **She wants outcome feedback but will mute notifications.** Pull-based case status + a weekly exception digest is her stated preference; verify the digest cadence and channel actually reach her (she is most reachable at ~07:00, and works from a phone).
 - **Sensitive-site handling has no route in the current design.** She needs a female-only / room-control flag and a private ASHA→specialist channel set **before** scheduling. Verify with ASHAs and the MO whether this is workable, and who is permitted to see a case flagged sensitive.
 - **Open design decisions this interview did not settle — both are gating for Release 1.** (a) **What screens a case before it reaches the specialist** — unaided ASHA judgement, MO review, or rules-based triage on the captured findings? This determines her caseload and the meaning of the urgent flag. (b) **Whether a case can be sent and answered without a live call attached** (store-and-forward as first-class, not fallback) — her stated preference, but it changes the core workflow model. Also unresolved: offline queue-and-resend behaviour, voice capture of findings in Hindi or Halbi, and the visibility model (raising ASHA + specialist only, versus block-wide).
 - **Single-source risk.** Everything above is one stakeholder’s account, given under time pressure. Triangulate with the ASHA (Sunita), a PHC Medical Officer and the District Leprosy Officer before treating any of it as settled — in particular the ASHA-side burden of the “mandatory findings” model, which was designed here entirely from the specialist’s side of the call.
-

5. RECORDED CONFLICT: IMAGE QUALITY REQUIRED FOR REMOTE ASSESSMENT VERSUS FIELD DEVICE AND BANDWIDTH REALITY

Position 1 — what she needs to assess remotely. Colour, margin definition and surface detail are clinically decisive — the patch border is “the discriminator between leprosy, vitiligo, pityriasis alba and fungal patch” (Q2, Q3). Her stated floor is 2–3 daylight images, close, in focus, with a scale reference, cropped to exclude the face (Q6), plus a mandatory structured findings set — sensation test with method and eyes-closed status, nerve palpation thickened/tender per nerve, simple motor testing (Q3, Q6, Q12). Below that floor she does not refuse; she defaults to “send them to district” — “an unsafe referral dressed up as a safe one” (Q6).

Position 2 — what the field actually offers. Sunita works an entry-level phone in a village on 2G or no signal; by her own estimate a third of the district cannot sustain a live video call at all (Q5, Q12). Photographs arrive indoors under tube light from about four feet, or with flash washing

out the pigment difference, or blurred, and retakes rarely return because the ASHA has already moved to the next hamlet — “the case just evaporates” (Q5). In the case she returns to most, video dropped four times, the call fell back to audio, and the reporting MO’s untrained nerve palpation (“normal, maybe slightly thick”) concealed a Type 1 reaction; the patient arrived eleven weeks later with fixed, irreversible clawing (Q5).

Her counter-proposal. Reorder the dependency rather than fight the network: make the structured findings mandatory and the image secondary — “those are text, text goes over 2G... make the findings mandatory and the image optional, you’ve currently got that backwards” (Q12). Build quality control into the point of capture, while the patient is still standing there, rather than a retake request three days later (Q5, Q12). Send a tight, high-quality crop of the border rather than a large low-quality frame of the whole area — which also removes the face, solving privacy in the same stroke (Q6, Q12). Queue captures offline and upload once signal returns, with voice capture in Hindi or Halbi for free text but structured fields kept mandatory for sensation and nerve results (Q12). Treat store-and-forward as the main road and reserve live video as an escalation held to the Tuesday/Friday windows for cases that genuinely cannot be settled on paper (Q5, Q8, Q12).

Consequences she described when images were unusable: the case “evaporates” because the retake never comes back; the referral defaults to district when data is missing, which “fewer than half” complete; and, in her worst remembered case, the combination of a failed video call and an untrained examiner’s mis-read nerve produced eleven weeks of undiagnosed reaction and a permanent claw hand (Q5).

Options to take forward:

- (a) Build capture-time coaching into the ASHA’s device — guided wide-site shot plus tight border shot, coin/scale reference, a focus check before the image is accepted — implementing her “quality control while the patient is still standing there.”
- (b) Make the structured findings set (sensation test with method and eyes-closed status; nerve palpation thickened/tender per nerve; simple motor) mandatory independent of image quality, so a case can be dispositioned on text alone over 2G, per her “make the findings mandatory and the image optional.”
- (c) Give “image inadequate” a first-class outcome with a short reason code and one specific retake instruction, defaulting every unresolved case to “unassessed — needs in-person” rather than silent reassurance, and keep the case open under her name until it has been physically reviewed.
- (d) Make store-and-forward the default workflow with offline queue-and-upload, reserving live video for the Tuesday/Friday escalation windows for cases that cannot be settled on the packet.

Recommendation: adopt (b) as the non-negotiable Release 1 baseline — she named it the one field she will defend against every simplification — paired with (a) and (d) as the surrounding workflow, since she presents these as one design rather than four independent choices; treat

(c) as the safety rule governing every case the other three fail to resolve. Bench-test the coached crop against her actual diagnostic threshold with real ASHA-captured images before committing, per the untested-resolution flag raised in Section 4.

APPENDIX A — CONSOLIDATED REQUIREMENTS REGISTER (FROM THIS INTERVIEW ONLY)

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-30	Deliver the full pre-call packet (who, story, examination, images, referral question) the evening before a scheduled consultation, readable within about four minutes.	Functional	Q6, Q8, Q13	Must have
FR-31	Support a minimum-floor packet (six data points plus one photograph plus the referral question) sufficient to return one of three dispositions — see urgently / wait for outreach camp / not leprosy, treat locally.	Functional	Q6	Must have
FR-32	Make the sensation-test field mandatory with method used and whether the patient's eyes were closed recorded; reject "sensation: normal" without the method.	Functional	Q6, Q12, Q13	Must have
FR-33	Capture nerve-palpation findings per named nerve, both sides — thickened Y/N and tender Y/N — as a mandatory structured field.	Functional	Q3, Q6, Q12	Must have
FR-34	Capture the referral question ("why am I being called, what do you want from me") at the top of every packet.	Functional	Q6	Must have
FR-35	Attach the reporting ASHA's identity to every case for longitudinal reliability tracking.	Functional	Q6, Q11	Should have
FR-36	Capture 2–3 daylight images per case, close, in focus, with a scale reference, cropped to exclude the patient's face.	Functional	Q6, Q12	Must have
FR-37	Provide capture-time quality coaching on the ASHA's device (wide site shot plus tight border shot, focus check)	Functional	Q12	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	before an image is accepted for upload.			
FR-38	Provide a sensitive-case flag and a private ASHA-to-specialist channel, settable before scheduling, for private-site or concealment-risk cases.	Functional	Q7	Should have
FR-39	Provide fixed weekly teleconsultation windows (candidate: Tuesday and Friday 15:30–17:00) with capacity capped below the theoretical maximum (5 of 6 slots booked, one held open).	Functional	Q8	Must have
FR-40	Require same-morning attendance confirmation from the ASHA; if the patient is not present within three minutes of the slot, requeue automatically to the next case.	Functional	Q8	Must have
FR-41	Surface a confirmed-attendance count to the specialist by lunchtime on a consultation day.	Functional	Q8	Should have
FR-42	Provide a checklist-driven urgent lane (tender nerve, new weakness, sudden lesion change, eye involvement) in which the packet reaches the specialist within two hours and is dispositioned by written opinion or a short voice call, not an unscheduled video call.	Functional	Q8, Q13	Must have
FR-43	When a scheduled slot is unfilled, surface the next reviewed store-and-forward case so the specialist can dispose of it immediately (queue backfill).	Functional	Q9	Should have
FR-44	Send a patient-preparation prompt to the ASHA ahead of a scheduled appointment (what will happen, how long, where to sit, what to wear).	Functional	Q9	Should have
FR-45	Record the specialist's opinion and recommended action, dated, attached to the patient's record, and delivered to the MO and ASHA in actionable language.	Functional	Q10, Q13	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-46	Fire an automatic follow-up task on the action's due date asking whether it happened; if unconfirmed, requeue it to the specialist's list.	Functional	Q10	Must have
FR-47	Show treatment-started status (Y/N, date) on the case, retrievable on demand (pull) rather than pushed.	Functional	Q11	Must have
FR-48	Provide a weekly digest of exceptions — recommended actions that did not happen.	Functional	Q11	Must have
FR-49	Provide calibration feedback — notify the specialist when a remote “not leprosy” opinion is later reversed by a confirmed diagnosis, and surface repeated discrepancies from one ASHA as a retraining signal rather than a blame signal.	Functional	Q11, Q12	Should have
NFR-17	Packet format and length must allow the specialist to read and understand a case within about four minutes.	Non-functional	Q6	Must have
NFR-18	Packets must arrive the evening before a scheduled slot, not minutes before it.	Non-functional	Q6, Q8	Must have
NFR-19	Field data capture must be offline-first — work done with no tower must never be lost, and must queue and upload once signal returns.	Non-functional	Q12	Must have
NFR-20	Images must be handled for low bandwidth — a small, high-quality crop is preferred over a large, low-quality full-frame image, sized to upload over 2G.	Non-functional	Q5, Q12	Must have
NFR-21	Outcome and status notifications must be pull-based plus a bounded weekly digest, not per-event push alerts.	Non-functional	Q11	Must have
NFR-22	Support voice capture of free text in Hindi or Halbi where typing is a barrier, without weakening the mandatory structured fields for sensation and nerve findings.	Non-functional	Q12	Should have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
NFR-23	The system must function reliably as a store-and-forward channel in areas that cannot sustain live video (about a third of the district).	Non-functional	Q5, Q12	Must have
NFR-24	Structured clinical fields must be usable by a low-literacy field worker on a small screen without adding excessive friction.	Non-functional	Q6, Q12	Should have
NFR-25	Asynchronous inputs (packets, digests) should be timed for the specialist's actual availability, around 07:00 before OPD.	Non-functional	Q8, Q11	Should have
NFR-26	The specialist's opinion must be durably retrievable and defensible later, not exist only as an ephemeral chat message.	Non-functional	Q4, Q10	Must have
BR-10	A remote teleconsultation opinion must never itself authorise starting MDT; only an in-person, trained confirmation may initiate treatment.	Business rule	Q3, Q12, Q13	Constraint on design
BR-11	Urgency must be determined by a fixed checklist (tender nerve, new weakness, sudden lesion change, eye involvement), never by whoever is most anxious that day.	Business rule	Q8, Q13	Constraint on design
BR-12	An inadequate image must never resolve as reassurance; the default outcome is "unassessed — needs in-person," and the case stays open under the specialist's name until physically reviewed.	Business rule	Q12	Constraint on design
BR-13	Image or data quality problems traced to an ASHA must be routed as a training/equipment signal, never used to grade or blame her.	Business rule	Q11, Q12	Constraint on design
BR-14	The specialist reviews packets for her own slots the evening before; slot capacity is deliberately held below the maximum to protect against cascading	Business rule	Q8	Constraint on design

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	delay.			
BR-15	Sensitive-site cases require a female-only / room-controlled handling path rather than a standard live video call.	Business rule	Q7	Constraint on design
CON-12	She is the sole leprosy-trained dermatologist for about 11 blocks and 1.8 million people; a second sanctioned post has been vacant for three years.	Constraint	Q1	Absolute
CON-13	Live video is unsustainable across roughly a third of the district, including outreach camp locations.	Constraint	Q5, Q8, Q12	Absolute
CON-14	Field devices are entry-level phones on 2G or no signal.	Constraint	Q12	Absolute
CON-15	Her clinical calendar is fixed (OPD, procedures, ward rounds, MDT clinic, district review, outreach camps), leaving only specific windows available for consultation.	Constraint	Q8	Absolute
CON-16	A three-hour bus journey each way separates most patients from the district hospital; fewer than half of those referred complete it.	Constraint	Q1, Q10	Absolute
CON-17	Certain findings — nerve palpation, sensation testing, slit-skin smear, skin texture — cannot be transmitted through any camera or call.	Constraint	Q3	Absolute
EC-05	Stigma from uncontrolled image circulation (marriage prospects, family standing, employment) must be designed against as a primary risk, not treated as a paperwork concern.	Emotional/cognitive	Q4, Q7	Must design for
EC-06	The system, not the specialist's memory, must carry follow-up so she is never "the database" for every opinion she gives.	Emotional/cognitive	Q10	Must design for
EC-07	She needs calibration feedback on her remote judgement without being overwhelmed by per-event notifications, which she will mute.	Emotional/cognitive	Q11	Must design for

ID	Requirement / observation (stated as a need)	Type	Source	Priority
EC-08	Patient and family reluctance to disclose or display sensitive-site lesions in front of others in the room must be designed for, not treated as an edge case.	Emotional/cognitive	Q7	Must design for

Design rules implied, in the stakeholder's own framing

- **Findings before pictures.** “Make the findings mandatory and the image optional — you’ve currently got that backwards.” The structured sensation and nerve data must never be gated behind a successful image upload.
- **Coach at the point of capture, not after.** “Every quality problem I’ve described is really a problem of feedback arriving three days late, after Sunita has walked to the next hamlet.” Quality control belongs on the ASHA’s handset while the patient is still standing there.
- **Store-and-forward is the main road, not the fallback.** “Build it that way round and the bandwidth problem mostly disappears.” Live video is an escalation reserved for the Tuesday/Friday windows, not the default channel.
- **A missing field is a silent unsafe referral, not a safe one.** “Every missing field pushes me toward an unsafe referral dressed up as a safe one.” Incomplete packets must be visible as incomplete, not quietly accepted.
- **An inadequate image must cost the system something.** “A photograph I couldn’t read must cost the system something. At the moment it costs nothing, which is exactly why those cases evaporate.” Default to “unassessed — needs in-person,” never to reassurance.
- **One speed does not fit this disease.** “Don’t build one speed. Slow and certain for diagnosis, fast and decisive for reactions — and the checklist decides which lane a case is in, not the ASHA’s anxiety and not mine.”
- **Protect the slot by underfilling it.** “If you fill every slot the first delay destroys the rest.” Schedule below capacity so one delay cannot cascade through the whole window.
- **An unrecorded opinion produces no action.** “Nobody chases what nobody wrote down.” Every opinion needs a date, an addressee and a system-driven check that the action happened.
- **Notify sparingly or lose the specialist entirely.** “If you send me a notification every time something happens to every case I’ve touched, I’ll mute it in a fortnight and then you’ve

lost me for the ones that matter.” Pull-based status plus a bounded weekly digest, not per-event alerts.

- **Never let quality feedback read as blame.** “The moment she thinks a bad photo reflects on her, she’ll stop sending the difficult cases — which are the only ones I want.” Retraining signals must be separated from anything that looks like grading.
- **The system supports the examination; it does not perform it.** “This thing supports the examination, it doesn’t perform it.” The governance boundary — no remote authorisation of MDT — must be enforced in the design itself, not left to convention.

STAKEHOLDER INTERVIEW LOG

SparshCare — Requirements elicitation · Interview 4 of 6

Label	Content
Analyst	Dr. Gunjan Singhal
Date of interview	17 August 2026
Stakeholder role	District Leprosy Officer (DLO), National Leprosy Eradication Programme (NLEP). Owns detection targets, monitoring and reporting for the district; reports to the State Leprosy Officer.
Stakeholder (persona)	Mr. S. Verma — 4 years as District Leprosy Officer in this district (12 years total in the state health service, including 3 years at the state leprosy cell in monitoring and evaluation)
Location / facility	District Leprosy Office, Bastar, Chhattisgarh
Project	SparshCare — leprosy screening, teleconsultation & referral system
AI tool used	Claude (Cowork) — model claude-opus-5, role-play persona “S. Verma, District Leprosy Officer”
Interview mode	Semi-structured, single session, 13 questions with follow-up probes
Duration (in-persona)	~50 min (not stated by the stakeholder; estimated from the depth of 13 questions with extensive probes)
Chat link / screenshot ref	SPARSHCARE / INT-04 / VERMA-DLO — [paste chat link or screenshot filename]

1. PURPOSE AND INTERVIEW APPROACH

Objective: this interview set out to learn, from the programme owner accountable for district-level NLEP performance, whether the indicators he reports upward can be trusted at the point they are produced; how quickly a genuine geographic cluster of new cases becomes visible to someone who can act on it; why referred suspects are or are not converting into confirmed, treated cases; and how much of the field-level data burden he is asking for is actually load-bearing against a real decision, versus inherited from forms nobody has ever questioned. It also probed, directly, the trade-off between his appetite for rich district-level reporting and the burden that appetite places on a field worker with forty houses a day and no signal — and, by the end, a second trade-off he had not anticipated between his own reporting ambitions and the exposure risk to the person named in the record.

Question flow used (deliberate sequence — rapport → current process → evidence → pain → constraints → conflict → solution ideas last):

- **Q1-Q2 (rapport / baseline)** — his accountabilities, reporting chain and deadlines, then the district's case numbers and disability-at-detection trend.
- **Q3-Q5 (current process and evidence)** — how one screening form actually travels from Kondagaon to his desk; which systems he reports into and what they give back; how he detects a wrong number today.
- **Q6-Q8 (pain)** — what happens to a poorly-performing block and how that shapes what travels upward; the “found nothing vs. sent nothing” zero-reporting ambiguity; how a real cluster (Ghatpara) was actually found, and how late.
- **Q9-Q10 (constraints and consequences)** — which decisions genuinely need real-time data versus which are fine monthly or slower; why a referred suspect cannot currently be tracked to treatment.
- **Q11 (constraints, elicited)** — who sees his reports at what resolution, and what a cluster response discloses to a village.
- **Q12 (conflict)** — the field-count and latency conflict put to him directly, resolved into a sorting principle, a set of must-do/must-never-do rules, and a stated build priority.
- **Q13 (solution ideas last)** — playback for confirmation, two corrections, one reinstated omission, and a committed minimum eight-field data set.

Tag key: Each captured line is marked with one or more of five categories — **[Functional]** (something the system must do) · **[Non-functional]** (a quality it must have) · **[Business rule]** (an existing organisational rule) · **[Constraint]** (a fixed environmental, economic or regulatory limit) · **[Emotional/cognitive]** (fear, trust, stigma, mental load). Each answer additionally carries a roll-up tag and is marked *need* (a problem to be solved) or *solution* (a proposed answer, to be traced back to the underlying need). In Appendix A these categories are carried forward as ID prefixes: FR = functional, NFR = non-functional, BR = business rule, CON = constraint, EC = emotional/cognitive.

2. QUESTION-BY-QUESTION LOG

Questions are recorded in the order asked. Answers are condensed evidence notes; quoted text is the stakeholder's own wording. Figures are as stated by the stakeholder and are listed for verification in Section 4.

Q1: Tell me about yourself — how long have you been with the programme, and what exactly are you held accountable for?

→ Probe: Who do you report to, and on what deadline?

A1:

- Background: MPH, 12 years in the state health service; 4 years as DLO in this district; previously 3 years at the state leprosy cell in the monitoring and evaluation section, receiving and compiling district reports. Says this gave him both sides of the paper — “I know exactly how the numbers get massaged on the way up.” **[Emotional/cognitive]**
- Not accountable for clinical care (that is the Medical Officers’). Accountable for programme performance on four things: **[Business rule]**
 - Annual new case detection rate — low numbers are read by the state as hidden cases and weak surveillance, not as success. **[Business rule]** **[Emotional/cognitive]**
 - Proportion of new cases with Grade-2 disability at detection — the indicator he “loses sleep over”; a visible deformity at first diagnosis means years undiagnosed. Child case proportion read similarly as active transmission. **[Business rule]** **[Emotional/cognitive]**
 - Treatment and referral completion — “a case detected but not put on MDT is a case I have converted into a statistic.” **[Business rule]** **[Emotional/cognitive]**
 - Campaign and active-detection coverage — contact tracing and prophylaxis, population covered against population targeted. **[Business rule]**
- Reporting chain: sub-centre/PHC → block (first week) → DLO → State Leprosy Officer → Central Leprosy Division → national report and WHO global returns. **[Business rule]**
- Deadline: consolidated district monthly report to the state within the first fortnight of the following month. Quarterly performance review; annual district return. **[Business rule]** **[Constraint]**
- Reality against that deadline: “I am, in practice, always reporting on a month that ended six weeks ago.” Some block returns arrive after his submission date, so he submits with estimates and revises later. **[Constraint]** **[Emotional/cognitive]**
- Two conditions he set at the outset: field data must become visible to the district at the moment of capture, not when someone downstream chooses to forward it; and every record must carry an accountable author in place of the register signature he relies on today. **[Functional]** **[Business rule]**

→ Tags: **[Functional]** **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | *need — states accountabilities and deadlines, not a solution*

Q2: How many new leprosy cases does the district record in a year? And what proportion already have a disability at the time they’re detected?

→ Probe: *Which way has that second number been moving? What would it take to move it?*

A2:

- District population ~2.4 million; self-described as moderately endemic — “not Bihar or Chhattisgarh, but certainly not leprosy-free.” **[Constraint]**
- New cases: 412 last financial year; 388 the year before; 341 before that (partial campaign year, not comparable). Detection rate ~17 per 100,000. **[Business rule]**
- Multibacillary proportion ~55% (read as ongoing transmission). Child cases 6.3% — 26 children. **[Business rule]**
- Grade-2 disability at detection: 6.8% last year (28 people); 5.9% the year before; 4.2% four years ago. State average nearer 4%. **[Business rule]**
- Trend is moving the wrong way. Rose sharply when active case detection stopped during the pandemic period and has plateaued at a bad level rather than recovering. **[Business rule] [Emotional/cognitive]**
- IMPORTANT CAVEAT HE VOLUNTEERED — the indicator is a ratio and behaves bidirectionally. Weak surveillance shrinks the denominator to self-presenting (already visible) cases and the percentage rises; a good active-detection campaign finds early paucibacillary cases and the percentage falls without any change in underlying disability. He accepts this is by design but says he cannot interpret the figure without knowing HOW each case was found — and “my paper forms do not reliably tell me the mode of detection.” Named as a gap he wants closed. **[Functional] [Business rule]**
- What would move it: (i) earlier recognition in the field — act on a pale or numb patch, not on a claw hand; (ii) shorten the interval between suspicion and confirmation — he suspects a substantial share of Grade-2 cases were seen months earlier and fell through the gap between ASHA suspicion and MO examination, with nobody accountable for that person in the interval; (iii) go where the cases are — send a survey team this month, not after the annual review. **[Functional] [Emotional/cognitive]**
- Currently discovers clusters retrospectively: “I sit with the annual data in April, I plot it, and I say ah, there was a cluster in that panchayat last August.” **[Constraint] [Emotional/cognitive]**
- Requires mode of detection recorded for every case, and a suspected case held as an OPEN ITEM from the ASHA’s flag until an MO either confirms or rules out — “that interval is where my disability indicator is being decided.” **[Functional] [Business rule]**

→ **Tags:** **[Functional] [Business rule] [Constraint] [Emotional/cognitive]** | *need — with a quantified baseline; treat the figures as claims to verify (see Section 4)*

Q3: How does data from a block like Kondagaon reach you today? Walk me through the journey of one screening form.

→ *Probe: How long does that take end to end? And how much of your own time goes into compiling it?*

A3:

- Kondagaon: ~80 km from HQ, hill terrain, 4 PHCs, 1 CHC, ~31 sub-centres, ~300 ASHAs. His furthest block. **[Constraint]**
- The chain, as he described it: **[Business rule]**
 -
 - 1. ASHA records a suspect on a photocopied survey sheet during a household round. **[Business rule] [Constraint]**
 -
 - 2. ANM transcribes suspects into the sub-centre register at the weekly/fortnightly meeting — not the same day; if the meeting is missed it waits. TRANSCRIPTION 1, “where names, ages and village spellings begin to drift.” **[Non-functional] [Constraint]**
 -
 - 3. PHC pharmacist or non-medical assistant compiles sub-centre figures into the PHC monthly format at the monthly meeting — TRANSCRIPTION 2. Critical: at this point the person becomes a tick in a column. If he never attended the PHC, “he appears nowhere at all. He simply evaporates.” **[Constraint] [Emotional/cognitive]**
 -
 - 4. Block Programme Manager adds up across 4 PHCs and the CHC — TRANSCRIPTION 3 — while chasing whichever PHC has not sent its format. **[Constraint]**
 -
 - 5. District data entry operator enters block figures into the district format and the state portal — TRANSCRIPTION 4. **[Constraint]**
- Four transcriptions between a health worker examining an arm and a figure he certifies to the state — “four opportunities for error, and several opportunities for improvement of figures by people whose performance is judged on those figures.” **[Functional] [Non-functional] [Emotional/cognitive]**
- Physical transport: post, hand-carry by whoever happens to be travelling to district HQ, or a WhatsApp photograph of the form which his staff then retype. **[Constraint]**
- End-to-end lag: 5–7 weeks for Kondagaon (up to 9 weeks in monsoon when roads close); 3–4 weeks for nearer blocks. Against a state deadline in the second week — “my deadline arrives before my data does.” So he submits on trend plus partials and files a revision. “Every DLO in this state does the same and every one of us will deny it in a meeting.” **[Constraint] [Emotional/cognitive]**
- His own time: 8–10 working days per month on the reporting cycle, excluding his staff — telephoning block officers, reconciling figures that do not add up (block says 42 referred,

PHC formats total 38), answering state queries he cannot answer because he holds only the aggregate somebody sent him. Roughly one third of his working month on “arithmetic and chasing.” **[Constraint] [Non-functional]**

- Closing admission: “I do not fully trust the number I have signed... it is the figure the system produced.” **[Emotional/cognitive]**

→ Tags: **[Functional] [Non-functional] [Constraint] [Emotional/cognitive]** | *need — the as-is process, quantified*

Q4: What systems do you report into today — the state portal, NLEP formats? And what do those give you back?

→ Probe: *What do you still end up doing by hand despite them?*

A4:

- Systems he reports into: **[Business rule]**
 - Nikusth — the NLEP case-based reporting portal. Case-based in principle (per-case diagnosis, classification, disability grading, treatment start). In practice entered at block/district level FROM PAPER in month-end batches: “the portal is case-based but the data reaching it is not fresh — it is a digitised paper register. Garbage in, elegantly formatted garbage out.” **[Constraint] [Non-functional] [Emotional/cognitive]**
 - HMIS — facility-based monthly aggregates; leprosy is a handful of rows inside a very large form. Aggregate only, “tells me nothing about a person.” **[Constraint] [Emotional/cognitive]**
 - IHIP / IDSP for surveillance — sits awkwardly, leprosy is not a weekly-alert outbreak disease. **[Constraint]**
 - State review formats — Excel workbooks circulated before each quarterly review, “always slightly different from last quarter’s, always due in three days.” **[Constraint] [Emotional/cognitive]**
 - The NLEP monthly (MPR) and annual formats themselves. **[Business rule]**
- What they return: validation of well-formed entries, proof of compliance, retrospective aggregates. “Remarkably little that I can act on.” No alert for an accumulating cluster; no exception list of referrals with no recorded outcome. “They are repositories, not instruments... built to satisfy the level above, not to serve the level entering the data” — so data entry is experienced as a tax rather than a tool. **[Functional] [Emotional/cognitive]**
- What he still does by hand: **[Constraint]**

- A parallel district Excel workbook — block-wise and month-wise, “my real management system,” because no portal gives him the cross-tabulation in the time he has. Notes every DLO keeps one and every one is slightly inconsistent with what was officially submitted. **[Constraint] [Emotional/cognitive]**
- Cluster detection by coloured pins on a wall map, one colour per quarter. “A wall, and pins. And it works better than any software I have been given, because I can see it.” Weakness: ~3 months stale and invisible to anyone outside his room. **[Functional] [Constraint] [Emotional/cognitive]**
- Referral closure by telephone — no query exists. DLO → block officer → PHC → MO’s register, matching a name an ASHA wrote six weeks earlier possibly spelled differently. Answer returns in 2–3 days, partial, and disbelieved. “Most months, I do not know.” **[Functional] [Constraint] [Emotional/cognitive]**
- Monthly reconciliation across block reports, PHC formats and portal contents. **[Constraint]**
- Every ad hoc state query — e.g. “child cases in tribal habitations this quarter” — is an evening, a calculator and a stack of paper. **[Constraint] [Emotional/cognitive]**
- Explicit warning: the new system must FEED the existing formats (Nikusth, HMIS, state Excel) rather than become a fifth thing to maintain — “if my staff must enter data twice, they will enter it badly once and not at all the second time. That is not cynicism, it is four years of observation.” **[Non-functional] [Business rule]**
- Root cause he names himself: “We do not follow people. We count events.” Wants a persistent person identity from first suspicion through confirmation, treatment and completion — “if your system follows the person, most of my other complaints solve themselves.” **[Functional] [Emotional/cognitive]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**

[Emotional/cognitive] | *need + solution — interoperability and longitudinal identity are the need; naming specific portals is solution-flavoured and needs feasibility verification*

Q5: You said reports arrive inaccurate or inflated. How do you know when a number is wrong?

→ Probe: What do you do about it? And what has it cost you when you acted on a number that turned out to be wrong?

A5:

- Detection is by judgement, not by system — “by smell, mostly. Which is not a defensible auditing method, but it is what I have.” His heuristics: **[Emotional/cognitive]**

- Internal arithmetic — sub-parts not adding to totals (mostly caught by his DEO). **[Non-functional]**
 - Implausible work rates — e.g. a block claiming ~300 ASHAs screened 480,000 people in a 14-day fortnight (>100 people per ASHA per day examining skin, in hill terrain). Usually a household population count copied as though everyone was examined. **[Non-functional] [Emotional/cognitive]**
 - Round numbers — exactly 95% coverage, exactly 1,200 households. “Real field data is untidy. Tidy numbers have been through a human being with an opinion about what the number should be.” **[Non-functional] [Emotional/cognitive]**
 - Ratios that break — 38 new cases with zero Grade-2 and zero child cases is not naturally occurring in his district; means disability is not being graded, or is being graded zero because zero looks good. Also MB/PB ratios out of line with neighbouring blocks. **[Functional] [Non-functional]**
 - Suspiciously smooth trends — 31, 32, 30 month after month means estimating from last month, not measuring. “Real detection is lumpy.” **[Non-functional]**
 - Timing — “numbers improve in the month before a state review. Every time.” **[Non-functional] [Emotional/cognitive]**
 - Sustained zero-reporting — four consecutive zero months from a sub-centre is absence of screening, not absence of disease. **[Functional] [Non-functional]**
- Every heuristic is applied by eye, weeks late, and only if he happens to look. He estimates he catches “perhaps half”; plausible inflation “sails straight through me and up to the state under my signature.” **[Non-functional] [Emotional/cognitive]**
 - Escalation when he suspects a figure: telephone the block officer (a revised figure with no explanation usually confirms his suspicion) → raise it publicly at the monthly block review (effective but costly in goodwill) → supervisory verification visit re-checking a sub-centre register against households. **[Business rule]**
 - Verification capacity: ~2 days per month against 30+ sub-centres across 5 blocks. “It is a token. Everybody knows it is a token.” **[Constraint] [Emotional/cognitive]**
 - COST — the decision he most regrets. Two years ago he had funds and staff for ONE special active case detection drive. On the figures he sent it to the block with a visibly rising trend and poor detection, over an adjacent block reporting steady modest numbers and “good coverage” (the exactly-95% figure). Eleven months later the annual plotting showed the cluster was in the second block: 14 cases in three adjacent panchayats, 4 with Grade-2 disability, one a child of nine. “I sent my one available team to the wrong block because I trusted a number that had been smoothed, and I had no way to know it had been smoothed.” **[Emotional/cognitive]**

- Ongoing costs: credibility with the state cell whenever he revises; and defensive decision-making — he now discounts every block report by “an instinct I cannot document,” i.e. programme decisions made on a hunch about which officer exaggerates. **[Emotional/cognitive]**
- His argument for many data fields, stated here: “a short form is unfalsifiable.” Corroborating fields make fabrication hard — rich data is “my only defence against being lied to.” He explicitly invited the analyst to push back on field-worker burden, “with evidence rather than agreeing with me politely.” **[Business rule]**
[Emotional/cognitive]
- Wants automated plausibility and ratio checks at entry and at aggregation, surfaced as exceptions; and wants the SHAPE of reporting behaviour — which units have gone quiet, record-created time versus submitted time, and records entered in a single batch late on the last day of the month. **[Functional]** **[Non-functional]**

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Constraint]**

[Emotional/cognitive] | *need — the specific checks he lists are candidate solutions derived from a real need, kept as examples rather than the requirement itself*

Q6: What happens to a block whose numbers look poor?

→ Probe: *So how does that affect what gets reported upward?*

A6:

- Escalating consequences, in his own order: block-wise performance displayed side by side on screen at the monthly review before all block officers (deliberate comparison) → written request for explanation → supervisory visit, “experienced as an inspection regardless of how I frame it” → entry in the performance record of the block officer, the MO, and colouring how ANMs and ASHAs are assessed → effect on fund flow and utilisation justification → whether that block gets the next campaign, the next extra staff, “the next thing anyone wants.” **[Business rule]**
- At the bottom of the chain, ASHA incentives are activity-linked, so the pressure lands on “the person with the least power and the smallest income in the entire structure.” **[Business rule]** **[Emotional/cognitive]**
- Effect on upward reporting, stated plainly: “the number that protects you travels upward, and the number that is true stays in the village.” A block officer reporting an honest bad figure with reasons (vacant ANM posts, roads closed) receives a letter and a visit; one reporting a smooth, slightly-improved figure receives nothing at all — “and nothing at all is the best available outcome.” Mostly not fraud: “estimation in the direction of safety.” **[Business rule]** **[Constraint]** **[Emotional/cognitive]**
- MOST DAMAGING SINGLE CONFUSION in his data: “my system cannot distinguish we looked and found nothing from we did not look.” Both arrive as a zero. “Those are

opposite facts and they demand opposite responses from me” — reduce surveillance there, versus go there immediately. He treats both as reassurance: “that is exactly backwards.” **[Functional] [Emotional/cognitive]**

- Admits he is inside the same machinery, not outside it: the state does to him what he does to his blocks. When his Grade-2 figure moved 5.9% → 6.8% he explained it to the state as a post-pandemic effect rather than as possibly-honest grading of previously under-graded cases — “which is true, but it was not the whole truth, and I chose it because it was the version that survived the meeting.” **[Emotional/cognitive]**
- Position on accountability: does not want it removed (“without comparison and consequence, nothing moves at all”) but says it attaches to the wrong object — “it punishes the appearance of poor performance, which means it punishes honesty, which means it destroys the data I need to manage with. I have been running the instrument that blinds me.” **[Business rule] [Emotional/cognitive]**
- FIRST MAJOR CONCESSION on the field-count conflict: “more fields, by themselves, will not fix that. Twenty fields filled defensively are worse than five fields filled honestly, because twenty give me false confidence.” Would trade a considerable number of fields for honest process capture (that a visit occurred, when, where, what was actually examined). Stresses he is not conceding the whole argument. **[Functional] [Business rule] [Emotional/cognitive]**
- Warning he raised himself: near-real-time visibility means he — and the state — will see failure faster than ever, and the field will immediately understand the system as surveillance pointed at them. The design must therefore give the block officer and the ASHA a reason to WANT the data accurate; if oversight is the only rationale, “they will defeat it, and they will be within their rights to.” **[Non-functional] [Emotional/cognitive]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**

[Emotional/cognitive] | *need — a root cause rather than a feature request; the incentive problem cannot be solved in software, and design must not assume it away*

Q7: If a block reports no suspected cases this month, how do you tell “they found nothing” from “they sent nothing”?

→ Probe: *What have you done when you suspected it was the second one?*

A7:

- Direct answer: “most of the time, I cannot tell.” Both arrive as the same blank in the same cell; nothing in the form distinguishes the two states. **[Functional] [Emotional/cognitive]**
- Weak proxies he uses, all reconstructed by eye and after the fact: **[Constraint]**

- Whether any OTHER report arrived from that unit — immunisation figures, antenatal registrations, stock indent. If they did, the reporting mechanism works, so a leprosy zero more plausibly means “did not screen.” If nothing arrived, it is an administrative failure, not a surveillance finding. **[Functional] [Constraint]**
- Whether an activity denominator exists (households visited) against which zero suspects can be judged plausible. **[Functional]**
- Whether the post is vacant — no ANM, unfilled ASHA position, maternity leave. “Half my zero-reports have a simple explanation,” usually discovered only AFTER he has queried the figures, “which is embarrassing.” **[Functional] [Emotional/cognitive]**
- How long the unit has been quiet — one zero month is noise, four consecutive is a finding about the reporting unit, not about the disease. **[Functional]**
- Whether neighbouring sub-centres reported suspects — geography arguing against the zero. **[Functional]**
- Requirement he states directly: “a zero must be an act, not an absence.” Calls it nil reporting from his state-cell days — somebody must actively certify “I screened, I found nothing.” And the worker must be able to say WHY: no screening conducted / post vacant / area inaccessible / festival week / screened and genuinely negative. “Distinct reasons, each meaning something different to me, none of them shameful if they are true.” **[Functional] [Business rule]**
- What he has done: telephone the block officer (who properly defends his staff) → request the underlying register → verification visit, taking the register and visiting 15–20 households from it. “Not to catch anybody. To find out what actually happened.” **[Business rule] [Emotional/cognitive]**
- KEY FINDING from one such visit — a sub-centre with five straight zero months while neighbours reported steadily. He expected fabrication. He found a well-regarded, genuinely working ASHA who had STOPPED RECORDING SUSPECTS because on three earlier occasions her referrals had not reached the PHC and she had been told at a meeting that her referrals were “not converting.” She understood a non-converting referral counted against her, so she quietly stopped making them. “My own incentive structure had taught a good worker to report zero.” Her zero was neither found-nothing nor sent-nothing — it was “learned not to.” **[Business rule] [Emotional/cognitive]**
- Twice he has found block figures that existed nowhere below the block level. **[Constraint] [Emotional/cognitive]**
- Self-criticism: after each such episode he added a column to a form — “which I now recognise was mostly for my own comfort.” **[Emotional/cognitive]**

- Capacity limit he volunteered: ~2 verification days a month against 30+ sub-centres in 5 blocks, and he samples units he already suspects. So his detection of bad reporting is biased toward blocks already judged bad, and “I am almost certainly blind to a well-behaved-looking block that has been quietly estimating for years.” **[Constraint]** **[Emotional/cognitive]**
- Wants mandatory reasoned nil reporting; quiet units surfaced proactively and ranked without him going looking; and administrative context held alongside the numbers — vacancy status, whether the worker was active elsewhere that month, when the record was actually created. “I do not primarily need a system that tells me the district found 412 cases. I need one that tells me where I am not looking.” **[Functional]** **[Business rule]**

→ Tags: **[Functional]** **[Business rule]** **[Constraint]** **[Emotional/cognitive]** | *need — one requirement stated almost as a rule (“a zero must be an act, not an absence”); adopt the rule, design the mechanism*

Q8: Tell me about a cluster you eventually found. How did you find it, and when?

→ *Probe: What would have been different if you’d known two months earlier?*

A8:

- The Ghatpara cluster — three adjacent panchayats along a single road, in the block he did NOT send the special drive to (see Q5). **[Emotional/cognitive]**
- Timeline: cases from June (2 in one panchayat), August (1), September (3), November (2), continuing; 14 by February. Each arrived inside an unremarkable block total of ~31 and was spread across five monthly reports. “Two out of thirty-one is invisible. The aggregate hid them perfectly.” **[Functional]** **[Constraint]**
- HOW IT WAS ACTUALLY FOUND: in April, at the quarterly review, a CHC Medical Officer remarked in passing over tea that he felt he was “seeing a lot of cases from that side lately.” Verma then re-plotted his pins sorted by VILLAGE rather than by block and saw 14 cases in a two-kilometre stretch. A contact survey team then found 3 more cases the district had never detected, plus household contacts requiring prophylaxis. **[Functional]** **[Constraint]** **[Emotional/cognitive]**
- Total elapsed: first case June → pattern visible April → team on the ground the following June. ELEVEN MONTHS. “The proximate cause of detection was a doctor’s casual impression during a tea break. Not my portal. Not my monthly report.” **[Functional]** **[Emotional/cognitive]**
- Structural cause he identifies: “my data is aggregated by administrative unit and reported by month. A cluster is a phenomenon of place and continuity.” 14 cases on one road over nine months, divided across five blocks and nine monthly reports, disappears. “My reporting geometry is designed to hide exactly the thing I most need to see.” **[Constraint]** **[Functional]**

- If known two months earlier — he separates programme claims from clinical ones, deferring individual progression to the MO: **[Emotional/cognitive]**
 - At two months there would have been 2–3 cases in one place, which already exceeds his threshold for sending a survey team. **[Functional]**
 - Household and neighbourhood contact tracing across the three panchayats would have converted most of the 14 from self-presenting (i.e. already visibly affected) to actively detected — “and that difference is the whole of my Grade-2 indicator.” **[Functional]**
 - Cannot honestly claim all 4 Grade-2 cases were preventable; some were probably established by June. But a case found in August rather than the following June is ~10 months of treatment started earlier and 10 fewer months of household exposure; one of the 17 was a nine-year-old, meaning transmission was live throughout the period he was reading unremarkable totals. Prophylaxis for contacts would have gone out ~10 months sooner. **[Emotional/cognitive]**
 - And the one special drive would have gone to the block that needed it — changing outcomes in two blocks, not one. **[Emotional/cognitive]**
- HIDDEN COST he flagged: because he found it late, 17 extra cases appeared in one quarter and the state read it as deteriorating control. He spent a review meeting defending his own belated competence. “Finding the truth late is punished as though it were the failure itself” — which returns to why nobody hurries to report bad news (Q6). **[Business rule] [Emotional/cognitive]**
- Smallest actionable signal he wants: not a monthly report — something unprompted saying “three cases within one kilometre within sixty days — look here.” Must watch continuously across the district AND ACROSS MONTH BOUNDARIES rather than resetting each reporting cycle. **[Functional] [Non-functional]**
- And on alert handling: “an alert that arrives on my dashboard among forty other things is not a system, it is a notification. I want an alert that becomes an item — assigned, dated, closed with a recorded action.” **[Functional]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**

[Emotional/cognitive] | *need — with a usable threshold candidate (≈3 cases / 1 km / 60 days) offered for validation; confirm epidemiologically with the Medical Officer and state cell*

Q9: Which decisions would you actually make differently with data in one day instead of three weeks?

→ Probe: *So which of your indicators genuinely need to be near-real-time, and which are perfectly fine monthly?*

A9:

- Called this the sharpest question of the interview and answered it as a triage rather than a wish list. Decisions genuinely changed by one-day data: **[Emotional/cognitive]**
 - WHERE TO SEND A TEAM THIS WEEK — 3 cases in proximity within 60 days. “Acting in week two is materially different from acting in month three.” **[Functional] [Non-functional]**
 - CHASING AN OPEN REFERRAL — the one where a single day matters most, and he said he had not appreciated it until answering these questions in sequence. A flagged suspect is findable only briefly: the ASHA knows the household, the family remembers the conversation. Three weeks later the man has migrated for work, the ASHA has done 400 more visits, and the name is a line in a register — “an archaeological record.” Seeing on Tuesday that 11 suspects flagged last week have no recorded outcome means somebody can act while those people are still traceable. “This, not my Grade-2 percentage, is where near-real-time saves people.” **[Functional] [Non-functional] [Emotional/cognitive]**
 - MID-CAMPAIGN REDEPLOYMENT — currently he learns how a 14-day drive went AFTER it ended and can only hold a post-mortem. Day-4 visibility of coverage by panchayat (30% versus 80%) lets him move teams while the drive is still running. “This alone would justify the system to the state.” **[Functional] [Non-functional]**
 - SENDING A SUPERVISOR TO A UNIT GONE QUIET — two weeks of silence addressed in week three is a recoverable month; found in next month’s report it is a lost one. **[Functional] [Non-functional]**
 - STOCKOUTS — MDT blister packs, prophylaxis supply. A same-week problem with a same-week solution; learning it from the monthly report means people were turned away. **[Functional] [Non-functional]**
- What all five share: “they are about individuals and places, and they decay with time.” Perishable operational facts. **[Non-functional] [Emotional/cognitive]**
- What is fine MONTHLY or slower — a substantial concession: **[Business rule]**
 - Grade-2 disability at detection — his most important indicator, and it does NOT need to be real-time. A ratio on ~28 cases a year; watched daily it would be noise “and I would over-react to it, which is worse than not seeing it.” Monthly for reference, QUARTERLY for judgement. “It is a measure of whether my system is working, not a trigger for anything I do on a Tuesday.” **[Non-functional] [Business rule] [Emotional/cognitive]**
 - Child case proportion and MB/PB ratio — quarterly. Small numbers, slow signals. **[Business rule]**

- Treatment completion — inherently a 6–12 month cohort measure; real-time is meaningless. What he DOES need near-real-time is the individual defaulter (this person missed collection) — again an operational trigger, not an indicator. **[Functional] [Business rule]**
- Annual detection rate and prevalence — annual. **[Business rule]**
- PIVOTAL PRINCIPLE HE ARRIVED AT: “Indicators are for judging. Triggers are for acting.” He concedes he had been demanding that his INDICATORS be real-time when what he needs real-time is a small set of OPEN ITEMS — unclosed referrals, geographic accumulations, quiet units, stockouts, defaulters. “Perhaps six things. Each one is a list of specific cases requiring action, not a percentage.” **[Functional] [Business rule] [Emotional/cognitive]**
- Stated requirement: a daily-updating WORKLIST of open items, plus a MONTHLY indicator set he can actually trust. Explicitly rejects “a live dashboard of forty percentages, which — I say this against my own instincts — I would look at twice and then stop looking at, as I have with every dashboard anyone has ever given me.” **[Functional] [Non-functional] [Emotional/cognitive]**
- SECOND MAJOR CONCESSION on the field-burden conflict, drawn from his own triage: if most indicators are monthly or quarterly, most of his fields need not be collected in the field at all. Disability grading, classification and treatment detail should come from the MO at confirmation, “when there is a clinician present and time to be accurate, rather than from an ASHA standing in a doorway... That is not her data to give me and I have been wrong to want it from her form.” The ASHA’s job is to flag the person, place and finding — “and to not lose them.” **[Functional] [Business rule] [Emotional/cognitive]**
- Two further design requirements: keep the WORKLIST and the REPORT structurally distinct (“if the two are mixed into one screen I will use neither properly”); and record his own response actions in the system so that in six months he can test whether responding to a cluster within three weeks actually reduced disability at detection — “if I cannot demonstrate that this system changes outcomes, the state will fund it for two years and then not renew it, and I will be back to the pins on the wall.” **[Functional] [Non-functional] [Emotional/cognitive]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Emotional/cognitive]** | *need — the single most useful answer for scoping; resolves the “everything must be real-time” pressure into six perishable triggers*

Q10: How do you know today whether a referred person actually reached the PHC and started treatment?

→ Probe: *And when you can’t tell, what do you do?*

A10:

- Structural cause: “there are two records, and they were never designed to meet.” The ASHA referral exists only as a tick in an aggregate “suspects referred” column with no record of WHO; the MO confirmation exists in the PHC leprosy register when a person presents. No shared identifier, no reference number. **[Functional] [Constraint]**
- Worse than a missing key — the loop cannot close even with goodwill on both sides: “the Medical Officer has no list of expected arrivals.” He cannot know the man before him is one of 11 suspects flagged in his catchment, because nobody gave him that list. The person may not mention being referred, and is recorded as an ordinary OPD attendance. “The referral is not lost through negligence. It is lost because the information travelled in only one direction, upward as a count, and never sideways to the person who needed it.” **[Functional] [Constraint] [Emotional/cognitive]**
- Ordinary matching friction on top: age recorded “about 35” and later 41; village spelled two ways; no telephone number or a neighbour’s phone; two men of the same name in one hamlet. **[Constraint]**
- THE HEADLINE FIGURE: ~1,100 suspects referred last year against 412 confirmed cases — a gap of ~700 people. “I do not know how many of those seven hundred were examined and correctly ruled out, and how many never arrived at a facility at all.” Some were looked at by a trained worker who thought “this may be leprosy” and then vanished from the programme entirely. “Some of them will walk into a district hospital in three years with a deformity, and I will record them as a new Grade-2 case, and I will have no idea that I already found them once.” **[Business rule] [Emotional/cognitive]**
- Same three-state collapse as Q6/Q7: examined-and-negative, did-not-attend, and no-record all arrive as the same silence. **[Functional]**
- INSTITUTIONAL CAUSE: the state does not ask for a referral completion rate. Suspects referred and cases confirmed sit in two separate rows and nobody has ever required them to be linked. “In four years as DLO I have never once had a true referral closure figure for my district. Not a poor one. None at all.” **[Business rule] [Emotional/cognitive]**
- What he does when he cannot tell: reports the two figures separately “and lets the reader draw whatever conclusion he likes. That is the truthful answer and it is not a proud one.” Workarounds tried: **[Emotional/cognitive]**
 - ASHA accompanies the patient — closure is near-total, but costs her most of a day plus travel she pays herself, against forty other duties. Used only for referrals that “look serious,” i.e. triage on her impression rather than any system. **[Constraint] [Emotional/cognitive]**
 - Referral slip with counterfoil — tried in two blocks, failed within four months. Slips lost, stubs not returned, more paper in an already drowning chain. He notes the idea was sound and the medium wrong: “it is precisely the problem a digital

record solves for free — the two halves of the slip stay connected without anyone carrying anything.” **[Functional] [Constraint]**

- MO confirmation camps in the village — his best intervention, because it removes the journey rather than tracking it. Where he does it “my closure problem disappears entirely,” which tells him the barrier is distance and lost wages, not patient reluctance. Expensive, so only a few times a year in selected places. **[Constraint] [Emotional/cognitive]**
- Otherwise he accepts the loss: “my district’s largest single leak of cases is a gap I have stopped measuring because I could not measure it” — described as the most corrosive thing he said. **[Emotional/cognitive]**
- Wants: a referral as an OPEN ITEM WITH A NAME, visible at both ends — the ASHA can see her referral is still unresolved and the MO can see who is expected in his catchment before they arrive (“at present the only person who knows both halves is nobody”). And closure with an EXPLICIT OUTCOME: confirmed / examined and ruled out / attended elsewhere / refused / migrated / not traceable. “Give me not traceable as a recordable outcome and you will have taught my programme something it does not currently know how to say. I would rather have four hundred honest not traceable entries than seven hundred people I cannot account for at all.” **[Functional] [Business rule]**

→ Tags: **[Functional] [Business rule] [Constraint] [Emotional/cognitive]** | *need — and, by his own later ranking in Q12, the single highest-value requirement*

Q11: Who all sees your reports, and at what level of detail? And when you send a team into a village after a cluster alert, what does the village conclude about who was identified?

→ Probe: *Has that ever created a problem for a family?*

A11:

- NOTE: this is the requirement he did not volunteer anywhere in the preceding ten answers. It had to be elicited, and it visibly changed his position. **[Emotional/cognitive]**
- Access as it stands: ASHA — her own households. ANM — sub-centre register with names, addresses, findings. MO — full clinical detail for his facility. Block officer — aggregates PLUS named line-listings (necessary for contact tracing and follow-up, “you cannot do contact tracing from a percentage”). DLO — aggregates officially, case-level line-lists routinely because he asks. State cell — aggregates plus case-level portal data. Central Leprosy Division — aggregates. WHO — fully anonymous. **[Business rule] [Non-functional]**
- His own observation on that gradient: anonymous at the top, progressively identified toward the person — so “the greatest concentration of identifiable information sits at exactly the level with the least data-protection capability.” **[Non-functional] [Emotional/cognitive]**

- HOW NAMED LINE-LISTS ACTUALLY TRAVEL TODAY (volunteered, unprompted):
[Constraint]
 - As WhatsApp photographs, sitting in block officers' phone galleries and in groups "which include people who have no business seeing them." [Constraint] [Non-functional]
 - As Excel files on the district data entry operator's desktop, "which has no password worth the name." [Constraint] [Non-functional]
 - Projected on a screen at monthly reviews with names visible, in a room of ~30 people, "to make a point about follow-up." [Constraint]
 - As printed campaign microplans listing named suspects, handed to survey teams made up of local people "who then walk into the village of the person named on the sheet they are carrying." [Constraint] [Emotional/cognitive]
 - "I have never once thought of any of that as a data-protection question. I thought of it as getting the work done." [Emotional/cognitive]
- What a cluster response discloses: "about as discreet as a brass band." A government vehicle, 4-5 staff in a village of 200 households, house-to-house along one specific lane because the alert was geographic. Everyone knows within an hour, and can see "which lane they concentrated on and which doors they spent longest behind." [Constraint] [Emotional/cognitive]
- Contact tracing is disclosive by design — it starts from an index case and asks about that household and its immediate neighbours, so observers can reconstruct who the index case was. "I have been designing my cluster response for epidemiological efficiency... and epidemiological efficiency and discretion are, it turns out, in direct opposition. I have optimised entirely for one and never once weighed the other." [Constraint] [Emotional/cognitive]
- On clinical reassurance: notes the MO would correctly say leprosy is curable and a treated person is not a source of infection, but "that is not what the village believes." The belief concerns impurity rather than infectivity and "no amount of correct clinical information touches it in the timeframe of a fortnight's campaign." [Emotional/cognitive]
- HARM — asked directly, he retrieved a specific case. After the Ghatpara survey, the panchayat's ASHA told his block officer some months later that a MARRIAGE NEGOTIATION FOR A YOUNG WOMAN in one of the visited households had been withdrawn. She was not the patient — her uncle was, and was already on treatment. The team's visit was seen, the matter was discussed, the other family withdrew. Reported secondhand, months late, never followed up. [Emotional/cognitive]

- Others of the same shape, all informal and long afterwards: a man doing daily-wage work at a shop who stopped being called after his household was visited; a family asked by relatives to keep separate utensils; a patient who moved to his brother's house in another district — which also removed him from treatment follow-up entirely.
[Emotional/cognitive]
- None of it is recorded anywhere: “my system counts cases detected and disability graded. It has no field, anywhere, for harm the programme caused.” So 17 cases found at Ghatpara is recorded as an unambiguous success. “I have been managing a programme with a revenue account and no expense account.” **[Functional]**
[Emotional/cognitive]
- EXPLICIT SELF-CORRECTION: every demand he had made — more fields, faster sync, wider visibility, district-level geographic mapping — is an increase in exposure when viewed from the position of the person in the record. More fields means more sensitive detail about a stigmatised condition on a phone that may be shared within a household. Faster sync means named case-lists in more hands sooner. “A village-level cluster map on my dashboard is, in a village of two hundred households, very nearly a list of names.” And: “I have not once, in this entire conversation, asked you who should NOT see this data. That is my omission, not yours.” **[Non-functional]** **[Emotional/cognitive]**
- On the ASHA specifically: she is a NEIGHBOUR — lives in the village, her children go to school with the patient's children, will still be there in ten years. “Whatever list you put on her phone, you are placing inside a dense social network that no software controls.” Not an argument for withholding (she cannot work blind and is the reason cases are found) but for distinguishing what she NEEDS to see from what she is ABLE to see: “there is a difference between your seven follow-ups this week and every leprosy case in your village since 2019, and I would have cheerfully asked you for the second.” **[Non-functional]** **[Business rule]** **[Emotional/cognitive]**
- Revises his Q5 argument: rich data does defend against fabrication, but “the cost is not only the ASHA's time, which is what I thought you would argue. The cost is borne by the person in the record, who is not in this room and was never consulted about how many fields I would like.” **[Emotional/cognitive]**
- What he requires: role-based minimum-necessary access ENFORCED IN SOFTWARE, not in a policy he signs — “I am the wrong person to be trusted with this judgement. Left to myself I will always request more, and I will always have a defensible programme reason for it, and I have just demonstrated that to you at length.” And on mapping: give him the coarsest resolution that still supports a decision, with finer detail behind a deliberate, logged step. “If a habitation-level count is enough to send a team, I do not need to see the houses — and I should not be given them merely because the data exists.” **[Non-functional]** **[Business rule]**
- Closing state: “I am somewhat unsettled. I came into this conversation certain that my problem was insufficient data. I am leaving less certain that more data, arranged the

way I would have arranged it, would have been a kindness to anybody.”
[Emotional/cognitive]

→ Tags: [Functional] [Non-functional] [Business rule] [Constraint]
[Emotional/cognitive] | *need — elicited, never volunteered; must not be dropped from the requirement set on the grounds that he raised it last*

Q12: Let me put a difficulty to you. Sunita is covering around forty houses a day on an entry-level phone with no signal, and she told me twenty-five fields per person is impossible.

Dr. Rao wants clinical detail but only what a non-clinician can reliably record. So — which of your fields drive an actual decision, and which exist because the form has always asked for them? And beyond the fields: what must this system do for you, and what must it never do?

→ Probe: (Conflict resolution / harvest question.)

A12:

- WITHDRAWS his opening position (“I would happily require many fields per person”) — and gives a data-quality reason rather than a sympathetic one: “a field she cannot fill reliably does not give me weak data, it gives me false data.” Twenty-five fields at forty households a day produces a form completed from evening recall, or completed with whatever value passes validation — from which he would then compute indicators and make confident decisions. “I am better off with six fields I can trust than twenty-five I cannot, because the six will not mislead me.” [Non-functional] [Emotional/cognitive]
- Offers a SORTING PRINCIPLE rather than a list, in three categories: [Business rule]
 - CATEGORY 1 — only a person standing there can supply it. This is the field worker’s, and should be nothing else: who (identity + household reference); where (precise capture, coarse display); when the observation happened; who observed; what she saw (structured, wording deferred to Dr. Rao); how the person came to attention; under-15 yes/no; contact of a known case yes/no. Roughly 7–8 items at first contact. “I am not going to pretend seven is nothing, but I believe it is defensible, and if Dr. Rao tells me one of them is unreliable in her hands I will drop it.” [Functional] [Business rule]
 - CATEGORY 2 — the system already knows it; NEVER ASK A HUMAN: block, PHC, sub-centre, worker identifier, entry timestamp, device — all derivable from who is logged in and where they are standing. “Every one of these appears on my current forms and every one is a transcription error waiting to happen.” Includes a notable insight: equity reporting (tribal habitation figures the state requires) can be DERIVED FROM CLASSIFIED GEOGRAPHY rather than by asking a person their category at the doorstep — “I get the same indicator without anyone having to ask anybody anything.” Caste, religion, occupation, education, income come OFF the form: “those columns exist on my formats, nobody at district level has ever analysed them, and I have never once made a decision from them.” [Functional] [Non-functional] [Business rule]

- CATEGORY 3 — the clinician's, at confirmation: disability grading, MB/PB classification, lesion count, nerve involvement, smear findings, regimen and treatment start date. "I have been demanding these from a form filled in a doorway. They belong to Dr. Rao, in a room, with time, and they will be BETTER there. I concede this entirely and I should have seen it years ago." **[Functional]**
[Business rule] **[Emotional/cognitive]**
- On legacy fields: "most of my form exists because it has always existed. Nobody ever removed a column. Every column is somebody's old requirement, and each one costs Sunita seconds she does not have, forty times a day." **[Non-functional]**
[Emotional/cognitive]
- THE ONE LINE HE HOLDS (expects resistance): the field worker must be able to record that she screened and found nothing. Acknowledges how it sounds — extra taps for a negative — but grounds it in the wrong-block decision: "a zero I cannot distinguish from an absence is worse than useless, it is actively misleading." TRADES IT DOWN: not per household, not per person — "one act per round, per day or per area." Wants to argue it out with her directly rather than have it dropped quietly in design. **[Functional]**
[Business rule]
- MUST DO: **[Functional]**
 - Never lose a person — a flagged suspect stays open until closed with a stated outcome, including "not traceable." **[Functional]**
 - Be honest about its own completeness — tell him what has not yet reached him, so unsynched is never mistaken for absent. **[Non-functional]**
 - Refuse silence — no unit can simply say nothing; every zero carries a reason. **[Functional]** **[Business rule]**
 - Watch continuously, across months and administrative boundaries, for accumulation in a place — and hand the result over as an ASSIGNED ITEM with a date and a closure, not a notification. **[Functional]** **[Non-functional]**
 - Show him the quiet units unprompted and ranked. **[Functional]**
 - Feed his existing obligations — Nikusth, HMIS, state formats. "If it becomes a fifth system requiring separate entry, it will fail, and it will deserve to." **[Non-functional]** **[Constraint]**
 - Keep the worklist and the report apart. **[Functional]**
 - Record what HE did in response, so he can demonstrate in two years whether faster response actually reduced disability at detection. "Without that I cannot defend the funding." **[Functional]** **[Emotional/cognitive]**
- MUST NEVER DO: **[Business rule]**

- “It must never make an honest bad number more expensive to report than a flattering one.” States this as HIS to fix, not the software’s — “if SparshCare gives me sharper instruments and I keep using them to punish the appearance of failure, I will have automated the same lie at higher resolution and greater speed” — but warns: “do not design as though the incentive problem is not there, because it is the largest thing in the room.” **[Business rule]**
[Emotional/cognitive]
- Never show a role more than its work requires — enforced in the software, not in a circular he signs. **[Non-functional]** **[Business rule]**
- Never make household-resolution mapping his default view. **[Non-functional]**
- Never turn the field worker’s screen into an instrument pointed primarily at her — “if her tool’s main function is to let me check on her, she will defeat it, and I said earlier she would be within her rights.” **[Non-functional]**
[Emotional/cognitive]
- Never add a field without naming the decision it drives. “Put that question to me every time. I will not ask it of myself.” **[Business rule]**
- Never require signal to record — “data captured only where there is network is data about network coverage, not about disease.” **[Non-functional]** **[Constraint]**
- Never close anything by silence. **[Functional]** **[Business rule]**
- THE FIELD-WORKER INCENTIVE he offers as a trade (answering his own earlier question about why an ASHA would want the data accurate): honest referral closure ends her being judged on a conversion she cannot influence. “She flagged the person correctly; the record shows she did her part; the failure sits where it actually occurred — with the distance, the wage lost in travelling, the camp I did not hold. She is credited for finding, not for outcomes beyond her control. That is worth something real to her, and it is the only argument I have that is in her interest rather than mine.” **[Business rule]**
[Emotional/cognitive]
- PRIORITISATION — if only one thing is built in the first release, he chooses the REFERRAL CLOSURE LOOP over the cluster alert: “the cluster alert makes me a better commander. The referral loop stops the leak of people I have already found once,” and it is the only one of the two that also helps the field worker. “If you can build just one thing, build the one that lets a person not be lost.” **[Functional]** **[Emotional/cognitive]**
- Process requirement he set: any field dropped because it proves unworkable in the field must be brought back to him BEFORE build, not after — “I have spent four years discovering my requirements at the annual review, and I would like to stop.” **[Business rule]** **[Emotional/cognitive]**

→ Tags: [Functional] [Non-functional] [Business rule] [Constraint]
[Emotional/cognitive] | *need — an explicit, negotiated trade-off with a stated build priority; treat as the primary scoping input*

Q13: Playback for confirmation: automatic rather than hand-compiled indicators, trustworthy because sourced from what happened at the point of care; distinguish a block with no cases from a block that hasn't reported; clusters visible while a team can still be sent; know whether referred people completed follow-up; and accept fewer fields if the remainder are ones you'd act on. Have I got that right — and if the ASHA could only record eight things, which eight would you choose?

→ Probe: (*Validation of understanding + commitment to a minimum data set.*)

A13:

- Confirms the playback as “substantially right,” with two corrections and one omission. [Emotional/cognitive]
- CORRECTION 1 — on trust. Automatic capture removes the four transcriptions, “that is half the problem. It does not remove the other half, which is that people report what protects them.” So trustworthiness has a second condition: the system must disclose WHAT HAS NOT REACHED HIM so a gap is never read as a finding; and he must stop making an honest bad figure costlier than a flattering one. Asks that the second be recorded as HIS COMMITMENT, not as a requirement on the delivery team. [Functional] [Non-functional] [Emotional/cognitive]
- CORRECTION 2 — on “fewer fields,” which he asked be worded carefully: “I am not asking for less data overall.” Clinical detail still arrives, from Dr. Rao at confirmation, better than a doorway would produce. “What changes is WHO IS ASKED, AND WHEN. Sunita’s form shortens. My dataset does not necessarily shrink. Please write it that way, or someone will conclude I gave up my reporting requirements, and I did not.” [Business rule] [Emotional/cognitive]
- OMISSION he insisted be reinstated — EXPOSURE: who can see a record at what resolution, and what a cluster response reveals to a village about who was identified. The WhatsApp line-lists, the map that is functionally a disclosure, the Ghatpara marriage. “If that falls out of your summary because I raised it last and reluctantly, then this conversation will have improved my dashboards and left the cost exactly where it has always been — with the person in the record, who is not here. Keep it in.” [Non-functional] [Emotional/cognitive]
- THE EIGHT — each stated with the decision it drives, applying his own “never add a field without naming the decision” rule to himself: [Functional]

1. WHO — a name and a persistent household reference. Drives: closing a referral against a person; and not re-discovering the same man in three years as a fresh Grade-2 case. **[Functional]**
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 2. WHERE — location captured precisely, displayed coarsely. Drives: cluster detection, and where to send a team. **[Functional] [Non-functional]**
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 3. WHEN THE ENCOUNTER HAPPENED — held separately from when it synced. Drives: every timing measure he owns, and the 60-day cluster window. **[Functional]**
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 4. WHO OBSERVED. Drives: accountability in place of the register signature, and credit to the finder. **[Functional] [Business rule]**
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 5. WHAT SHE SAW — structured, minimal, “in Dr. Rao’s wording rather than mine.” Drives: whether this is a referable suspicion at all. **[Functional]**
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 6. HOW THE PERSON CAME TO ATTENTION — survey / self-presented / contact / school / campaign. Drives: interpretation of the Grade-2 indicator, “which is currently uninterpretable,” and whether campaigns are worth their cost. **[Functional] [Business rule]**
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 7. UNDER FIFTEEN — yes or no. Drives: the child-case indicator; active transmission. (Not exact age — “about 35 versus 41 has never changed a single decision I have made.”) **[Functional]**
-
 8. CONTACT OF A KNOWN CASE — yes or no. Drives: prophylaxis, and whether a cluster is one household or a neighbourhood. **[Functional]**
- DELIBERATELY OUTSIDE THE EIGHT, with reasons given so it is not read as evasion: the REFERRAL OUTCOME is not a doorstep field but a later state of the same record, closed by whoever resolves it, with “not traceable” available as an honest answer; and NIL REPORTING is one act per round, not a per-person field. “Neither belongs in her eight taps.” **[Functional] [Business rule]**
 - THE NINTH HE COVETS AND DECLINES TO REQUIRE — a means of re-contact (telephone number). “It would collapse my not-traceable problem more than anything else on the list. But a phone in a village is often a household phone, or a husband’s phone. Putting a stigmatised diagnosis at the end of a number I do not control could expose a woman to consequences I have no way to see and no way to undo.” So: OPTIONAL, OFFERED RATHER THAN REQUIRED, AND NEVER A VALIDATION ERROR. “Six months ago I would have made it mandatory and called it good data discipline.” **[Functional] [Constraint] [Emotional/cognitive]**

- Acknowledges his position has changed materially during the interview and invited that to be stated in the report. [Emotional/cognitive]
- Final process condition: if the field worker and the MO reduce the eight to six, he is to be told WHICH TWO AND WHY before build. "I will not defend a field I cannot justify. But I would rather lose it in an argument than discover its absence at the annual review." [Business rule] [Emotional/cognitive]

→ Tags: [Functional] [Non-functional] [Business rule] [Constraint]

[Emotional/cognitive] | need — validated playback with a committed minimum data set and two explicit corrections to the analyst's wording

3. TOP 3 PAINS I HEARD

1. Referral and suspect closure is structurally impossible — the programme's largest known leak of already-found people.

- The ASHA's referral and the MO's confirmation are two records with no shared identifier, and the MO holds no list of expected arrivals, so the loop cannot close even with goodwill on both sides.
- Last year: ~1,100 suspects referred against 412 confirmed — ~700 people whom he cannot classify as examined-and-ruled-out versus never-attended.
- In four years he has never had a referral closure figure at all, partly because the state has never asked for one. He has stopped measuring it: "a gap I have stopped measuring because I could not measure it."
- He himself ranked this above cluster detection for the first release.

(Q10, Q12, Q13)

2. Clusters are invisible until months after a team could have made a difference.

- Data is aggregated by administrative unit and by reporting month; a cluster is a phenomenon of place and continuity, so it dissolves across blocks and monthly reports.
- Ghatpara: 14 cases along a 2 km stretch accumulating over nine months, each invisible inside a block total of ~31; found in April via a Medical Officer's passing remark over tea, with a team on the ground eleven months after the first case. A contact survey then found 3 more undetected cases.
- Compounded by the wrong-block decision in Q5 — his one available special drive went to a block with a smooth, flattering figure instead of the true cluster block; 4 of the late-detected Ghatpara cases had Grade-2 disability and one was a child of nine.

(Q2, Q5, Q8)

3. He cannot trust the figures he certifies, and cannot distinguish absence of disease from absence of reporting.

- Four transcriptions sit between the field observation and his signature; 5–7 weeks end-to-end from Kondagaon (9 in monsoon) against a state deadline that falls earlier, so he submits estimates and revises; 8–10 of his own working days each month are lost to chasing and reconciliation.
- He detects bad numbers only by eye and estimates he catches about half, with ~2 verification days a month sampled from blocks he already suspects.
- Underlying all of it: a silent zero means both “assessed and negative” and “not assessed” — opposite facts demanding opposite responses, both read by the system as reassurance.

(Q3, Q5, Q6, Q7)

Honourable mentions (real, but second tier)

- The performance-management structure makes an honest bad figure more costly to report than a flattering one, at every level including his own — arguably the root cause of pain 3. He stated this himself and accepted it as his to fix rather than the system’s, but asked that the design not assume it away. (Q6)
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4. CONTRADICTIONS AND THINGS TO VERIFY

Items 1–3 are positions the stakeholder changed during the interview; the remainder are claims, assumptions or dependencies that need a second source before they enter the requirement baseline.

- **Field count** — his position moved during the interview, from “I would happily require many fields per person” and “a short form is unfalsifiable” to a committed 7–8 items for the field worker with clinical detail moved to the MO. Verify the eight against Sunita (40 houses/day, entry-level phone, no signal) and against Dr. Rao on what a non-clinician can record reliably. He has agreed to be told which fields are cut and why before build, not after.
- **Latency** — he opened demanding near-real-time indicators and closed by separating triggers (near-real-time worklist, ~6 open-item types) from indicators (monthly reference, quarterly judgement). Confirm the six trigger types with him in writing, and confirm he will not later request the live percentage dashboard he predicts he would abandon.
- **Access and exposure** — he asked for district-level village mapping and richer case detail, then conceded a village-level dot map is functionally a disclosure and that he is “the wrong person to be trusted with this judgement.” Verify the actual smallest map

resolution that still supports sending a team (habitation-level count?) and whether coarse-by-default with logged drill-down survives a real cluster response. Corroborate with the MO and the state cell, not with him alone.

- **All figures quoted from memory** — 412 / 388 / 341 new cases; 6.8% / 5.9% / 4.2% Grade-2 at detection; 6.3% child cases; ~55% MB; ~1,100 suspects referred; ~17 per 100,000; population ~2.4 million; state average ~4%. Reconcile against Nikusth, HMIS and the state annual return. He has already stated his personal Excel is inconsistent with what he officially submitted — expect up to three versions of the same year.
- **The Grade-2 indicator is bidirectional** — better active detection can lower the percentage without any change in underlying disability, and weak surveillance can raise it. Any future claim that SparshCare improved this indicator must be controlled for mode of detection and case volume. Confirm the target definition with the state cell before treating 6.8% as a baseline to beat.
- **Does the state actually require a referral completion indicator?** — he says it does not; suspects referred and cases confirmed sit in separate rows and were never meant to be linked. If correct, his highest-priority requirement has no reporting mandate behind it, which affects funding and mandatory-field decisions. Verify with the State Leprosy Officer.
- **“Zero” semantics against the existing formats** — verify whether Nikusth and HMIS can represent assessed-negative versus not-assessed at all, or whether a three-state model will be flattened on export. Also verify whether mandatory reasoned nil reporting can be imposed without a state instruction.
- **Integration is asserted, not confirmed** — feeding Nikusth, HMIS, IHIP and the state Excel formats needs technical verification (API? bulk upload? manual export only?). He is not the right source; escalate to state IT / NHM. He was unambiguous that a fifth parallel system requiring separate entry will fail.
- **Vacancy and HR data** — he says roughly half his zero-reports are explained by vacant posts and wants vacancy status shown beside the figures. That data sits in HR/NHM systems, not the leprosy programme. Confirm availability before committing to it.
- **Harm is unrecorded and currently unverifiable** — the withdrawn marriage negotiation, the lost daily-wage work, the separate utensils, the patient who migrated and left follow-up. All secondhand, months late, in no register. Cannot be baselined from existing records; measuring it at all needs a deliberate method and consent. Flag to the programme rather than filing as anecdote.
- **His own quality assurance is biased** — ~2 verification days a month, sampled from blocks he already suspects, catching “perhaps half.” Treat all historic block-level data as of unknown and unevenly distributed quality; do not use it as a clean baseline.

- **“The ASHA learned not to refer” needs corroboration** — his account of a worker who stopped referring because her referrals were “not converting” is a single verification visit. If the behaviour is widespread, referral volume is itself suppressed, the ~1,100 figure is an undercount, and the closure problem is larger than stated. Test in the field-worker interviews.
 - **Offline sync semantics unresolved** — two points he raised twice and that remain unanswered: where a record sits during days without network in Kondagaon, and how he is to distinguish incomplete data from complete data on his screen. He lists sync-completeness transparency as a must-do. Verify with the technical team.
 - **The optional ninth field (contact number)** — he wants it offered but never required and never a validation error, citing household and husband-controlled phones and risk to female patients. Verify how much of the not-traceable problem it would actually solve, and route the privacy decision to the programme rather than to him.
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5. RECORDED CONFLICTS

5A. District reporting appetite versus field data-entry burden

Position 1 — the district’s appetite for data. He opened this interview arguing for many fields per person: “a short form is unfalsifiable,” and rich, corroborating data is “my only defence against being lied to” (Q5). He explicitly invited the analyst to push back on field-worker burden with evidence rather than agree with him (Q5).

Position 2 — the field-worker burden put to him directly. In Q12 the analyst put a specific difficulty to him: Sunita covers around forty houses a day on an entry-level phone with no signal, and twenty-five fields per person is impossible for her; Dr. Rao wants only what a non-clinician can reliably record.

His counter-proposal / concession. He withdrew his opening position, on data-quality rather than sympathy grounds: “a field she cannot fill reliably does not give me weak data, it gives me false data... I am better off with six fields I can trust than twenty-five I cannot” (Q12). He replaced the field-count argument with a sorting principle in three categories — fields only a person standing there can supply (roughly 7–8 items, becoming his committed eight in Q13); fields the system already knows and must never ask a human (block, PHC, sub-centre, worker ID, timestamp, device, and equity/habitation category derived from geography); and fields that belong to the clinician at confirmation, not the doorstep (disability grading, MB/PB classification, lesion count, nerve involvement, smear findings, regimen, treatment start date) (Q12). He held one line even after conceding the rest — mandatory reasoned nil reporting — but traded it down from a per-person field to “one act per round, per day or per area” (Q12), and offered the field worker a direct incentive in return: she is credited for flagging a case, not for a referral outcome beyond her control (Q12).

Options to take forward:

- (a) Adopt his committed eight-field minimum for the field worker's first contact (who, where, when observed, who observed, what she saw, mode of detection, under-15 flag, contact-of-known-case flag), each tied to the specific decision it drives (Q13).
- (b) Populate Category 2 fields (block, PHC, sub-centre, worker identifier, timestamp, device, and equity/habitation classification) automatically from login and location context; never present them to a human as data-entry fields (Q12).
- (c) Defer all clinical and classification detail (disability grading, MB/PB, lesion count, nerve involvement, smear, regimen, treatment start date) to the MO at confirmation, per Category 3 (Q12, Q13).
- (d) Implement nil reporting as one certifying act per round/day/area, with a reason code, rather than a per-person or per-household field, satisfying his "a zero must be an act" rule without adding to per-person burden (Q7, Q12, Q13).

Recommendation: adopt (a)–(d) as the baseline field-worker scope, and carry forward his own process rule as a design governance rule: any field the field-worker or clinician interviews prove unworkable must be returned to him, with the specific fields and reasons named, before build — not discovered later at the annual review (Q12, Q13).

5B. Accountability-driven performance management versus honest reporting

Position 1 — comparison and consequence are necessary. He does not want performance accountability removed: "without comparison and consequence, nothing moves at all" (Q6). The current escalation ladder — comparative display at monthly review, written request for explanation, supervisory visit, entry in the performance record, effect on fund flow and future campaign allocation — is the mechanism that makes anything move in the district (Q6).

Position 2 — the same structure punishes honesty and suppresses the truth. "The number that protects you travels upward, and the number that is true stays in the village" (Q6). An honest bad figure with reasons draws a letter and a visit; a smooth, flattering figure draws nothing — "the best available outcome" (Q6). He has direct evidence this is not hypothetical: a well-regarded, genuinely working ASHA stopped recording suspects after being told her referrals were "not converting," because she understood a non-converting referral counted against her — "my own incentive structure had taught a good worker to report zero" (Q7). He names this the root cause underneath his trust problem in Q3, Q5 and Q6: "it punishes the appearance of poor performance, which means it punishes honesty, which means it destroys the data I need to manage with. I have been running the instrument that blinds me" (Q6).

His counter-proposal / concession. He does not ask the software to solve the incentive problem, and is explicit that it cannot: "if SparshCare gives me sharper instruments and I keep using them to punish the appearance of failure, I will have automated the same lie at higher resolution and greater speed" (Q12). He reframes the fix as his own commitment rather than a system requirement — he must stop making an honest bad figure costlier to report than a flattering one (Q13) — while still warning the design team not to assume the incentive problem

away, “because it is the largest thing in the room” (Q12). His one concrete design-level mitigation is the field-worker incentive trade: credit the ASHA for correctly flagging a case, not for a referral outcome she cannot control (Q12).

Options to take forward:

- (a) Do not attempt to solve the incentive/accountability problem in software; scope it explicitly as an organisational and programme-management responsibility, per his own framing, and record his stated commitment rather than a system requirement (Q12, Q13).
- (b) Build the credit/attribution model he proposed: the ASHA’s record shows she found and flagged the person correctly; where the outcome depended on factors she cannot control (distance, lost wages, an MO camp that did not happen), the failure is attributed there, not to her (Q10, Q12).
- (c) Build the three-state zero (assessed-negative / not-assessed / no-record) and mandatory reasoned nil reporting so a silent zero can no longer be read either way as reassurance, reducing — without eliminating — the space in which under-reporting can hide (Q6, Q7, Q12).
- (d) Flag to the state programme that a system making failure visible faster must be paired with a change in how failure is treated at review, or, in his own words, the people it makes newly visible “will defeat it, and they will be within their rights to” (Q6).

Recommendation: build (b) and (c) now, inside SparshCare’s scope. Escalate (a) and (d) to the state programme as a precondition for the whole system’s long-run credibility — he was explicit, more than once, that this is a problem design alone cannot fix, and that a sharper instrument aimed at the same broken incentive will only produce a faster, higher-resolution version of the same distortion.

APPENDIX A — CONSOLIDATED REQUIREMENTS REGISTER (FROM THIS INTERVIEW ONLY)

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-50	Capture mode of detection for every case (survey / self-presented / contact / school / campaign), since the Grade-2 indicator cannot be interpreted without it.	Functional	Q2, Q13	Must have
FR-51	Hold a suspected case as an open item from the ASHA’s flag until an MO confirms or rules it out; the interval	Functional	Q2, Q10, Q12	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	itself must be tracked.			
FR-52	Provide single point-of-contact capture with no re-transcription downstream, underpinned by a persistent longitudinal person identity.	Functional	Q3, Q4	Must have
FR-53	Run automated plausibility and ratio-exception checks at entry and at aggregation (implausible work rates, round numbers, broken MB/PB or Grade-2 ratios, suspiciously smooth trends), surfaced as exceptions.	Functional	Q5	Should have
FR-54	Capture reporting-behaviour metadata — record-created time versus submitted/sync time, and detection of single-batch late-month entry.	Functional	Q5, Q7, Q13	Should have
FR-55	Record a three-state zero — assessed-negative / not-assessed / no-record — and never allow a silent, undifferentiated zero.	Functional	Q6, Q7, Q12	Must have
FR-56	Require mandatory reasoned nil reporting, one certifying act per round/day/area (not per person), with reason codes (no screening / post vacant / area inaccessible / festival week / screened and negative).	Functional	Q7, Q12, Q13	Must have
FR-57	Proactively surface and rank quiet or non-reporting units, without requiring the DLO to go looking for them.	Functional	Q7, Q12	Must have
FR-58	Run continuous rolling spatio-temporal cluster detection across administrative and month boundaries, at a low threshold (candidate ≈ 3 cases / 1 km / 60 days, pending epidemiological validation).	Functional	Q8, Q9	Must have
FR-59	Deliver alerts as assigned, dated, closable action items with an owner and a recorded response action, not passive notifications.	Functional	Q8, Q12	Must have
FR-	Provide a daily-updating worklist of	Functional	Q9,	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
60	open items (unclosed referrals, geographic accumulations, quiet units, stockouts, defaulters), kept structurally separate from the indicator/report view.		Q12	
FR-61	Log the DLO's own response actions against alerts, to support later outcome evaluation (e.g. whether faster response reduced disability at detection).	Functional	Q9, Q12	Should have
FR-62	Provide a bidirectional referral loop — a named open item visible to both the referring ASHA and the receiving MO, including an MO-side expected-arrivals list.	Functional	Q10, Q12, Q13	Must have
FR-63	Provide an explicit referral closure outcome vocabulary: confirmed / examined and ruled out / attended elsewhere / refused / migrated / not traceable.	Functional	Q10, Q13	Must have
FR-64	Auto-populate system-derived fields (block, PHC, sub-centre, worker identifier, entry timestamp, device) from login and location context; never ask a human.	Functional	Q12	Must have
FR-65	Derive equity / tribal-habitation reporting from classified geography rather than asking a person their category at the doorstep.	Functional	Q12	Should have
FR-66	Support a minimum eight-item field-worker data set at first contact (who, where, when observed, who observed, what she saw, mode of detection, under-15 flag, contact-of-known-case flag), each tied to a named decision.	Functional	Q13	Must have
FR-67	Capture clinical and classification detail (disability grading, MB/PB classification, lesion count, nerve involvement, smear findings, regimen, treatment start date) only from the MO at confirmation, never from the field worker.	Functional	Q12, Q13	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-68	Provide an optional, non-blocking contact/re-contact number field — never mandatory, never a validation error.	Functional	Q13	Should have
FR-69	Never allow any open item (referral, alert, nil report) to close by silence; every item requires an explicit terminal state.	Functional	Q12	Must have
NFR-27	Eliminate transcription-induced data drift across the reporting chain (accuracy).	Non-functional	Q3	Must have
NFR-28	Make field data visible to the district at the moment of capture, not on downstream forwarding (timeliness).	Non-functional	Q1, Q3	Must have
NFR-29	Interoperate with and feed existing systems (Nikusth, HMIS, IHIP/IDSP, state Excel formats) rather than requiring duplicate manual entry.	Non-functional	Q4, Q12	Must have
NFR-30	Provide auditability / tamper-evidence for reported figures.	Non-functional	Q5	Should have
NFR-31	Support two latency tiers — a near-real-time worklist of triggers, and a monthly-to-quarterly trusted indicator set.	Non-functional	Q9	Must have
NFR-32	Disclose sync-completeness — what has not yet reached the district — so a gap is never misread as a finding.	Non-functional	Q4, Q13	Must have
NFR-33	Be offline-first; never require live network signal to record a screening or a nil report.	Non-functional	Q12	Must have
NFR-34	Enforce role-based, least-privilege access in software, not in a signed policy.	Non-functional	Q11, Q12	Must have
NFR-35	Log every drill-down below the coarse default geographic resolution.	Non-functional	Q11	Should have
NFR-36	Default all geographic displays to a coarse resolution (e.g. habitation-level counts) rather than household-resolution mapping.	Non-functional	Q11, Q12	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
BR-16	Preserve the existing reporting hierarchy and indicator ownership (sub-centre/PHC → block → DLO → State Leprosy Officer → Central Leprosy Division → national/WHO) with its fixed monthly/quarterly/annual submission cycle.	Business rule	Q1	Constraint on design
BR-17	Recognise that key indicators (notably Grade-2 disability proportion) are ratio-based and bidirectional; interpretation requires mode-of-detection context, not the raw percentage alone.	Business rule	Q2	Constraint on design
BR-18	Preserve supervisory verification and escalation as the mechanism for suspected bad data (telephone query → public review → verification visit), while the system adds proactive detection on top.	Business rule	Q5, Q6, Q7	Constraint on design
BR-19	Apply a field-justification rule — no field may be added to any form without naming the decision it drives.	Business rule	Q12, Q13	Constraint on design
BR-20	Apply a data-supplier rule — each field is supplied by whoever can reliably observe it, at the appropriate moment (field worker at first contact; system at capture; clinician at confirmation).	Business rule	Q12	Constraint on design
BR-21	Referral closure is not currently a state-mandated indicator; suspects-referred and cases-confirmed are unlinked reporting rows under existing rules.	Business rule	Q10	Verify
CON-18	Kondagaon block is remote and hill-terrain — ~80 km from HQ, roads closing in monsoon, ~31 sub-centres and ~300 ASHAs to cover.	Constraint	Q3	Absolute
CON-19	End-to-end reporting lag runs 5–7 weeks (up to 9 in monsoon) against a state deadline that falls earlier in the cycle.	Constraint	Q1, Q3	Absolute
CON-	No reliable unique person identifier	Constraint	Q10	Absolute

ID	Requirement / observation (stated as a need)	Type	Source	Priority
20	exists in the field; matching depends on name, approximate age and village spelling, all subject to drift.			
CON-21	DLO verification capacity is fixed and small — approximately 2 days per month against 30+ sub-centres across 5 blocks, and is sampled from units already suspected.	Constraint	Q5, Q7	Absolute
CON-22	The existing performance-management incentive structure actively distorts the source data the system will ingest; this cannot be designed away.	Constraint	Q6, Q12	Absolute
CON-23	Field data is captured and viewed on shared household or family-controlled phones, by field workers embedded in the communities they report on, making any captured record a potential stigma-disclosure risk.	Constraint	Q11, Q13	Absolute
EC-09	The DLO certifies figures upward that he does not fully trust, and personally absorbs the credibility cost whenever he revises them.	Emotional/cognitive	Q1, Q3, Q6	Must design for
EC-10	Late detection of a real cluster (Ghatpara) is experienced as personal and professional failure, compounding the disability and delay it caused.	Emotional/cognitive	Q8	Must design for
EC-11	The DLO's own incentive structure has previously taught a good field worker to under-report ("learned not to") rather than face being blamed for non-converting referrals.	Emotional/cognitive	Q7	Must design for
EC-12	Unsettled recognition, reached only when asked directly, that more data collected and displayed on his terms is a stigma-exposure risk to the person in the record, not an unambiguous good.	Emotional/cognitive	Q11	Must design for

Design rules implied, in the stakeholder's own framing

- **"A zero must be an act, not an absence."** Every non-report must be a deliberate, reasoned entry — never a silent blank that could mean either "found nothing" or "sent nothing."
- **"Indicators are for judging. Triggers are for acting."** Keep the near-real-time worklist of open items and the trusted monthly/quarterly indicator set structurally separate; do not merge them into one screen.
- **"Never add a field without naming the decision it drives."** Every captured item must trace to a specific decision he — or the field worker — actually makes.
- **"Data captured only where there is network is data about network coverage, not about disease."** Capture must be offline-first, always.
- **"Never show a role more than its work requires."** Enforce minimum-necessary, role-based access in the software itself, not in a circular anyone signs.
- **"It must never make an honest bad number more expensive to report than a flattering one."** His own commitment for the programme side of the system, but a standing warning to the design: do not build as though the incentive problem is not there.
- **"If it becomes a fifth system requiring separate entry, it will fail, and it will deserve to."** Feed Nikusth, HMIS, IHIP and the state Excel formats; never require duplicate manual entry.
- **"Give me the coarsest resolution that still supports a decision, with finer detail behind a deliberate, logged step."** Coarse-by-default geography; drill-down is an audited action, not the default view.
- **"I would rather have four hundred honest not traceable entries than seven hundred people I cannot account for at all."** Honest failure states — including "not traceable" — must be first-class, recordable outcomes, never treated as shameful or suppressed.
- **"Do not design as though the incentive problem is not there, because it is the largest thing in the room."** Any dashboard that makes failure visible faster must be matched by a change in how failure is treated at review, or it will be defeated by the people it exposes.

STAKEHOLDER INTERVIEW LOG

SparshCare — Requirements elicitation · Interview 5 of 6

Label	Content
Analyst	Dr. Gunjan Singhal
Date of interview	17 August 2026
Stakeholder role	Patient / person screened and referred — not a system user
Stakeholder (persona)	Ramesh, 41 — marginal farmer, village near Kondagaon (Bastar, Chhattisgarh); suspect case flagged by ASHA
Location / facility	Village near Kondagaon, referral catchment of Kondagaon PHC, Bastar, Chhattisgarh
Project	SparshCare — leprosy screening, teleconsultation & referral system
AI tool used	Claude (Cowork) — model claude-opus-5, role-play persona “Ramesh, patient/beneficiary”; interview conducted in English standing in for Hindi/Halbi
Interview mode	Semi-structured, single session, 13 questions with follow-up probes
Duration (in-persona)	~35 min equivalent
Chat link / screenshot ref	SPARSHCARE / INT-05 / RAMESH-BENEF — [paste chat link or screenshot filename]

1. PURPOSE AND INTERVIEW APPROACH

Objective: to understand the experience of the person being screened and referred — the cost of a referral journey in money, time and standing; the mechanics through which stigma actually spreads inside a small community; what would make the trip to the PHC worth making at all; and what a notification to his phone may safely say. Ramesh is the counterpart to Interviews 1–4: everything the ASHA, the Medical Officer, Dr. Iyer and the District Leprosy Officer propose is, in the end, tested against whether a person like him will act on it. A deliberate interviewing constraint was observed throughout: the interviewer never named the disease first — Ramesh was left to name it, or avoid naming it, himself, which he does only indirectly, through euphemism and the remembered case of a neighbouring villager.

Question flow used (deliberate sequence — rapport → current process → evidence → pain → constraints → conflict → solution ideas last):

- Q1 Rapport, family and village context — the two structural constraints everything else sits on: lost income and total social observation.
- Q2 Current care-seeking process: home remedies first, decision-making within the household, the distance and cost of the existing journey to the PHC, and prior failed trips.

- Q3–Q4 Evidence: the ASHA’s screening visit itself, and what he understood, and feared, from it.
- Q5 Pain: disclosure at home — why he told nobody, including his wife.
- Q6–Q7 Constraints: the true cost of a PHC visit, and what actually makes him hesitate.
- Q8 Evidence: two remembered village cases, and how stigma actually spreads.
- Q9 Pain: what a wasted trip does to his willingness to try again.
- Q10 Constraint: the phone as a shared, low-literacy household device.
- Q11–Q12 Solution ideas: his own conditions for going, and for accepting a video teleconsultation.
- Q13 Conflict and read-back: playback of the whole account, and his do-not-record list.

Tag key: Each captured line is marked with one or more of five categories — **[Functional]** (something the system must do) · **[Non-functional]** (a quality it must have) · **[Business rule]** (an existing organisational rule) · **[Constraint]** (a fixed environmental, economic or regulatory limit) · **[Emotional/cognitive]** (fear, trust, stigma, mental load). Each answer additionally carries a roll-up tag and is marked *need* (a problem to be solved) or *solution* (a proposed answer, to be traced back to the underlying need). In Appendix A these categories are carried forward as ID prefixes: FR = functional, NFR = non-functional, BR = business rule, CON = constraint, EC = emotional/cognitive.

2. QUESTION-BY-QUESTION LOG

Q1: Ramesh ji, tell me a little about yourself — your work, your family, your village.

→ *Probe: How does a normal day go for you in the season?*

A1:

- 41 years old. Owns roughly two acres — paddy in kharif, chana afterwards “if the rain has been kind”. When there is no work on his own land he goes for wage labour (road work, or on Sukhram’s larger holding) at ₹250–300 a day. **[Constraint]**
- Household of six: wife, elderly mother, three children — two sons (one still small) and a daughter aged 17. His voice softens on the daughter and then closes up; this recurs throughout the interview. **[Emotional/cognitive]**
- In season he is up before light, waters and weeds, comes back to eat, returns to the field. Labour contracts run roughly 8 a.m. to evening. **[Constraint]**

“If I do not go one day, that day is gone. There is no one who gives it back to me.”

“The village is not big. Everyone knows everyone’s business. That is the trouble with it.”

→ Tags: **[Constraint]** **[Emotional/cognitive]** | *need* — establishes the two structural constraints every later requirement sits on: a day of care costs a day of income, and social observation is total.

Q2: When someone at home falls ill, what do you do? Where do you go, and who in the family decides?

→ Probe: *How far is that, and how do you get there?*

A2:

- First line of care is home remedies and the bazaar shop: “the man there gives tablets, you tell him what it is and he gives” — two days’ medicine for ₹20–30. A doctor is considered only if that fails. **[Business rule]** **[Constraint]**
- Decision-making: he claims authority (“I decide ... it is my pocket, so”) but describes his mother pushing the temple and a man “who reads and gives a thread”, and his wife pushing the hospital. **[Business rule]** **[Emotional/cognitive]**
- Journey to the PHC on the Kondagaon side: 20–25 minutes’ walk to the road, then a wait for the shared jeep of anywhere up to an hour, then ~40 minutes’ ride. ₹50 each way; doubled if his wife accompanies him. **[Constraint]**
- Unprompted, he raises prior failed trips: he has already lost a day, the fare, and come home with nothing — twice. **[Constraint]** **[Emotional/cognitive]**

“Two times I have gone and the doctor was not sitting ... I lost the day, I lost the fare, I brought nothing home. That is the thing nobody thinks about when they say ‘go, go, get it checked.’”

→ Tags: **[Constraint]** **[Business rule]** **[Emotional/cognitive]** | *need* — the referral is not into a vacuum: it competes with an informal chemist, a faith healer and a two-to-three-hour round trip that has already failed him twice.

Q3: I’m told the didi came by recently. Walk me through that — what did she do, and what did she say to you?

→ Probe: *Where were you sitting? Was anyone else there?*

A3:

- The ASHA came house-to-house for the youngest child’s polio drops and her antenatal register. She saw the patch while he was washing at the hand pump, took his arm, and ran a pen-tip sensation test. **[Functional]**
- He answered yes to feeling the pen-tip — then admits the answer was not reliable. **[Emotional/cognitive]**

- She told him it was not from the sun and that he must show it at the PHC, and she wrote his name in her book. She did not say the name of the disease — he credits her for that. **[Business rule] [Emotional/cognitive]**
- The screening happened in the open, four steps from a neighbour's wife filling a pot, with the Sahu boys present. **[Non-functional]**
- Immediate anxiety attaches to the register, not the illness — what is in it, and who reads it further up the chain. **[Non-functional] [Emotional/cognitive]**

"Do you feel? Do you feel here?"

"I do not know if I felt. I said yes."

"She should have come inside. One minute inside the house, and nobody would know anything."

"What is in that book? Where does it go? She writes and then somebody at the block reads it, and then who else reads it —"

→ **Tags: [Functional] [Non-functional] [Emotional/cognitive]** | *need — two distinct requirements surface: screening must happen out of public view, and the patient must be able to understand where his record travels. Also a data-quality flag: a sensation result given under social pressure in a public place is not clinically trustworthy.*

Q4: What did you understand from what she told you?

→ *Probe: What did you think it might be? Did anything she said worry you?*

A4:

- He read her fear from her face, not her words. **[Emotional/cognitive]**
- He caught the internal contradiction in the reassurance — told it was nothing, but to go quickly. **[Emotional/cognitive]**
- His entire disease model is one remembered case — Budhram, from the next village: fingers bending and going, foot affected, people refusing to take money from his hand, his wife's family taking her back, him sitting at the edge of the temple. **[Emotional/cognitive]**
- He resents the euphemism and reads it correctly as confirmation. **[Functional] [Emotional/cognitive]**

"I understood that she was frightened of it. That is what I understood. Not from the words — from her face."

"She said it is nothing, do not worry, but go quickly. If it is nothing, why quickly?"

"That is the whole of what I know about it."

"Nobody has said it to me plain ... Everybody is being very careful with me. That itself tells me something."

→ **Tags:** [Functional] [Emotional/cognitive] | *need — the patient holds an inaccurate and terrifying model of the disease (late-stage disability only). Reassurance without information actively increases fear. Naming the condition privately and plainly is a requirement, not a courtesy.*

Q5: After she left, what did you do? Did you say anything about it at home?

→ *Probe: Why not?*

A5:

- "I put my sleeve down." He worked until dark, ate, lay down and did not sleep. [Emotional/cognitive]
- He told nobody — including his wife. When she asked why the ASHA had held his arm, he said she was asking about the boy's drops. [Non-functional] [Emotional/cognitive]
- The reason is his daughter. She is 17; her maternal uncle has opened talks with a family in Bhanpuri — "good people, the boy has work". Nothing is fixed, and disclosure would end it the same day, without even a reason given. [Emotional/cognitive]
- He describes the stigma as inherited and long-lived — ten years from now, people will still say "that house, the father had it". [Emotional/cognitive]
- He also fears involuntary distancing inside his own home: not rejection, but small unconscious acts, like the children's plates being kept separate. [Emotional/cognitive]

"I put my sleeve down."

"If it comes out of this house that there is that sickness in the father — that is finished. Finished the same day. They will not even give a reason, they will just stop coming."

"Ten years from now they will still say, that house, the father had it."

"She would not throw me out, she is not like that. But she would look at me a different way. She would take the children's plates and keep them separate. Not even meaning to. Her hands would do it."

→ **Tags:** [Non-functional] [Emotional/cognitive] | *need — concealment is a rational protective strategy, not ignorance. Any system feature that assumes household disclosure (family screening, home visits, shared devices) collides directly with this.*

Q6: She said you should go to the PHC and meet the doctor. If you went, what would that day actually involve — how would you get there, and what would it cost you?

A6:

- He costs it out precisely, as someone who has already done the sum: ₹100 return fare + ₹20–30 for food + ₹300 forgone wage = ₹420–430 per attempt — “for one arm that does not even pain me”. Doubled if the doctor is absent. The ₹300 wage figure is the top of the ₹250–300 daily range given in Q1 — the one he uses for a lost day in peak season. **[Constraint]**
- He states plainly that money is not the worst part. **[Emotional/cognitive]**
- The PHC waiting area is a public space drawing people from surrounding villages, with a known “corner room” that signals which service a person is queuing for. **[Non-functional]**
- Facility staff are a second leak: the compounder, the ward boy, all locally resident — an innocent remark at home in the evening becomes village-wide knowledge within a week. **[Non-functional] [Constraint]**

“Somebody from my village will be there ... And they will say — Ramesh, what have you come for? What do I answer? ‘Nothing.’ Then they watch which room I go into. There is a room there, a corner room. People know which room is for what.”

“The compounder, the ward boy — they are all from around here. One of them says something at his own house in the evening, to his own wife, not even with any bad intention, and by the next week it is in three villages.”

“I would be paying to destroy my own house ... Better the sleeve stays down.”

→ **Tags: [Constraint] [Non-functional] [Emotional/cognitive]** | *need — the true disincentive is exposure, not expenditure. Note two leakage vectors the software cannot fix alone: identifiable room/queue signage, and locally-resident facility staff.*

Q7: What makes you hesitate about going?

→ *Probe: What is the part you’re most unsure about?*

A7:

- Beyond cost and observation, he fears the irreversibility of being named: right now it is a patch he can call the sun; once a doctor names it, that cannot be taken back. **[Emotional/cognitive]**
- Part of him hopes avoidance will work — do not go, and maybe it goes away by itself. **[Emotional/cognitive]**
- The deepest uncertainty is efficacy: whether anything can even be done, given that Budhram “went to hospitals also” and still lost his fingers. **[Emotional/cognitive]**
- Worst-case calculus, stated explicitly: he loses his daughter’s marriage and still has the disease. **[Emotional/cognitive]**

- The gap that would change it: nobody has told him plainly that it can be finished. **[Functional] [Emotional/cognitive]**

“Right now it is a patch. I can call it the sun ... The day a doctor puts the name on it, that thing is in my house forever. It cannot be taken back out. Even if he says something else after — the name has already been said.”

“do not go, and then maybe it goes away by itself”

“Whether anything can even be done. Because if it is that, then what? Budhram went to hospitals also, they say. And still his fingers went.”

“Then I have lost my daughter’s marriage and I still have the disease.”

“Nobody has told me otherwise. Nobody has said to me, Ramesh, this can be finished. If somebody said that to me plainly, and I believed them — that would be a different thing. But nobody has said it.”

→ **Tags: [Functional] [Emotional/cognitive]** | *need — the single highest-leverage intervention in the whole interview: a credible, plain-language statement that early treatment means a complete cure with no deformity. Without it, avoidance is the rational choice.*

Q8: Has anyone in your village ever had something like this? What happened to them, and to their family?

→ *Probe: And how did people come to know about it?*

A8:

- Case A — a woman in the para behind the school, years ago. Kept in the back room of the house. Her family quietly stopped being invited to marriages and feasts. Her brother had to marry outside the area, “far, where they did not know”, and largely stopped visiting. **[Emotional/cognitive]**
- Case B — Budhram. Bent fingers, dragging foot. Shopkeepers made him put money on the ground and washed their hands after he left. His wife’s brothers came and took her and the children. He sat at the temple side. **[Emotional/cognitive]**
- How word spread — the key insight of the interview. For Budhram, his own body told the market. For the woman, it was the health system’s own footprint: the same health worker with the bag visiting only that house, every month, was itself the signal that gave her away. **[Business rule] [Non-functional]**
- His summary: nobody ever announces it; it comes out in pieces — a register, a visit, which room, a word at somebody’s chulha in the evening. **[Non-functional] [Emotional/cognitive]**

- He applies it to the interview itself: that is why he is talking carefully here too, because the analyst is also somebody who came, and somebody saw them come.
[Emotional/cognitive]

“not said out loud ... just, the invitation does not come to them”

“far, where they did not know”

“The health people kept coming to that house. Every month, that same man with the bag, going in at that house and no other house. People are not stupid. They sit and they watch the lane. Why is he going there? Then somebody saw the packet of medicine. Then the ANM said something to somebody’s mother.”

“Nobody ever announces it. Nobody stands up and says, this man has it. It comes out in pieces. A register. A visit. Which room you went into. A word at somebody’s chulha in the evening.”

“That is why I am sitting here talking carefully with you also. Because you are also somebody who came, and somebody saw you come.”

→ Tags: [Business rule] [Non-functional] [Emotional/cognitive] | need — critical design insight: disclosure happens through the observable pattern of health-system contact, not through any announcement. Monthly home delivery of MDT and repeat home visits are themselves the leak. Field staff visiting the patient must be indistinguishable from routine activity.

Q9: Suppose you lost a day’s wage, travelled all that way, and the doctor wasn’t there. What would you do?

A9:

- Not hypothetical for him — it has happened twice, for a knee complaint. He was told to come tomorrow; he sat under a tree first because he did not want to spend the return fare straight away; he went home. [Constraint] [Emotional/cognitive]
- His wife asked and he said the doctor was not there, and she said nothing — that silence was worse than if she had shouted. [Emotional/cognitive]
- He never went a third time. The knee still catches now in the cold. [Constraint] [Emotional/cognitive]
- Applied to this condition, where mobilising himself is far harder: it takes two weeks of building himself up before he can even get his legs to walk to the road. [Emotional/cognitive]
- If he arrives after that effort and finds nobody sitting, he will not go again — not the next week, not the next month; he will pull the sleeve back down, say it was the sun, and wait to see what his hand does. [Functional] [Emotional/cognitive]

“That silence — that is worse than if she had shouted.”

“still catches now in the cold”

“This thing, where I have to make up my mind for two weeks before I can even get my legs to walk to the road. Where I have decided, all right, today I will find out, God’s will. And I reach and there is nobody sitting.”

“Then I will not go again, sahib. Not the next week, not the next month. That much courage does not come twice. I will pull the sleeve down and I will say it was the sun. And I will wait and see what my hand does.”

→ Tags: [Constraint] [Functional] [Emotional/cognitive] | need — one failed trip is terminal for this care pathway, not merely a delay. A referral must be issued against confirmed, checked provider presence for a named slot, never as a generic instruction to attend the PHC.

Q10: If a message came to your phone about this, who else might see it, or read it out to you?

→ Probe: What if the message said what it was about?

A10:

- Basic keypad handset. He recognises callers “by the shape” of the name. He reads slowly — “some words ... if it is long, no.” [Constraint]
- Messages are handed to whoever can read — usually his son in class nine, otherwise the shopkeeper or whoever happens to be standing there. [Constraint] [Emotional/cognitive]
- The phone is not personal: it sits on a shelf at home while he is in the field; the children take it; his wife uses it when her mother calls. [Constraint]
- His strongest reaction of the interview came at the idea of a content-bearing SMS naming the disease, opened and read aloud by his son. [Emotional/cognitive]
- A direct rebuke to well-meaning design: people who send helpful reminders do not think about whose hands the message passes through. [Emotional/cognitive]
- What he will accept: a contentless appointment reminder — day and time only, nothing else — that his son can read aloud in front of the whole house without concern. [Functional] [Non-functional]
- What he prefers: a voice call, spoken directly to him, so nothing stays behind on the phone for anyone to find later. [Functional]

“A message comes, you hold it out to whoever can read.”

“You will write the name of that disease and send it into my house, in my pocket, and my son will open it? My son will read that about his own father, standing there, out loud, before he even understands what he is reading?”

“You people — you sit somewhere and you think, we will send him a reminder, very good, very helpful. You do not think about whose hands the paper passes through.”

“Let it say nothing. ‘Your appointment is on Thursday at nine.’ Nothing else. No name of the disease, no name of the department. Let my son read that out in front of the whole house ... I don’t care.”

“Let somebody just call me and say it in my ear. A voice I can put the phone down after. Nothing stays behind on the phone. Nothing for anyone to find later.”

→ Tags: **[Constraint]** **[Functional]** **[Non-functional]** **[Emotional/cognitive]** | need + solution — the need is that nothing persisting on the handset may identify the condition, and that comprehension must not depend on his reading ability. Voice call and contentless SMS are his proposed solutions; treat them as strong candidates, not as the requirement itself.

Q11: What would need to be true for you to feel that going is worth it — and that it stays private?

→ Probe: Who, other than the doctor, would you be alright with knowing?

A11:

- He counted three conditions on his fingers — clearly pre-thought. **[Emotional/cognitive]**
- ONE — a plain explanation, in Halbi or Hindi, sitting down, the way he would tell his own brother, including whether a medicine finishes it, in words he can repeat to his wife later. His only current picture is Budhram’s hand, and nobody has tried to correct it. **[Functional]** **[Emotional/cognitive]**
- TWO — certainty the doctor is present: not “he should be there” but confirmation from a person, not a paper, naming the day, the time, and that his name is written. He then volunteered teleconsultation unprompted, so nobody sees him or asks what he came for. **[Functional]** **[Business rule]**
- THREE — no home delivery of medicine. He wants to collect it himself, even from further away, somewhere nobody knows his face. **[Functional]** **[Business rule]**
- Disclosure circle, in his words: the doctor to begin with, and nobody else; the ASHA by default (since she already knows), conditional on her entering the house to screen and on her book not sitting open at the block office. **[Non-functional]** **[Emotional/cognitive]**
- His wife — yes, but on his terms and his timing, once he has the full thing to tell her, not hearing it first from a neighbour. **[Non-functional]** **[Emotional/cognitive]**
- Excluded absolutely: the sarpanch, the anganwadi, the ward boy, the man who does the register — nobody in the village. **[Non-functional]**

- Conditional commitment: if those three things were true, he would go on Thursday. [Emotional/cognitive]

"In Halbi or Hindi, sitting down, the way you would tell your own brother. And if there is a medicine that finishes it — say that. Say it in words I can repeat to my wife later. Because right now the only picture in my head is Budhram's hand, and if that picture is wrong, somebody has to come and take it out of my head. Nobody has tried."

"Not 'he should be there.' Is there. Somebody who knows — a person, not a paper — telling me: Ramesh, Thursday, nine o'clock, he is sitting, your name is written. Then I will spend the four hundred. Then I will lose the day. I am not a miser, sahib, I only cannot afford to throw it away twice."

"If he could look at my arm from there and say what it is, and I do not have to sit in that corridor at all ... That would be the best thing. Nobody sees me, nobody asks me what I came for."

"It must not come to my house every month in a man's bag. Because that is what killed that woman's family ... Let me collect it. Let me go and take it from somewhere in the town where nobody knows my face. Even if it is further. I will manage the going."

"she already knows, so what is the use of saying no"

"Let me tell her. Me. In my own time, when I have the full thing to tell her, when I can say 'and it will be finished in this many months.' Not her hearing it from the neighbour's wife at the hand pump. If she hears it from outside first, I will never get that back."

"Nobody in the village. Not the sarpanch, not the anganwadi, not the ward boy, not the man who does the register. Nobody."

"If those things were true — I would go on Thursday."

→ **Tags: [Functional] [Non-functional] [Business rule] [Emotional/cognitive]** | need + solution — stated as three needs (plain-language curability information; verified provider availability; no identifiable home delivery). Teleconsultation and anonymous town pickup are his proposed solutions to those needs — valuable, but validate against programme rules before adopting.

Q12: If a doctor from the district spoke with you on a video call instead of you travelling, how would that feel to you? Who would you want sitting beside you?

→ Probe: And who would you not want in the room?

A12:

- He accepts the value at once — it saves the day, the fare and the eyes at the PHC — and says so without pretending otherwise. [Emotional/cognitive]

- Diagnostic confidence reservation: whether a patch can be seen properly on a small screen in poor indoor light, and what happens if a one-minute look gets it wrong. **[Non-functional]**
- Aftercare reservation: a screen going dark after hard news leaves him alone with it; somebody must be there afterwards to ask what the doctor meant. **[Functional]** **[Emotional/cognitive]**
- Who he wants beside him: the ASHA, and only her. She frightened him but knows the words and can put hospital Hindi into plain speech. He was emphatic it must be one person only. **[Functional]** **[Emotional/cognitive]**
- Venues excluded: the anganwadi and the panchayat bhavan, because people sit there all day watching who goes in. **[Non-functional]**
- Devices excluded: anyone else's phone — whose it is, what stays on it, who scrolls through it in the evening. **[Non-functional]** **[Constraint]**
- People excluded: the ward boy; the mitanin from the other para ("her mouth is loose"); his neighbour, however good he is to him; and his children — specifically not his daughter, who must not hear anything before he has decided what to tell her. **[Emotional/cognitive]**
- His actual ask, in full: a room with a door that shuts, one didi, and the doctor's face — nothing more. **[Functional]** **[Non-functional]**

"and I will not pretend otherwise"

"Can he see it properly in that small screen? A patch on the skin — the light here is not good inside the house. If he looks for one minute on a phone and says something, and it is wrong —"

"A man's face on a screen is not the same as a man in front of you. If he tells me a hard thing and then the screen goes off, who is left? I am sitting in the room alone with it. So ... somebody must be there afterwards. Somebody I can turn to and say, what did he mean by that."

"she talks fast and she frightened me, but she is not a bad woman, and she knows the words. If the doctor speaks in that hospital Hindi I will nod and understand half. She can put it in plain speech."

"But only her. One person."

"that is where people sit all day with nothing to do but watch who goes in"

"whose phone is it, what stays inside it afterwards, who scrolls through it in the evening"

"her mouth is loose, everyone knows it"

"however good he is to me"

“Not my daughter. Whatever is going to be said, she is not to hear it before I have decided what to tell her.”

“Somewhere with a door, sahib. That is all I am asking. A room with a door that shuts, one didi, and the doctor’s face. I have never asked for anything more than that.”

→ **Tags:** [Functional] [Non-functional] [Constraint] [Emotional/cognitive] | *need — teleconsultation is acceptable only as a bundle: a private lockable venue that is not a community building, a trusted interpreter physically present, a trusted device that retains nothing, and a human who remains after the call ends. Delivered without the bundle it fails.*

Q13: Playback: “The illness frightens you less than people finding out, because of what it could mean for your daughter and your family. You’d go if you were certain the doctor would be there, because a wasted day costs wages you can’t spare. You’d want it explained in your own language, and to know it can be fully cured. And nothing should come to your phone that tells anyone else what this is about. Have I understood you? And is there anything you would not want written down anywhere at all?”

A13:

- He confirmed the playback in full. Nobody has said things back to him before — everyone has told him what to do; the analyst is the first to repeat it so he can hear whether it is right. [Emotional/cognitive]
- He singled out the point that actually moves him: the certainty that the doctor is sitting there. Say that part again when he is refusing to go, and it will move his legs. [Functional] [Emotional/cognitive]
- His do-not-record list, given carefully:
 - - (a) His name and the disease name on the same line, anywhere a local person can see it — not in the ASHA’s book “that rides around in her bag”, not on a block-office list “where the peon reads everything while he is drinking his tea”, not on a paper pinned to a wall. [Non-functional]
 - - (b) No patient-held card naming the condition: it is in his pocket, and one day it falls out, or his wife is looking for money in his shirt. [Non-functional]
 - - (c) No mark, paint or number on the house — “I have seen that done for other things.” [Non-functional]
 -

- (d) His daughter's name nowhere near the record — not as “daughter of”, not in any family column, nothing. She has done nothing; let her name stay clean. **[Non-functional] [Emotional/cognitive]**
- A final requirement, volunteered as he stood to leave: the diagnosis itself must be delivered by a person, seated, with time — not a paper, not a message on the phone, not the didi calling out from the gate. He can carry a bad thing if a man tells it to him properly. **[Functional] [Business rule]**
 - He closed with a conditional commitment, pulling his sleeve back down over the patch as he had at the start: if somebody tells him the doctor is sitting, he will come on Thursday. **[Emotional/cognitive]**

“You have understood ... Nobody has said it back to me before. Everyone has told me things — go here, do that, don't worry. You are the first one who has repeated it to me so I can hear whether it is right.”

“Say the middle part again to me one day when I am refusing to go — the part about being certain the doctor is sitting. That is the one that will move my legs.”

“Then it is in my pocket and one day it falls out, or my wife is looking for money in my shirt.”

“I have seen that done for other things.”

“Not as ‘daughter of’, not in any family column, nothing. She has done nothing. Let her name stay clean.”

“Not a paper handed to me, not a message on the phone, not the didi calling out from the gate. A person, in a room, who has time to sit while I take it in. Even if what he has to say is bad. I can carry a bad thing if a man tells it to me properly.”

“If somebody tells me the doctor is sitting, I will come on Thursday.”

→ **Tags: [Functional] [Non-functional] [Business rule] [Emotional/cognitive] | need**
— the record-design constraints are stated as absolutes and should be treated as acceptance criteria. Note the last one is a channel rule that overrides convenience: diagnosis is never delivered by SMS, app notification or third party.

3. TOP 3 PAINS I HEARD

1. Exposure through the health system itself — not the disease.

His greatest fear is that contact with services will reveal his status: an ASHA screening him at the hand pump in front of neighbours, his name in a register that travels to the block office, a known “corner room” at the PHC, locally-resident staff who talk at home in the evening, and monthly home delivery of medicine that the whole lane can watch arrive. The consequence he names is concrete and immediate — his 17-year-old daughter's marriage proposal collapses “the same day”, and the household carries

the stigma for a decade. This outranks cost, distance and the illness itself, and it is why he has told nobody, including his wife. (Q1, Q3, Q5, Q6, Q8, Q11, Q12, Q13)

2. Wasted trips — ₹420–430 and a lost day per attempt, with no guarantee anyone will be there.

The journey is a walk, an open-ended wait for a jeep and a 40-minute ride, at ₹100 return fare, plus ₹20–30 for food, plus a ₹300 forgone wage — his peak-season figure, at the top of the ₹250–300 daily range he first gives in Q1 — ₹420–430 total per attempt. He has been turned away twice before for a knee complaint and never went back; the knee still troubles him. He states flatly that one failed attempt will end his engagement with this too: “That much courage does not come twice.” A referral with no confirmed appointment is, for him, worse than no referral. (Q1, Q2, Q6, Q9)

3. No one has told him, in his own language, that it can be cured.

His only reference case is a man who lost his fingers and “went to hospitals also”. Nobody has named the condition to him plainly, and the reassurance he did get (“it is nothing, but go quickly”) read to him as confirmation that it is serious. Without a credible statement that early treatment means a full cure with no deformity, avoidance is his rational choice — because a diagnosis he cannot act on still destroys his family’s standing. He said the counterfactual himself: “If somebody said that to me plainly, and I believed them — that would be a different thing.” (Q4, Q7, Q11)

4. CONTRADICTIONS AND THINGS TO VERIFY

Items to resolve with the ASHA, the PHC medical officer, the district leprosy officer, or in follow-up patient interviews — before any of this is written as a requirement.

- **Sensation test result is unreliable.** He told the ASHA “yes, yes I feel”, then admitted: “I do not know if I felt. I said yes.” Verify with the ASHA and PHC whether patient-reported sensation from a public, socially-pressured screening is being recorded as clinical fact. If so, screening data quality is compromised at source and re-testing under private conditions is required.
- **He wants privacy AND plain speech — test where the boundary sits.** He resents euphemism (“Everybody is being very careful with me. That itself tells me something”) yet cannot tolerate the disease name being spoken or written where anyone might perceive it. Not a contradiction in principle (plainness in private, silence everywhere else) but the exact boundary must be tested with real draft message copy and real venue options.
- **Teleconsultation is both his preferred option and one he doubts.** He volunteered it as “the best thing”, then questioned whether a skin patch can be assessed on a small screen in poor indoor light. Verify with the district dermatologist / leprosy medical

officer what image and lighting quality is clinically sufficient, and whether tele-assessment can be diagnostic or only triage.

- **Teleconsultation device ownership is unresolved.** He owns only a basic keypad handset (Q10) — no camera, no video capability — yet by Q12 he is setting detailed conditions for a video teleconsultation (private room, trusted device that retains nothing, one escort) without the interview ever establishing whose device — the ASHA's, a facility tablet, or something else — would actually host that call. Verify with the ASHA or PHC before treating his Q12 conditions as implementable as stated.
- **The ASHA is simultaneously the threat and the chosen ally.** He blames her for screening him in the open and for the register, yet names her as the one person he would allow beside him on a video call. Verify whether this is genuine trust or resignation (“she already knows, so what is the use of saying no”), and whether he would accept a different trained escort instead.
- **Decision authority is claimed but shared.** He says “I decide” while describing his mother steering toward a faith healer and his wife toward the hospital. Verify how much delay the faith-healing route typically introduces before a referral is acted on, and whether the household is a legitimate intervention point at all given his refusal to disclose at home.
- **Wife-disclosure timing conflicts with contact tracing.** He insists he must tell her himself, only once he can say “it will be finished in this many months” — i.e. after treatment is confirmed. Standard practice requires early household examination. This is a live design conflict; verify the actual protocol with the district leprosy officer before assuming either side.
- **His drug-supply preference may conflict with programme practice.** He refuses monthly home delivery of MDT and wants anonymous pickup in town, “even if it is further”. Verify whether NLEP / district practice permits collection at an alternate facility or private pharmacy, whether unmarked packaging is feasible, and what the adherence trade-off is.
- **All cost and availability figures are self-reported.** ₹250–300 daily wage, ₹50 each way, ₹420–430 per attempt — verify against local rates. Critically, verify the “doctor absent twice” claim against PHC staffing and OPD records: if absence is systemic (camps, deputation, single-doctor posting), an appointment-confirmation feature may not be implementable as designed.
- **He assumes a contentless SMS is safe — it may not be.** “Your appointment is on Thursday at nine” satisfies him, but repeated messages from the same unfamiliar sender, read aloud by his son, may still form a pattern that invites questions. Verify by testing draft copy with three or four comparable patients, and check whether the SMS sender ID or header would itself carry a programme name.

- **“No name in any register” cannot be taken literally.** Case notification to district and national registries is a statutory and programme requirement. His actual stated need is narrower and achievable: no locally-accessible record may link his name to the diagnosis. Verify the minimum recordable dataset with the district authority and confirm that role-based access control — not non-recording — is the answer.
 - **Single respondent, non-representative.** This is one 41-year-old male head-of-household with a marriageable daughter, which shapes his entire fear profile. Verify with at least two further patient interviews — ideally a woman, an unmarried young adult, and an elderly patient — before treating any of the above as general requirements.
-

5. RECORDED CONFLICT: PROGRAMME’S NEED TO RECORD AND REPORT VERSUS THE BENEFICIARY’S NEED FOR NON-DISCLOSURE

Position 1 — the programme’s need to record and report. NLEP requires a line-listed record of every suspect and confirmed case (name, village, screening date, findings) that travels from the ASHA’s register up through block, district and national reporting; treatment follow-up requires monthly contact; case notification to statutory registries is a mandatory programme requirement (see §4); SMS or app-based reminders are the default, low-cost channel for confirmed appointments.

Position 2 — Ramesh’s need for non-disclosure. No locally-accessible record may put his name and any reference to the disease on the same line — not in the ASHA’s book that rides in her bag, not on a block-office list the peon reads over tea, not on a paper pinned to a wall. No patient-held card naming the condition. No mark, paint or number on the house. His daughter’s name nowhere near the record, in any form. Disclosure limited to the doctor first, the ASHA conditionally (she already knows, but her book must not sit open at the block office), and his wife only on his own timing. Any message to his phone must carry no identifying content — “Your appointment is on Thursday at nine,” nothing else. The diagnosis itself must be delivered in person, seated, by a human, never by paper, SMS, app notification or a third party.

His counter-proposal. Restrict access rather than restrict recording: let the record exist where the programme needs it, but keep it off anything a local person — ASHA’s notebook, block list, wall paper, patient-held card — can casually see. Replace monthly home delivery of MDT with an anonymous collection point in town, even if further away. Replace content-bearing SMS with a contentless appointment line or, preferably, a voice call that leaves nothing on the phone. Bundle any teleconsultation with a private, lockable room and a single trusted escort of his choosing.

Options to take forward:

- (a) Maintain full statutory line-listing at district/national level as legally required, but enforce strict role-based access control so that no locally-accessible register or list ever

displays his name and diagnosis together — this answers his “book that rides in her bag” fear without breaking mandatory reporting.

- (b) Adopt a contentless SMS as the default notification channel (“appointment on [day] at [time]” only), with a voice call offered as an alternative — no disease name, department name or programme name in any message that persists on the handset.
- (c) Explore an alternate MDT collection point, or private-pharmacy pickup, in place of mandatory monthly home delivery, subject to district/NLEP adherence-monitoring rules — verify feasibility before promising it to him or to any patient.
- (d) Where the household is a legitimate intervention point (for example, contact examination), let the patient control the timing and manner of disclosure to his own family rather than a field worker or the system disclosing on a fixed schedule, provided the delay does not exceed whatever limit the district leprosy officer sets for household contact examination.

Recommendation: adopt (a) and (b) as baseline design rules for all patients, not only Ramesh, since neither costs the programme anything to grant and both directly answer his stated fears. Treat (c) and (d) as verify-with-district-authority items — cross-reference the corresponding bullets in §4 — before committing to either in writing.

APPENDIX A — CONSOLIDATED REQUIREMENTS REGISTER (FROM THIS INTERVIEW ONLY)

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-70	Conduct any physical screening test (e.g. sensation testing) in a private, out-of-sightline setting, never in an open public space	Functional	Q3, Q6	Must have
FR-71	Let the patient know, in plain terms, where his screening record travels and who can read it	Functional	Q3	Should have
FR-72	Deliver a plain-language, private explanation of the condition — including that it can be cured — before or instead of a euphemistic “it’s nothing, but go quickly” message	Functional	Q4, Q7, Q11	Must have
FR-73	Issue a referral only against a confirmed, verified provider slot for a named day and time — never a generic instruction to “come to the PHC”	Functional	Q9, Q11	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-74	Offer a voice-call reminder, in Hindi or Halbi, as an alternative to any written message	Functional	Q10	Should have
FR-75	Default any SMS/app notification to a contentless appointment reminder — day and time only, no disease or department name	Functional	Q10, Q13	Must have
FR-76	Support remote teleconsultation so the patient's arm/lesion can be assessed without travelling to the PHC	Functional	Q11, Q12	Should have
FR-77	Offer an alternate, unmarked medicine collection point instead of mandatory monthly home delivery	Functional	Q11	Should have (verify)
FR-78	Deliver any diagnosis in person, seated, by a human with time to sit with the patient — never by paper, SMS, app notification or a third party calling from outside	Functional	Q13	Must have
FR-79	Let the patient control the timing and manner of disclosure to his own household, rather than the system or a field worker disclosing on a fixed schedule	Functional	Q5, Q11	Should have
FR-80	Let the patient nominate a single trusted escort (e.g. the ASHA) for a teleconsultation, and exclude named individuals and venues	Functional	Q12	Should have
FR-81	Flag a sensation-test result taken in a public, socially-pressured setting as provisional, pending re-test under private conditions	Functional	Q3	Should have
FR-82	Exclude any family member's name (e.g. "daughter of...") from a record that also carries the diagnosis	Functional	Q13	Must have
FR-83	Never issue a patient-held card, chit or paper that names the condition	Functional	Q13	Must have
FR-84	Never mark, paint or number the patient's house in connection with the diagnosis	Functional	Q13	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
NFR-37	Screening and consultation venues must be private — out of sightline of neighbours, not conducted at the doorstep or a public water point	Non-functional	Q3, Q6, Q12	Must have
NFR-38	Facility layout and signage must not make a specific room identifiable as the leprosy/skin service	Non-functional	Q6	Must have
NFR-39	Locally-resident facility staff contact with a given patient must not form a visible pattern (e.g. repeated visits to one house)	Non-functional	Q6, Q8	Must have
NFR-40	Any persisted notification (SMS, app) must carry no condition name, department name or programme name	Non-functional	Q10, Q13	Must have
NFR-41	Message comprehension must not depend on the patient's reading fluency — short, simple or spoken content only	Non-functional	Q10	Must have
NFR-42	Notification content must remain safe to be read by any other household member who happens to open it	Non-functional	Q10	Must have
NFR-43	Teleconsultation venue must be a private, lockable room, not a community building such as an anganwadi or panchayat bhavan	Non-functional	Q12	Must have
NFR-44	Any device used for teleconsultation must retain nothing of the call afterwards	Non-functional	Q12	Must have
NFR-45	Image and lighting quality for remote skin assessment must be sufficient for the intended clinical use (diagnostic vs triage-only)	Non-functional	Q12	Should have (verify)
NFR-46	No locally accessible record may display the patient's name and diagnosis on the same line or page	Non-functional	Q13	Must have
BR-22	Statutory case notification to district/national registries is mandatory; must be reconciled with	Business rule	Q13 (see §4, §5)	Verify

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	his “nobody in the village” request via role-based access, not non-recording			
BR-23	Existing MDT drug distribution model is monthly home delivery by a visiting health worker — identified by the stakeholder as the primary observable leak mechanism	Business rule	Q8, Q11	Constraint on design
BR-24	The ASHA’s field register/notebook and any block-office list currently function as physically accessible, unsecured records	Business rule	Q3, Q13	Constraint on design
BR-25	Sensation testing by the ASHA is the frontline field diagnostic step; its reliability when performed under social/public pressure has not been verified	Business rule	Q3	Verify
BR-26	The existing informal care pathway (bazaar chemist, faith healer) is consulted before any formal referral is acted on	Business rule	Q2	Constraint on design
CON-24	A round trip to the PHC costs ₹420–430 per attempt (₹100 return fare + ₹20–30 food + ₹300 forgone wage at his peak-season rate), doubled if his wife accompanies him or the doctor is absent	Constraint	Q2, Q6	Absolute
CON-25	Daily wage is ₹250–300; a lost work day is unrecoverable income with no compensation mechanism	Constraint	Q1	Absolute
CON-26	One-way journey mechanics: 20–25 minutes’ walk, up to an hour’s wait for a shared jeep, ~40 minutes’ ride	Constraint	Q2	Absolute
CON-27	Device available is a basic, shared keypad handset with no personal ownership and limited reading fluency	Constraint	Q10	Absolute
CON-28	The village is small enough that social observation of visits and movements is near-total	Constraint	Q1, Q6, Q8	Absolute
CON-	A single wasted trip (provider absent)	Constraint	Q9	Absolute

ID	Requirement / observation (stated as a need)	Type	Source	Priority
29	is functionally terminal to further engagement with this care pathway			
EC-13	Fear that being diagnosed collapses his daughter's pending marriage negotiation immediately, with stigma persisting for the household for a decade or more	Emotional/cognitive	Q5, Q8	Must design for
EC-14	Disease model is a single terrifying remembered case of late-stage disability; euphemistic reassurance is read as confirmation and increases fear	Emotional/cognitive	Q4, Q7	Must design for
EC-15	Absent explicit, credible curability counselling, avoidance is his rational choice	Emotional/cognitive	Q7	Must design for
EC-16	Anticipates involuntary emotional distancing within his own household even without formal disclosure	Emotional/cognitive	Q5	Must design for
EC-17	Trust in the ASHA is ambivalent — resented as a source of exposure, yet the only person accepted as an escort; may be resignation rather than genuine trust (see §4)	Emotional/cognitive	Q3, Q8, Q12	Must design for

Ramesh is not a system user; he never touches SparshCare directly. The functional requirements above (FR-70–FR-84) are therefore not actions he performs in the system — they are requirements placed on the system on his behalf, inferred by the analyst from his account of what would make a referral survivable and worth acting on.

Design rules implied, in the stakeholder's own framing

- “Somewhere with a door, sahib. That is all I am asking.” — a private, lockable venue is an absolute precondition for screening and teleconsultation, not a nice-to-have.
- “Let it say nothing ... Nothing else.” — a notification's silence is itself the requirement: day and time only, never the reason.
- “Is there. Somebody who knows ... telling me.” — a confirmed, human-verified appointment is what moves him; a hopeful instruction to “come and see” does not.

- “It must not come to my house every month in a man’s bag.” — the delivery channel is the leak; design the supply chain, not just the message, for invisibility.
- “A person, in a room, who has time to sit while I take it in.” — a diagnosis is delivered by a human, never by a message, a paper, or someone calling from a gate.
- “That much courage does not come twice.” — design for exactly one chance; a broken promise ends engagement with the pathway entirely.
- “If somebody said that to me plainly, and I believed them — that would be a different thing.” — plain-language curability counselling, said and believed, is the single highest-leverage feature in this entire interview.

STAKEHOLDER INTERVIEW LOG

SparshCare — Requirements elicitation · Interview 6 of 6

Field	Value
Analyst	Dr. Gunjan Singhal
Date of interview	17 August 2026
Stakeholder role	IT / System Administrator, District Health Society
Stakeholder (persona)	Priya — sole System Administrator for the district's field health systems; at least four years in the role (per her own account of one uninterrupted holiday in four years)
Location / facility	District Health Department IT unit, Bastar, Chhattisgarh
Project	SparshCare — leprosy screening, teleconsultation & referral system
AI tool used	Claude (Cowork) — model claude-opus-5, role-play persona "Priya, IT / System Administrator"
Interview mode	Semi-structured, single session, 13 questions with follow-up probes
Duration (in-persona)	~50 min
Chat link / screenshot ref	SPARSHCARE / INT-06 / PRIYA-SYSADMIN — [paste chat link or screenshot filename]

1. PURPOSE AND INTERVIEW APPROACH

Objective: to learn, from the one person who will actually operate SparshCare's infrastructure day to day, whether the system as conceived can be run in this district at all — specifically: whether offline sync can be trusted to preserve data under a deterministic and accountable conflict rule; whether access control and audit can answer, on demand, who saw a given patient's leprosy status and when; what support capacity a rollout of this scale genuinely requires against a team of 1.5 people; how the system should behave when an external dependency (SMS gateway, video platform, identity service) fails; and, in her own words, what she would refuse to put into production.

Question flow used (deliberate sequence — rapport → current process → evidence → pain → constraints → conflict → solution ideas last):

- Q1 Scoping and rapport: new build versus documentation, operational versus data/security scope, her two "decide-early-or-sink" items, and a stated red line.
- Q2 Role and baseline: systems she runs, user and device volumes, team size, and the administrator-to-user ratio that governs every later design choice.

- Q3–Q4 Evidence from a prior rollout: what broke first after go-live, and what the resulting support calls actually sound like.
- Q5 Device and identity lifecycle: handset changes, attrition, revocation and caseload handover.
- Q6 Data-quality evidence and role boundary: a duplicate-record incident, and who is qualified to resolve a clinical conflict.
- Q7–Q9 Access control and audit, in depth: who can see what today, how quickly access is revoked, whether “who viewed this record” can be answered, and what a leak of this specific condition would mean.
- Q10 Integration failure behaviour: the SMS gateway, the video platform, the identity service, and downstream data consumers.
- Q11 Support capacity and after-hours failure: what happens at nine in the evening in a village, and the workarounds that unavailability produces.
- Q12 Deliberate conflict probe: security and audit controls versus field-worker speed.
- Q13 Solution ideas last: must-haves to operate the system, refusal conditions, and what she would want tested before go-live.

Tag key: Each captured line is marked with one or more of five categories — **[Functional]** (something the system must do) · **[Non-functional]** (a quality it must have) · **[Business rule]** (an existing organisational rule) · **[Constraint]** (a fixed environmental, economic or regulatory limit) · **[Emotional/cognitive]** (fear, trust, stigma, mental load). Each answer additionally carries a roll-up tag and is marked *need* (a problem to be solved) or *solution* (a proposed answer, to be traced back to the underlying need). In Appendix A these categories are carried forward as ID prefixes: FR = functional, NFR = non-functional, BR = business rule, CON = constraint, EC = emotional/cognitive.

2. QUESTION-BY-QUESTION LOG

Q1: Scoping — is this a new build or documentation of an existing system, and is the focus operational or data?

A1:

- New build; both operational and data/security aspects in scope. **[Business rule]**
- Priya raised two “decide-early-or-sink” items unprompted: (a) offline-first architecture is mandatory, not optional — field staff work hours without usable data; (b) confidentiality of leprosy status is a legal and ethical absolute, because disclosure causes social destruction, not just a compliance incident. **[Non-functional] [Constraint] [Business rule]**

- Stated red line: no design in which a record can silently lose a field with no way to recover the previous value. **[Non-functional] [Business rule]**
- She asked in return for (i) the list of user classes and volumes before any permissions discussion, and (ii) an early decision on whether the system must integrate with the state health stack (ABHA / ABDM / FHIR / national leprosy programme reporting). **[Business rule] [Constraint]**

→ Tags: **[Constraint] [Non-functional] [Business rule]** | *need — scope framing*

Q2: Tell me about your role — which systems do you run for the district health department, and who do you support?

→ Probe: *How many field users, and how many people on your team?*

A2:

- Title: System Administrator, District Health Society. In practice the only technical person between the field and any server. **[Constraint]**
- Systems run: state HMIS reporting layer; NIKSHAY (TB); immunisation and maternal health registers on the state platform; a legacy locally-hosted district hospital patient database (vendor-maintained, responsive only near contract renewal). **[Constraint]**
- Also owns the “plumbing” nobody counts as a system until it fails: SMS gateway account, Zoom licences for the teleconsultation pilot, device provisioning for ~600 field tablets/phones, and the backup routine. **[Constraint] [Non-functional]**
- ~1,100 ASHAs on the roll; ~700 actively using any digital tool in a given month. ~90 ANMs. ~40 medical officers across PHCs/CHCs. ~12 block programme managers. ~8 district-tier users (programme officers, DHO office, statistical assistant). **[Constraint]**
- ASHAs are the dominant load: mostly personal (not department) low-cost Android phones, low storage, shared with family; many functionally low-literacy in English, basic Hindi reading; frequent handset changes (sold, broken, stolen, handed to a migrating family member). **[Constraint] [Non-functional]**
- Team size: one. Plus a non-technical data entry operator at the district office. “Block IT coordinators” exist on paper in 14 blocks; realistically ~3 blocks have someone able to reset a password without calling her. **[Constraint]**
- Her framing: ~1,100 users to 1.5 administrators is the constraint that governs every design choice — any security feature that generates support calls must be judged against that ratio. **[Constraint] [Non-functional] [Emotional/cognitive]**

→ Tags: **[Constraint] [Non-functional] [Emotional/cognitive]** | *need — capacity/context constraint*

Q3: Walk me through the last field app rollout you were part of. What happened after it went live?

→ Probe: What broke first? And what did you end up doing about it?

A3:

- State-mandated, vendor-built nutrition and growth-monitoring app, ~2.5 years ago; district was a pilot. ~400 ASHAs trained in 3-hour block sessions; week-one adoption ~70% of trained users — leadership considered it a success. **[Business rule]**
- First failure (~day 10): authentication. Vendor-generated username plus a password set at training; no self-service reset, no working OTP fallback (SMS silently failed in low-connectivity blocks). 60–80 reset requests per week; Priya spent roughly two hours a day for four months on resets. **[Functional] [Non-functional]**
- Second failure (worse, slower to surface): optimistic sync with server-side “most recent timestamp wins”, merging at record level rather than field level. Device clocks were wrong or manually altered, so “most recent” was fiction; an ANM editing one field overwrote the whole record, discarding offline entries made by the ASHA the same morning. **[Functional] [Non-functional] [Business rule]**
- Undetected for ~7 weeks. Discovered only when a block programme manager questioned an implausible drop in her severe-acute-malnutrition counts. Estimated a few thousand growth measurements lost, unrecoverable — no version history, and the client cleared local data after a “successful” sync. **[Functional] [Non-functional]**
- Remediation: vendor added supervisor-mediated reset so ANMs could reset their own ASHAs — cut Priya’s reset volume by ~70% overnight; she rates this the single highest-value change made. Sync fix took ~11 months and a version upgrade; in the interim she ran a nightly export to a separate store as a shadow backup because she did not trust the primary system. **[Functional] [Non-functional] [Emotional/cognitive]**
- Lasting damage was cultural, not technical: field staff learned that “the app loses your work.” Two years on, some ASHAs still run parallel paper records. Her conclusion — trust is far more expensive to rebuild than to keep. **[Emotional/cognitive] [Non-functional]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Emotional/cognitive]** | *need*
— *evidenced by prior failure*

Q4: What do the support calls actually sound like? What do field workers call you about, and how do they reach you?

→ Probe: How long does one password reset take, start to finish? What is that worker doing in the meantime?

A4:

- No helpdesk exists. Contact is via her personal WhatsApp number, circulated in training slides several rollouts ago and now propagated to hundreds of contact lists. Some requests route via block coordinators or ANMs; some arrive as paper slips at the district office. **[Constraint]**
- Reports are not diagnostic. Typical message: “Didi, it’s not opening.” Often a blurred screenshot of the wrong screen; frequently a voice note (which she prefers — more information than any form field). Users do not distinguish app failure, no network, expired data pack, and full device storage; all four arrive as the same sentence. **[Non-functional] [Constraint] [Emotional/cognitive]**
- Approximate mix during the nutrition app period: ~50% login/password; ~20% phone changed or app disappeared; 10–15% “I entered the data and now it’s gone”; remainder genuine bugs, storage, connectivity, OS auto-update breaking permissions. **[Non-functional] [Constraint]**
- Peak support load is 21:00–23:00 — ASHAs do household visits by day and data entry at night after domestic work. There is nobody on duty at that hour but her. **[Constraint]**
- Technical time: ~2 minutes. The slow part is identifying the user — e.g. 31 users named “Sunita”, requiring block/sub-centre questions or an ID card photo. **[Functional] [Non-functional]**
- Credential delivery is itself a control failure: the new password is typed into WhatsApp and then lives permanently in a chat log on a family-shared phone. She asked explicitly for this to be designed out. **[Non-functional] [Constraint] [Emotional/cognitive]**
- Real elapsed time: median ~1.5 days from first message to being back in the app — because she is handling fifteen other things, and because a second round trip is common (mistyped password, re-lockout). **[Non-functional] [Constraint]**
- Meanwhile the worker does not stop working; she does her visits on paper. What stops is the record of the work. She then faces two days of backlog entry from paper slips, at night, in one sitting — precisely the condition in which data quality collapses (approximate dates, half-remembered detail, some entries never made). Where payment or rating depends on entries, there is an incentive to enter something rather than nothing. **[Functional] [Non-functional] [Emotional/cognitive]**
- Her framing: the cost of a password reset is not two minutes of admin time — it is 1.5 days of degraded data plus an incremental loss of trust. **[Non-functional] [Emotional/cognitive]**

→ Tags: **[Functional] [Non-functional] [Constraint] [Emotional/cognitive]** | *need — with one solution offered (supervisor-mediated / device-bound login)*

Q5: What happens when a worker changes her phone, or leaves the job and hands it over?

→ Probe: What happens to the data still sitting on the old handset?

A5:

- Not an edge case: ~15–20% of active field users change handset per year; ASHA attrition ~10% annually, higher in some blocks. Roughly 20–30 device or identity events per month, handled by one administrator. **[Constraint]**
- Current practice is unsystematic: she resets the credential, the worker installs on the new phone, and the old handset is simply left as-is. No remote wipe, no session revocation, no device registry. A stolen device cannot be invalidated; in the nutrition app the session token remained valid for 90 days. **[Functional] [Non-functional] [Constraint]**
- Two things persist on the old device. (a) The offline local database — a copy of real patient records for her whole catchment (roughly 200–1,000 households depending on block), which for SparshCare would include leprosy status, treatment stage and contact-tracing links. (b) The unsynced queue — entries that never reached the server. **[Constraint] [Non-functional]**
- Direct design tension flagged: security demands immediate wipe of the old device; data integrity forbids wiping before the queue is confirmed drained, because the old handset may hold the only copy of some records. Someone must decide in writing which wins and under what conditions. **[Non-functional] [Business rule]**
- Handover is the harder case. Operationally the caseload must transfer to the incoming ASHA (who is on treatment, who is overdue). But that is a bulk transfer of sensitive records to a person the patient has never met and never consented to. Today it happens informally via a paper register; digitised it becomes a much larger, faster disclosure with no human in the loop. She wants MO authorisation for caseload reassignment, logged as a disclosure event rather than a routine admin action. **[Functional] [Business rule] [Emotional/cognitive]**
- Common undocumented case: the worker changes phone without telling her, so two devices hold the account. Credential sharing is also normal field behaviour (given to an ANM or neighbour to “enter the data for me”), which sets a hard ceiling on what any audit trail can prove — a login does not reliably equal a person, and she wants the design to acknowledge that rather than pretend otherwise. **[Constraint] [Business rule] [Emotional/cognitive]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | *need + explicit unresolved conflict*

Q6: Tell me about a time data came back from the field wrong, or twice, or didn't come back at all. How did you find out?

→ Probe: How did you decide which version was correct? Who made that call — you, or a clinician?

A6:

- Immunisation register, ~18 months ago: duplicate child records — same name, same mother's name, same village, two system IDs, each with a partial dose history. A fuzzy match across the block found ~1,400 probable duplicate pairs against a base of ~40,000 children. **[Functional] [Constraint]**
- Cause: no reliable identifier. Aadhaar inconsistently captured for infants; phone numbers reused and shared across households; names spelled multiple ways with inconsistent Devanagari/Roman transliteration. When search failed, the app made creating a new record easier than finding the existing one — so users created. **[Functional] [Non-functional] [Business rule]**
- Compounding harm: both records then accumulated real clinical data (one dose recorded under record A at the sub-centre, the next under record B at an outreach camp). Neither record is complete and neither is wrong — a real history split in half, not junk to be deleted. **[Functional]**
- Detection: not by the system. No duplicate detection, no alert, no dashboard indicator. It surfaced through denominator inflation — the fully-immunised percentage fell while ANMs reported more sessions, and one block programme manager challenged her own numbers. Her general observation: in her entire career, software has never once reported a data integrity problem to her; a human questioning an implausible figure has been the only detection mechanism. **[Functional] [Non-functional] [Emotional/cognitive]**
- She owned the mechanical part: candidate pair matching, whether two records were the same child, and the merge script — with nothing deleted, only superseded, and both originals retained. **[Functional]**
- She explicitly refused to decide which dose history was clinically true, because choosing wrong means either an under-protected child or a duplicate dose. Reconciliation lists (both versions plus the conflict) went to each block's medical officer, who decided — sometimes by sending the ANM to check the mother's own immunisation card, often the most reliable record available. Took months; some pairs remain unreconciled. **[Functional] [Business rule] [Emotional/cognitive]**
- Boundary stated for SparshCare: she is accountable for data being intact, traceable and mergeable; she is not qualified to decide which clinical fact is true. A conflict over treatment stage or classification must go to the MO or leprosy programme specialist. Her requirement is only that the system present a clean conflict with both versions and provenance, and never silently pick one. **[Functional] [Business rule]**

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Emotional/cognitive]** | *need*
— *plus explicit role boundary (defer to MO)*

Q7: Across the systems you run today, who can see what? How is that decided, and who approves it?

→ *Probe: How quickly can you take someone's access away when they leave, and how would you know they'd left?*

A7:

- Access is largely as configured years ago and never revisited; the model differs per system, and the inconsistency is itself a problem. **[Business rule]** **[Constraint]**
- State HMIS: role templates defined by the state, not modifiable locally. A broad “district user” role grants near-total district visibility and is held by ~11 people, at least two of whom need only the monthly report export. Granularity does not exist, so users receive one needed permission plus forty others. **[Business rule]** **[Constraint]**
- Hospital database: no least-privilege concept. Shared logins (“reception”, “lab”) used by whoever is on shift, so the audit log identifies a terminal, not a person. Raised in writing three times; vendor says the fix requires an unfunded version upgrade. Her assessment: that log would not survive a serious investigation. **[Non-functional]** **[Business rule]** **[Constraint]**
- Where she has control, geographic scoping works (ASHA sees her catchment; ANM her sub-centre; block manager her block). The vertical dimension does not: within scope, users see the whole record, because no field has ever been classified by sensitivity. There is no concept of “you may see that this person needs a visit but not the diagnosis” — which is precisely the distinction SparshCare depends on. **[Functional]** **[Non-functional]** **[Business rule]**
- Approval in principle is by DHO signature; in practice a block manager sends a WhatsApp message and she creates the account, with paperwork following sometimes. Her stated reason is asymmetric pressure — refusing leaves a health worker idle for two weeks, and no one has ever been reprimanded for granting access too quickly. There is no periodic access review; she has never been asked to produce a list of who has access to what. **[Business rule]** **[Constraint]** **[Emotional/cognitive]**
- Technical revocation takes under a minute. Median real gap between departure and revocation is months; some accounts have never been revoked — she believes an audit today would find active credentials for people who left two or three years ago. **[Functional]** **[Non-functional]** **[Constraint]**
- Root cause: no offboarding process routes through IT. HR processes a transfer or resignation without notifying the administrator; she learns incidentally (e.g. a meeting remark in August about a doctor who moved in March). For ASHAs, leaving is often not

an event at all — activity simply tapers, and “inactive” is indistinguishable from “busy with harvest” for the first two months. **[Business rule]** **[Constraint]**

- Her workaround: flag accounts with no activity for 90 days. Blunt in both directions — catches genuine leavers late, and disables ASHAs who were ill or had a broken phone, who then call her. **[Functional]** **[Non-functional]** **[Emotional/cognitive]**
- Two requirements she wants for SparshCare: access grants that expire by default (tied to a posting, with a review date, so the system forces the question), and the ability to answer “who has looked at this patient’s record” in under five minutes without writing a query. **[Functional]** **[Non-functional]**
- She notes a limit on IT’s power here: revocation tooling is near-worthless unless offboarding notification becomes a mandated process, which requires authority above her level. **[Business rule]** **[Constraint]**

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Constraint]**

[Emotional/cognitive] | *need — with two solutions offered (expiring grants, access query)*

Q8: If I asked you tomorrow, “who has viewed this particular person’s record?” — could you answer me?

→ Probe: What would it take? And what happens to you if you can’t answer?

A8:

- No — not in any form that would hold up where it mattered. **[Non-functional]** **[Emotional/cognitive]**
- Writes are partially reconstructable: created_by / updated_by / timestamps exist in the data. But most systems overwrite rather than append, so she could name who last modified a record, not everyone who ever did. **[Functional]** **[Non-functional]**
- Reads cannot be answered at all. Nothing she runs logs a view. Someone opening a record, reading the treatment history and closing it leaves no trace — and reads are what matter for a confidentiality breach, since leaks happen by looking, not by editing. **[Non-functional]** **[Constraint]**
- Best available fallback is web server access logs: retention is 30 days on some systems, 7 on others, and possibly nil on one. They would yield an IP, a timestamp and a URL — not a reliable human identity (shared logins), not whether the record was actually read, and nothing older than a month. **[Non-functional]** **[Constraint]**
- Requirements she specified to make the answer possible: append-only access logging that captures reads as well as writes, recording user, device, record, fields returned and timestamp; retention in years rather than days, because disclosure allegations surface long after the event; storage separate from the application database so compromising

the app cannot alter the trail; and a query interface usable by a non-engineer, because a control that needs her to write SQL at 23:00 will not be exercised in time. **[Functional]**
[Non-functional]

- Accountability: there is no data protection officer and no privacy team in the district. Formally the department is accountable; practically, any question (“what do the logs show?”) lands on her as the holder of the admin password. She assesses that the department could not currently meet its breach-reporting obligation under the DPDP Act, because the scope of a breach that cannot be detected cannot be reported. She has stated this in writing once; it produced no budget line. **[Business rule]** **[Constraint]**
[Emotional/cognitive]
- Her stated personal concern is not the penalty but the human outcome — having to tell an affected family that she does not know how the disclosure happened. This is why she pressed audit trails over features throughout the session. **[Emotional/cognitive]**
- Hard requirement: “who has accessed this record” must be a supported query in version one. Read logging cannot be retrofitted, because history not captured cannot be recovered. **[Functional]** **[Non-functional]** **[Business rule]**

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Constraint]**
[Emotional/cognitive] | *need — stated as a non-negotiable*

Q9: For this particular condition, what would a leak actually mean — for the department, and for you?

→ Probe: *What are you obliged to do if it happens?*

A9:

- Clinical framing deferred to the MO, but noted: leprosy is curable, treatment is free, and infectiousness ends quickly on multidrug therapy — none of which has affected the social reality in these blocks. Stigma is intact. **[Emotional/cognitive]** **[Business rule]**
- Consequences for the individual: collapse of marriage prospects or pressure on an existing marriage; social exclusion at shops and in the village; children excluded from play; loss of work, particularly domestic or food-handling work, without the employer stating a reason; segregation in eating and washing within her own home. Possible family-law implications — she flagged this as a question for a lawyer, not for her. **[Emotional/cognitive]** **[Constraint]**
- Programme-level second-order effect, which she considers the decisive one: if people believe diagnosis becomes public, they stop presenting. Late presentation with established disability is exactly the outcome the programme exists to prevent. A single leak suppresses case-finding across a district for years, because the calculation people make is rational. The only thing offered in exchange for disclosure is a promise of

confidentiality; if that promise breaks publicly, nothing remains to offer.

[Emotional/cognitive] [Business rule] [Constraint]

- Consequences for the department: media cycle, inquiry, probable suspension of the digital programme pending investigation (i.e. reversion to paper and the loss of two years), and financial penalty as data fiduciary under the DPDP Act, which would be significant against a district health budget. She observes that institutional consequence tends to land on whoever is closest to the machine. **[Constraint] [Emotional/cognitive]**
- Her working understanding (which she asked to be confirmed by someone with legal standing): notify the Data Protection Board and the affected individuals under the DPDP Act, with no materiality threshold allowing her to sit on a small incident, and tight timelines; separately, departmental reporting to the state; and, if it is a cyber incident, CERT-In reporting on a six-hour clock. **[Business rule] [Constraint]**
- Her operational objection: every one of those obligations requires stating scope — which individuals, which fields, over what period. Notification is not abstract; it means someone physically travels to a village and tells a woman her leprosy status may have been exposed. With current logging she could not produce that list, so she would be choosing between over-notifying (causing large harm needlessly) and under-notifying (leaving people exposed), with no evidence either way. **[Non-functional] [Constraint] [Emotional/cognitive]**
- Condition-specific hazard she raised: the breach notification is itself a disclosure event. An SMS to a shared family phone, or an ASHA delivering it verbally in a courtyard, causes the exact harm being reported. A designed notification channel must exist before an incident, not be improvised during one. **[Functional] [Emotional/cognitive]**
- Her summary position: the audit log is not a compliance checkbox — it is the instrument that determines whether a bad day is contained or catastrophic; the difference between “43 records were viewed by this account on these dates, here they are” and “we don’t know.” **[Non-functional] [Emotional/cognitive]**
- Two governance gaps she asked to be closed in this build: an incident response runbook (who decides, who notifies, through what channel), and a named person with legal authority to make the notification call at speed — a six-hour clock and a district that needs three days to convene a meeting are incompatible. **[Business rule] [Constraint]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | *need — risk and obligation framing*

Q10: Tell me about a time an integration failed on you — a message gateway, an identity service, a video platform. How did you find out, and what did the users see?

→ Probe: *What should they have seen instead? And which other systems will expect data out of this one?*

A10:

- SMS gateway (state-empanelled aggregator): ~8 months ago delivery began failing for one carrier's subscribers — roughly a third of volume — silently. The API returned 200 and "submitted" for every message; delivery receipts were not being consumed. Found 11 days later when an ANM mentioned poor turnout at an outreach session. Users saw nothing: the patient does not know a message was due, and the worker assumes it arrived. Dashboard was green throughout. **[Functional] [Non-functional] [Constraint]**
- Zoom (teleconsultation pilot): visible failure — spinner, then "unable to connect", on a patient's phone in a PHC waiting room during a booked MO slot. Causes mixed: genuine bandwidth, and scheduling logic that never checked host availability against Zoom, so two consultations were booked into one host account. The ANM told patients the system was broken, which was accurate. Several patients had travelled an hour; two did not return. **[Functional] [Non-functional] [Emotional/cognitive]**
- ABHA: not an outage but an availability assumption. Verification depends on an OTP to a linked mobile, which in these blocks is frequently wrong, the husband's, or long disused. The app treated failed verification as a hard stop, so registration failed with no fallback path; workers, trained that ABHA was mandatory, turned patients away. Her characterisation: not an integration failure but a design that assumed the happy path was the only path. **[Functional] [Non-functional] [Business rule]**
- SMS: the worker should see delivery state rather than send state — "delivered" / "not delivered, call her" — with undelivered messages surfacing as a work item; and the administrator should be alerted when any carrier's delivery rate falls below baseline. **[Functional] [Non-functional]**
- Zoom: never offer a slot the system cannot honour (real availability checking); on failure at point of use, present a defined in-app fallback — reschedule, voice call, or flag for physical referral — and record the failure, because a consultation that did not happen is a clinical event, not merely a technical one. **[Functional] [Non-functional]**
- ABHA: must be optional at capture. Create the patient on a local system identifier, mark the record unverified, allow linkage later. She named this her strongest position of the session: never block care on an identity service. **[Functional] [Business rule] [Emotional/cognitive]**
- General architectural principle requested: every external dependency has a defined degraded mode; the degraded mode is visible to the user rather than silent; and no external dependency may block a clinical action. With Zoom, SMS and ABHA all unavailable, an ASHA must still be able to record a visit and a dose given. **[Functional] [Non-functional]**
- Downstream consumers she expects (and warns are usually discovered late): national leprosy programme reporting (case notifications, treatment cohorts — currently

compiled by hand); state HMIS aggregate indicators; district review dashboards; and, if ABHA is in scope, FHIR-shaped records pushed to the ABDM health information exchange. **[Business rule] [Constraint]**

- Her caution on ABDM: “interoperable” and “confidential” are in tension — once a leprosy diagnosis enters a national exchange, her role-based controls end at the boundary and she cannot audit what happens beyond it. **[Non-functional] [Constraint] [Emotional/cognitive]**
- The consumer nobody lists: disability and welfare schemes requesting extracts for pension or certificate eligibility. Today such a request arrives as a phone call from a tehsil office and would be met with an ad-hoc spreadsheet export — an uncontrolled disclosure pathway that must be either designed as a formal, consented, logged interface or explicitly refused. **[Functional] [Business rule] [Constraint]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | *need + solutions offered (delivery receipts, degraded modes)*

Q11: When something goes wrong at nine in the evening in a village, how does anyone come to know, and who fixes it?

→ Probe: *What are your working hours, realistically? What have people done when you weren't reachable?*

A11:

- Nobody comes to know. Server-level outage would page her via a basic uptime check, but the failures that actually affect field users are individual — her login, her storage, her sync queue — and are invisible from the centre. There is no client-side error reporting; if the worker does not message, the event does not exist. **[Functional] [Non-functional] [Constraint]**
- Discovery therefore depends on a woman with a low-cost phone deciding whether to disturb the administrator at 21:00. The confident ones message; the quiet ones stop using the app. She cannot distinguish “this block has low usage because they are busy” from “something broke for all of them in June” — an asymmetry she believes has cost more than any outage. **[Non-functional] [Emotional/cognitive]**
- Official hours 10:00–17:00, six days. Actual: answering WhatsApp ~07:00–23:00, because that is when field staff work. No on-call rota — there is nobody to rotate with. One uninterrupted holiday in four years, during which she still checked messages. When genuinely unreachable, nothing is handled; messages accumulate and are worked oldest-last, because recent ones are loudest. **[Constraint] [Emotional/cognitive]**
- Workaround 1 — revert to paper, and many never fully return. (The benign case.) **[Business rule] [Constraint]**

- Workaround 2 — credential sharing: using a colleague's or the ANM's login because the work must be recorded against a real deadline. This is the most common workaround and it destroys the evidentiary value of the audit log; forty entries from one account across three villages cannot be distinguished from fraud versus four women helping each other. **[Non-functional] [Business rule] [Emotional/cognitive]**
- Workaround 3 — photographing records and sending them on WhatsApp ("it's not saving, here's the data, please enter it"). She has received patient information this way many times. For SparshCare this means leprosy status sitting in WhatsApp backups on shared family handsets — current routine practice, not a hypothetical vector. **[Non-functional] [Constraint] [Emotional/cognitive]**
- Workaround 4 — delegating entry to a family member (typically a son who is better with phones), who then sees everything the worker sees. **[Non-functional] [Constraint] [Emotional/cognitive]**
- Her causal argument: none of this is carelessness. Each workaround is a rational response to a system that failed while the obligation to record remained. The workaround is the symptom; unavailability is the disease. A system that fails at 21:00 with no recourse is a system that generates WhatsApp photographs of leprosy diagnoses, whatever the security document says. **[Emotional/cognitive] [Non-functional]**
- Consequently she frames her support requirements as security requirements: self-service or supervisor-mediated recovery so no administrator is needed at night; queue-and-retry so there is no lost work to photograph; and client-side error reporting so failures reach her without depending on the user complaining. **[Functional] [Non-functional]**
- Two asks raised: budgeted first-line support capacity (at least one trained person per block for resets and triage), and acceptance of client-side telemetry — crashes, failed syncs, stuck queues — acknowledging it is another data flow to secure, and inviting pushback now rather than in a later privacy review. **[Functional] [Constraint]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | *need — root cause of insecure workarounds*

Q12: Tension: Sunita covers forty houses a day on a shared phone and finds logging in painful — but you need to prove who recorded what and who saw it. How do you see that balance?

→ Probe: *What's the least friction you could accept and still stand behind the audit trail?*

A12:

- Her position has changed over four years: better security education is not the answer. If a control is painful the worker will defeat it, and will be right to — her obligation is to the forty households, not to the audit log. Every friction-heavy control Priya has

imposed has been worked around within a month. **[Non-functional]**
[Emotional/cognitive]

- The design constraint is therefore not “how much friction will she tolerate” but “what control has she no reason to circumvent.” Anything else is theatre that ends up quoted in an incident report. **[Non-functional]** **[Business rule]**
- She will not trade away the audit trail — not for compliance reasons, but because if a read cannot be attributed, the affected woman is offered nothing but an apology. She argues the two needs are only opposed if authentication is built badly. **[Non-functional]** **[Emotional/cognitive]**
- Proposed model: separate proving the device from proving the person. Bind the account to a registered device (enrolled once with ANM or MO in the loop, cryptographic key on the handset) — silent, and free after day one. Add a short local PIN chosen by the user, or biometric where supported, recoverable by her ANM rather than by the administrator. **[Functional]** **[Non-functional]**
- Session length should reflect real risk, not an office policy template: unlock once per working session, remain unlocked for a few hours of active use, with a short idle timeout (~2 minutes) so a handset left on a charger in a shared room locks itself. The threat is a family member picking up the phone, not a sophisticated attacker. **[Non-functional]** **[Constraint]**
- Step-up authentication should be attached to sensitive data, not to every screen: recording a visit or a dose requires no extra challenge; opening the leprosy status field, exporting data, or viewing a patient outside the worker’s catchment re-prompts and is logged prominently. Friction then lands a handful of times a day rather than forty. **[Functional]** **[Non-functional]**
- Her stated floor: one PIN entry at session start, plus re-prompt on sensitive access, plus a short idle lock. Below that she would not sign off, because she would be endorsing an audit trail she does not believe. **[Non-functional]** **[Emotional/cognitive]**
- Caveat 1 — device binding shifts rather than dissolves the attribution problem: if the phone and PIN are handed to a neighbour, the log says Sunita and is wrong. Mitigation is anomaly flagging (e.g. forty records from one account across three villages in an hour) treated as signal, not proof. She insists the log’s honest claim is “this device, unlocked with this PIN, at this time” — strong evidence, not certainty — and that the system’s own language should be accurate about that. **[Non-functional]** **[Constraint]** **[Emotional/cognitive]**
- Caveat 2 — the shared phone is the real vulnerability and no authentication design fixes it, because the local database sits on a family-used handset. She therefore wants the local store encrypted with a key derived from the user credential, and a hard cap on how much data resides on the device (active caseload only, not block history). “Least

data on the most exposed surface” — the point she says she would fight for if budget forced a choice. **[Non-functional]** **[Constraint]** **[Emotional/cognitive]**

- Two decisions she wants owned: whether the build supports sensitivity as a property of the data field rather than the screen (schema implications, very hard to add later), and who sets session duration — warning that a generic security template will impose a 15-minute timeout that locks a worker out mid-courtyard. **[Non-functional]** **[Business rule]**

→ Tags: **[Functional]** **[Non-functional]** **[Business rule]** **[Constraint]**

[Emotional/cognitive] | *solution proposed — validate against security review and with ASHAs*

Q13: What must this system do for you to run it properly — and what would make you refuse to put it into production at all?

→ Probe: *What would you want to see tested before go-live?*

A13:

Must-haves to operate it:

- Offline-first with field-level merge and full change history; nothing overwritten silently, nothing deleted locally until server acknowledgement, every version recoverable. **[Functional]** **[Non-functional]**
- Append-only read logging, stored separately from the application database, retained for years, queryable by a non-engineer. **[Functional]** **[Non-functional]**
- Device registration as a first-class object — visible enrolled devices, revocable, with a defined behaviour for a revoked device holding an undrained queue. **[Functional]** **[Non-functional]**
- Field-level sensitivity in the schema, so leprosy status is protected independently of which screen displays it. **[Functional]** **[Non-functional]**
- Encryption at rest on the handset keyed to the user credential, plus a hard cap on caseload data held on the device. **[Non-functional]**
- Supervisor-mediated credential recovery, so a reset never requires the district administrator at night. **[Functional]** **[Non-functional]**
- Client-side error and sync telemetry, so failures reach IT without depending on a user complaint. **[Functional]** **[Non-functional]**
- Defined, user-visible degraded modes for every external dependency, with no clinical action blocked by one. **[Functional]** **[Non-functional]**
- Access grants that expire by default. **[Functional]** **[Business rule]**

Refusal conditions (would object in writing and decline to run):

1. ABHA verification as a hard requirement for patient registration — she will not run a system that turns people away from leprosy treatment because an OTP did not arrive. **[Business rule] [Emotional/cognitive]**
 2. No read audit trail — accepting personal accountability for an obligation with no instrument to meet it. **[Non-functional] [Emotional/cognitive]**
 3. Sensitive fields visible by default to every role, with confidentiality handled by hiding a screen rather than by the data model. “Presentation-layer confidentiality is not confidentiality.” **[Non-functional] [Business rule]**
 4. Sync that can lose a write — record-level last-write-wins with no history. She has watched that destroy several thousand measurements and will not sign for it twice. **[Non-functional] [Emotional/cognitive]**
- Flagged but not a refusal: if no support capacity is budgeted beyond her, she will state plainly that the department is accepting a single point of failure who has a family and occasionally falls ill. **[Constraint] [Emotional/cognitive]**
 - Not a demo — a demo proves nothing, since everything works with two records and full signal in a conference room. **[Non-functional]**
 - Conflict testing with real scenarios: two devices offline editing the same patient, one with an incorrect system clock, reconnecting in both orders — inspecting the resulting record and its history; plus a case where they edit different fields and both survive. **[Functional] [Non-functional]**
 - Airplane-mode endurance: a full working day offline across 40+ households, then sync. Repeated with the app force-killed mid-sync, and with the device running out of storage mid-sync — the last being where field devices actually live. **[Non-functional] [Constraint]**
 - Every integration tested in its failure state, not only its success state: SMS gateway returning success without delivering; Zoom unreachable; ABHA timing out. She wants to see what the field worker sees and what the recovery path is. **[Functional] [Non-functional]**
 - Adversarial permissions testing — deliberate attempts to reach a record outside scope by manipulating an ID in a request rather than by navigating the UI; authorisation enforced server-side in all cases. **[Non-functional]**
 - An audit-log dry run, which she named as the test she cares most about: pick a patient at random and let her answer “who has accessed this record and when” cold, in front of the analyst, within five minutes. If it cannot be done in a controlled exercise it cannot be done during an incident. **[Non-functional] [Emotional/cognitive]**

- Load testing on a real device — the cheapest Android in circulation among the district's ASHAs, not a developer handset. **[Non-functional] [Constraint]**
- Closing request: the requirements document must carry the operating ratio through — 1,100 users, one administrator, no on-call. Specifications tend to define the system and assume the operator; a design that needs a competent administrator awake at 21:00 does not work in this district. **[Constraint] [Emotional/cognitive]**
- Closing boundary restated: clinical questions — classification, treatment stages, what a safe reconciliation looks like when two records disagree on diagnosis — go to the MO or the leprosy programme specialist, not to her. **[Business rule]**

→ Tags: **[Functional] [Non-functional] [Business rule] [Constraint]**
[Emotional/cognitive] | *need + acceptance criteria (test plan)*

3. TOP 3 PAINS I HEARD

1. Silent sync loss with no accountable arbiter for the conflict

Record-level “most recent timestamp wins” merging, run against unreliable device clocks, silently discarded offline entries in the last rollout — undetected for seven weeks, a few thousand growth measurements unrecoverable because there was no version history and the client cleared local data after a “successful” sync. It was found only because a block manager questioned an implausible number, never by the software. The lasting damage was cultural: field staff learned “the app loses your work”, and two years on some ASHAs still keep parallel paper records. Her clearest boundary follows directly from this: she will own the mechanical merge, but she is not qualified to decide which clinical fact is true when two versions of a record disagree — that decision must go to the MO or leprosy programme specialist, with both versions and provenance shown, and nothing silently picked. (Q3, Q6, Q13)

2. No read audit trail — she cannot answer “who has viewed this record”, and the exposure is personal as well as institutional

Nothing she runs today logs a view; only patchy web server access logs exist, with 7–30 day retention, shared logins on the hospital side, and no field-level detail. She assesses that the department could not currently meet its DPDP breach-scope reporting obligation, because a breach that cannot be detected cannot be reported — and she has said so in writing once, without result. Her stated personal concern is not the compliance penalty but the human outcome: having to tell an affected family that she does not know how a disclosure happened. This is why she pressed audit trails over features throughout the session, and why “who has accessed this record” as a supported, non-engineer-usable query is one of her four hard refusal conditions for go-live. (Q7, Q8, Q9, Q13)

3. One administrator for ~1,100 users, with no on-call — which converts every outage into an insecure workaround

Team of one, plus a non-technical data entry operator; peak support demand is 21:00–23:00 when ASHAs do data entry after domestic work; median password reset takes 1.5 days elapsed and costs roughly 1.5 days of degraded, retrospectively-entered data. Credentials are currently delivered over WhatsApp into family-shared chat logs. When she is unreachable, users share credentials, photograph patient records into WhatsApp, delegate entry to family members, or revert to paper — 20–30 device or identity events a month are handled manually, with no device registry, remote wipe or session revocation. Her own argument is the one to carry forward: none of this is carelessness, it is a rational response to a system that fails while the obligation to record remains — the workaround is the symptom, unavailability is the disease. (Q2, Q4, Q5, Q11)

Honourable mentions (real, but second tier)

- Silent integration failure where the field worker is told nothing: the SMS gateway returned “submitted” for messages that were never delivered, undiscovered for 11 days; the dashboard stayed green throughout (Q10).
- Duplicate patient/child records with no reliable identifier — Aadhaar inconsistently captured, names transliterated multiple ways — producing ~1,400 probable duplicate pairs in one register alone, detected only by denominator inflation (Q6).
- Access control with geographic scoping but no vertical (field-level) scoping — within a permitted catchment, a user sees the whole record, because no field has ever been classified by sensitivity (Q7).
- Identity-service availability treated as a hard stop rather than an assumption to design around — ABHA OTP failures turned patients away from registration entirely (Q10).

4. CONTRADICTIONS AND THINGS TO VERIFY

- **Wipe versus do-not-lose** — Security requires immediate wipe of a lost/replaced handset; data integrity forbids wiping before the unsynced queue is confirmed drained, since the old device may hold the only copy. Her preferred resolution — a revoked session may still push pending writes but may read nothing — needs a technical feasibility and cost decision. Unresolved.
- **Security friction versus ASHA usability** — Priya’s floor (PIN at session start + step-up on sensitive fields + ~2-minute idle lock) is her judgement, not the field’s. It has not been tested with ASHAs and may conflict with what the ASHA stakeholder says. Verify with ASHA interviews and a usability trial.
- **Audit trail versus credential sharing** — She wants attribution, and simultaneously states that credential sharing is normal field behaviour that no design fully prevents. The audit log can therefore only ever claim “this device, unlocked with this PIN” — confirm that legal/compliance stakeholders accept an evidentiary standard weaker than user-level certainty.

- **Caseload continuity versus consent** — Transferring a departing ASHA's caseload is operationally necessary but is a bulk disclosure to someone the patient never consented to. Her proposal (MO authorisation, logged as a disclosure) needs programme and legal sign-off.
- **Interoperability versus confidentiality** — If ABDM/FHIR export is in scope, her role-based controls end at the district boundary and she cannot audit downstream access. Requires an explicit decision on what is shared, and whether upward reporting can be aggregate-only.
- **Inactivity-based deprovisioning** — Her 90-day rule disables genuine users who were ill or had a broken phone, while catching real leavers late. Any expiry-by-default mechanism must be validated against real ASHA activity patterns, including seasonal work.
- **Legal obligations stated as a working understanding, not a confirmed one** — Her DPDP / CERT-In understanding (Board and individual notification, no materiality threshold, six-hour cyber-incident clock) was explicitly flagged by her as requiring confirmation by someone with legal standing. Do not carry into requirements unverified.
- **Family-law implications of a leprosy diagnosis** — She referred this to a lawyer, not herself. Verify before using it as a justification in the business case.
- **Clinical items she deferred to the Medical Officer / leprosy programme specialist** — infectiousness and treatment facts; what a clinically safe reconciliation looks like when two records disagree on classification or treatment stage; which fields are genuinely sensitive versus merely clinical.
- **Quantitative figures are estimates from memory** — 1,100 ASHAs / ~700 active, 90 ANMs, 40 MOs, ~600 devices; ~1,400 duplicate pairs in ~40,000 children; 15–20% annual handset change; ~10% ASHA attrition; 60–80 resets per week. Confirm from records before using as a baseline.
- **Block IT coordinator coverage is unconfirmed** — whether “block IT coordinators” exist in 3 of 14 blocks or more affects the feasibility of supervisor-mediated support.

Open decisions to route to others (not the administrator's to make)

- **Named owner and signed-off document for the roles/permissions matrix** — she asked for a person, not “we'll configure it”, warning that access design otherwise defaults to whoever builds the login screen and ossifies.
- **Formal data protection accountability** — whether anyone holds it, or whether it defaults to the sysadmin by omission.

- **Retention periods** — for patient records and, separately, for audit logs (which normally must outlive the record). She has never received a written answer.
- **Offboarding notification as a mandated process** — whether departure can be made to route through IT automatically; requires authority above hers.
- **Scope of upward reporting** — whether reporting to state / national leprosy programme may be aggregate-only, or requires line-listed identifiable data.
- **ABDM linkage and first-line support budget** — whether ABDM linkage is a requirement or an option, and whether budget exists for first-line block-level support capacity.
- **Incident response runbook and notification authority** — whether a runbook is in scope, and who has authority to make a breach notification call at speed.

5. RECORDED CONFLICT: SECURITY AND AUDIT CONTROLS VERSUS FIELD-WORKER SPEED

Put to her directly in Q12, and reinforced by her access-control (Q7) and support-burden (Q4, Q11) answers.

- Security/audit position: the system must be able to prove who recorded and who viewed a given record, at the level of an individual person, to meet confidentiality obligations for the most stigmatising data the department holds.
- Field-worker position (as she frames it on the ASHA's behalf): Sunita covers roughly forty houses a day on a shared, low-cost phone; logging in is friction against that workload; her obligation is to the households in front of her, not to the audit log; and any control that is painful will be defeated within a month, regardless of the security document.
- Her counter-proposal, developed over four years of watching controls fail: separate proving the device from proving the person. Bind the account to a registered device (silent, free after day one); add a short local PIN or biometric, recoverable via the ANM rather than the administrator; keep the session unlocked for a working session with a short idle timeout (~2 minutes); and attach step-up re-authentication only to sensitive actions (viewing leprosy status, exporting data, viewing outside the worker's catchment), not to every screen. Her stated floor is one PIN entry at session start, plus step-up on sensitive access, plus a short idle lock — below that she will not sign off.
- She is explicit that this narrows rather than eliminates the gap: if the phone and PIN are handed to a neighbour, the log is wrong but plausible, so the honest evidentiary claim is “this device, unlocked with this PIN, at this time” — strong evidence, not certainty — mitigated by anomaly flagging (e.g. many records from one account across several villages in an hour) treated as a signal, not proof.

Options to take forward:

- (a) Adopt her proposed floor — device-bound account, single PIN at session start, step-up re-authentication on sensitive fields only, ~2-minute idle timeout — as the version-one authentication model, validated with ASHAs in a usability trial before build sign-off.
- (b) Go further and require stronger per-action authentication (e.g. biometric-only, no shareable PIN) to close the attribution gap more tightly — likely fails her “no reason to circumvent” test and risks the same abandonment pattern seen with the nutrition app; not recommended without field evidence to the contrary.
- (c) Defer session-timeout policy to a generic enterprise security template — explicitly warned against by her, since a template 15-minute timeout would lock a worker out mid-courtyard; not recommended.
- (d) Formally align legal/compliance expectations to the honest evidentiary standard — “this device, unlocked with this PIN, at this time” rather than certain personal identity — before build, so the audit trail is not silently oversold.

Recommendation: adopt (a) as the authentication design baseline, combined with (d) so legal and compliance sign off on the true evidentiary standard in advance; reject (b) and (c).

APPENDIX A — CONSOLIDATED REQUIREMENTS REGISTER (FROM THIS INTERVIEW ONLY)

ID	Requirement / observation (stated as a need)	Type	Source	Priority
FR-85	Field-level, deterministic sync merge — no record-level “most recent timestamp wins”; every field change merged and versioned independently	Functional	Q3, Q13	Must have
FR-86	Full change history retained on every record; nothing overwritten silently; every prior value recoverable	Functional	Q1, Q3, Q13	Must have
FR-87	No local deletion of an entry until server acknowledgement of receipt	Functional	Q1, Q13	Must have
FR-88	Clean conflict presentation when two versions of a clinical fact disagree — both versions plus provenance shown, never auto-resolved	Functional	Q6, Q12	Must have
FR-89	Clinical conflicts (treatment stage, classification) routed to the MO / leprosy programme specialist for decision; the system never resolves these itself	Functional	Q6, Q13	Must have
FR-	Supervisor-mediated (ANM/MO)	Functional	Q3,	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
90	credential and recovery capability, so a reset never requires the district administrator		Q4, Q11, Q13	
FR-91	Device registration as a first-class, visible, revocable object per user	Functional	Q5, Q13	Must have
FR-92	Defined behaviour for a revoked device holding an undrained sync queue — proposed: may still push pending writes, may read nothing	Functional	Q5, Q12	Should have
FR-93	MO authorisation required, and logged as a disclosure event, for bulk caseload reassignment between field workers	Functional	Q5	Must have
FR-94	Append-only access (read and write) logging capturing user, device, record, fields returned and timestamp	Functional	Q8, Q13	Must have
FR-95	Supported query, usable by a non-engineer, answering “who has accessed this record and when” in under five minutes	Functional	Q7, Q8, Q13	Must have
FR-96	Access grants that expire by default, tied to a posting with a review date	Functional	Q7, Q13	Should have
FR-97	Delivery-state visibility for outbound SMS (“delivered” / “not delivered, call her”), with undelivered messages surfaced as a work item, and administrator alerts when a carrier’s delivery rate drops	Functional	Q10	Should have
FR-98	Real host/slot availability checking before a teleconsultation booking is offered — never offer a slot the system cannot honour	Functional	Q10	Must have
FR-99	Defined in-app fallback on teleconsultation failure at point of use — reschedule, voice call, or flag for physical referral — recorded as a clinical event	Functional	Q10	Must have
FR-100	ABHA/identity verification optional at capture: patient created on a local identifier, record marked unverified,	Functional	Q10, Q13	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	linkage possible later; registration and care never blocked by identity-service failure			
FR-101	Client-side error and sync telemetry (crashes, failed syncs, stuck queues) reported automatically to IT without depending on a user complaint	Functional	Q11, Q13	Should have
FR-102	A formal, consented, logged export interface for legitimate third-party data requests (e.g. disability/welfare scheme extracts), replacing ad-hoc spreadsheet exports	Functional	Q10	Should have
FR-103	Assisted duplicate-record match/merge tooling — candidate pairs surfaced, nothing deleted, only superseded, both originals retained	Functional	Q6	Should have
FR-104	A pre-designed, tested notification channel for breach/disclosure communication to affected individuals, avoiding shared-phone SMS or verbal courtyard delivery	Functional	Q9	Should have
NFR-47	Offline-first architecture as a mandatory baseline, not an optional mode	Non-functional	Q1	Must have
NFR-48	System and support model must remain workable at the real operating ratio of ~1,100 users to 1.5 administrators, with no on-call rota	Non-functional	Q2, Q13	Must have
NFR-49	Self-service or supervisor-mediated account recovery — no requirement for night-time administrator availability	Non-functional	Q4, Q11, Q13	Must have
NFR-50	Encryption at rest on the handset, keyed to the user credential	Non-functional	Q5, Q12, Q13	Must have
NFR-51	Hard cap on the volume of caseload data resident on a device — active caseload only, not full block history	Non-functional	Q5, Q12, Q13	Must have
NFR-52	Field-level data sensitivity as a schema property, not a screen/presentation-layer control, so confidentiality is	Non-functional	Q7, Q12, Q13	Must have

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	enforced regardless of which screen displays a field			
NFR-53	Read-audit log storage separate from the application database, so compromising the application cannot alter the trail	Non-functional	Q8	Must have
NFR-54	Read-audit log retention measured in years, not days	Non-functional	Q8, Q13	Must have
NFR-55	Authentication model: device binding (enrolled once, cryptographic key) plus a short local PIN or biometric, recoverable via the ANM rather than the administrator	Non-functional	Q12	Should have
NFR-56	Session persists through a working session (hours of active use) with a short idle timeout (~2 minutes) rather than a fixed generic timeout	Non-functional	Q12	Should have
NFR-57	Step-up (re-)authentication attached only to sensitive actions (viewing leprosy status, exporting data, viewing outside catchment), not to every screen	Non-functional	Q12	Should have
NFR-58	Anomaly flagging on access patterns (e.g. many records from one account across multiple villages within an hour) treated as signal, not proof, given normal credential sharing	Non-functional	Q5, Q11, Q12	Should have
NFR-59	Authorisation enforced server-side in all cases, resistant to record-ID manipulation in requests	Non-functional	Q13	Must have
NFR-60	Every external dependency (SMS gateway, video platform, identity service) degrades gracefully and visibly; no clinical action is ever blocked by a dependency failure	Non-functional	Q10, Q13	Must have
BR-27	Roles and permissions must be defined per system with a named, accountable owner and a signed-off document, not left to default to whoever builds the login screen	Business rule	Q7	Constraint on design

ID	Requirement / observation (stated as a need)	Type	Source	Priority
BR-28	Geographic/organisational scoping (catchment, sub-centre, block) is necessary but not sufficient; sensitivity must also be scoped vertically within a record, not only horizontally by area	Business rule	Q7	Constraint on design
BR-29	Access approval must route through a defined authority (e.g. DHO/MO sign-off) accompanying account creation, not an informal request with paperwork following later	Business rule	Q7	Constraint on design
BR-30	Offboarding/departure should be a mandated process that notifies IT directly, rather than something the administrator learns incidentally	Business rule	Q7	Verify
BR-31	Caseload handover/reassignment between field workers requires MO authorisation and must be logged as a disclosure event, not treated as a routine admin action	Business rule	Q5	Constraint on design
BR-32	Retention periods must be explicitly set and documented, separately, for patient records and for audit logs (audit logs must normally outlive the record)	Business rule	Q8	Verify
BR-33	A named incident-response runbook and a named person with legal authority to make breach-notification decisions at speed must exist before go-live	Business rule	Q9	Verify
CON-30	Support capacity is fixed at one full-time administrator plus one non-technical data entry operator for ~1,100 field users; only ~3 of 14 blocks have any locally IT-capable coordinator	Constraint	Q2	Absolute
CON-31	~20–30 device or identity change events occur per month (handset change, attrition), all currently handled manually by one person	Constraint	Q5	Absolute
CON-32	The field device is typically a personal, family-shared, low-cost Android	Constraint	Q2, Q5,	Absolute

ID	Requirement / observation (stated as a need)	Type	Source	Priority
	handset with low storage — the environment any security or storage design must assume		Q12	
CON-33	No dedicated data protection officer or privacy team exists in the district; any accountability question defaults to the sysadmin	Constraint	Q8, Q9	Absolute
CON-34	Peak field support demand falls at 21:00–23:00 with no on-call rota and nobody but the sole administrator reachable	Constraint	Q4, Q11	Absolute
CON-35	A login/account cannot be reliably equated with a person in this environment, because credential and device sharing is normal, observed field behaviour	Constraint	Q5, Q11, Q12	Absolute
CON-36	Statutory breach-notification obligations (DPDP Act, CERT-In six-hour clock) apply to the department regardless of current detection capability, and cannot currently be met	Constraint	Q9	Absolute
EC-18	Cultural trust in the system, once lost to a sync failure, persists for years (parallel paper records) and is far more expensive to rebuild than to preserve	Emotional/cognitive	Q3	Must design for
EC-19	The administrator's personal exposure — being the one who must tell an affected family "we don't know" how a disclosure happened — drives her prioritisation of audit trails over features	Emotional/cognitive	Q8, Q9	Must design for
EC-20	Field workers' rational workaround behaviour under system unavailability (credential sharing, WhatsApp photographs of patient data, delegating entry to family) must be designed for as a security requirement, not dismissed as carelessness	Emotional/cognitive	Q11	Must design for

Design rules implied, in the stakeholder's own framing

- “Two things you decide early or you sink on” — offline-first is not optional, and confidentiality is a legal and ethical absolute, not a compliance line item.
- “This device, unlocked with this PIN, at this time” — the audit log's honest claim is strong evidence, not certainty; the system's own language should say so accurately rather than overclaim identity.
- “Least data on the most exposed surface” — the point she says she would fight for if budget forced a choice.
- “Presentation-layer confidentiality is not confidentiality” — sensitivity must be a property of the data field, not of whichever screen happens to display it.
- Never block a clinical action on an external dependency — with the video platform, SMS gateway and identity service all unavailable, an ASHA must still be able to record a visit and a dose given.
- Design to “what control has she no reason to circumvent”, not to how much friction a worker will tolerate — the only test that predicts whether a control survives contact with a forty-house day.
- Carry the operating ratio through the requirements document: 1,100 users, one administrator, no on-call. A design that needs a competent administrator awake at 21:00 does not work in this district.
- Trust is far more expensive to rebuild than to keep.
- Clinical questions — classification, treatment stage, safe reconciliation between disagreeing records — go to the Medical Officer or the leprosy programme specialist, not to IT.